

Oscillations

Question1

The displacement of a particle, executing simple harmonic motion with time period T , is expressed as $x(t) = A \sin \omega t$, where A is the amplitude. The maximum value of potential energy of this oscillator is found at $t = T/2\beta$. The value of β is ____ .

JEE Main 2026 (Online) 28th January Morning Shift

Answer: 2

Solution:

The potential energy (U) of a particle executing simple harmonic motion (SHM) is given by :

$$U = \frac{1}{2} m\omega^2 x^2$$

Substituting the displacement equation $x(t) = A \sin \omega t$:

$$U(t) = \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t$$

The potential energy U is maximum when the displacement x is maximum ($x = \pm A$).

This occurs when :

$$\sin^2 \omega t = 1 \Rightarrow \sin \omega t = \pm 1$$

This occurs for a particle starting from the mean position ($t = 0$) is at :

$$\omega t = \frac{\pi}{2}$$

Using the relationship between angular frequency ω and time period (T), $\omega = \frac{2\pi}{T}$

$$\Rightarrow \left(\frac{2\pi}{T}\right)t = \frac{\pi}{2} \Rightarrow t = \frac{T}{4}$$

It is given that $t = \frac{T}{2\beta}$,

$$\frac{T}{4} = \frac{T}{2\beta} \Rightarrow \beta = 2$$

Therefore, the value of β is 2 . So, the correct answer is 2 .

Question2

The velocity of a particle executing simple harmonic motion along x -axis is described as $v^2 = 50 - x^2$, where x represents displacement. If the time period of motion is $\frac{x}{7}$ s, the value of x is ____ .

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Answer: 44

Solution:

For a particle executing SHM with amplitude A and angular frequency ω , the velocity v at a displacement x from the mean position is given by relation,

$$v^2 = \omega^2 (A^2 - x^2)$$

The given equation is $v^2 = 50 - x^2$

$$v^2 = 1 (50 - x^2)$$

By comparing the two equations,

$$\omega^2 = 1 \Rightarrow \omega = 1 \text{ rad/s}$$

$$\Rightarrow A^2 = 50 \Rightarrow A = \sqrt{50} \text{ units}$$

The time period T is related to the angular frequency ω by the formula:

$$T = \frac{2\pi}{\omega}$$

Substituting $\omega = 1$:

$$T = 2\pi \text{ seconds}$$

$$\Rightarrow T = 2 \times \left(\frac{22}{7}\right) = \frac{44}{7} \text{ seconds} = \frac{x}{7} \text{ seconds}$$

$$\Rightarrow x = 44$$

Therefore, the correct answer is 44 .

Question3

The kinetic energy of a simple harmonic oscillator is oscillating with angular frequency of 176 rad/s. The frequency of this simple harmonic oscillator is _____ Hz. [take $\pi = \frac{22}{7}$]

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Options:

A.

88

B.

14

C.

28

D.

176

Answer: B

Solution:

In simple harmonic motion, the kinetic energy oscillates with angular frequency 2ω , where ω is the angular frequency of the oscillator.

Given:

$$2\omega = 176 \text{ rad/s}$$

So,

$$\omega = \frac{176}{2} = 88 \text{ rad/s}$$

Now, frequency is related to angular frequency by:

$$\omega = 2\pi f$$

Therefore,

$$f = \frac{\omega}{2\pi} = \frac{88}{2 \times \frac{22}{7}}$$

$$f = \frac{88}{\frac{44}{7}} = 88 \times \frac{7}{44}$$

$$f = 2 \times 7 = 14 \text{ Hz}$$

So, the correct answer is:

14

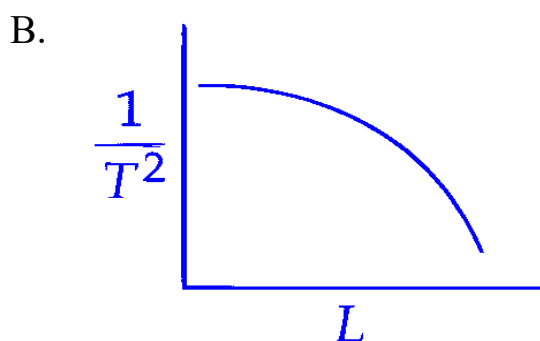
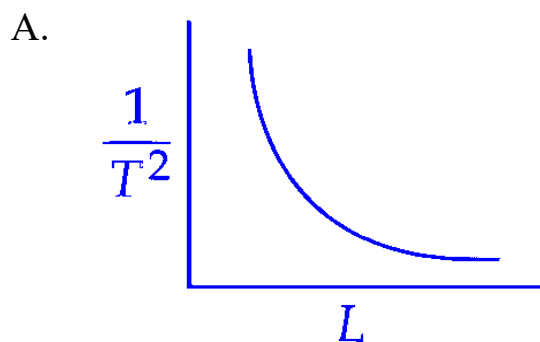
Option B.

Question4

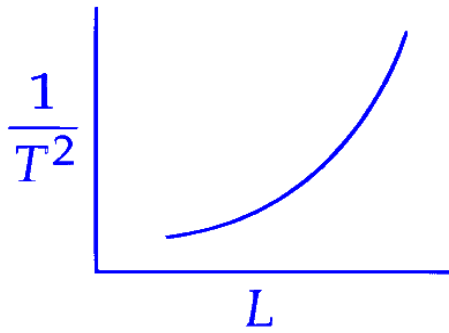
Using a simple pendulum experiment g is determined by measuring its time period T . Which of the following plots represent the correct relation between the pendulum length L and time period T ?

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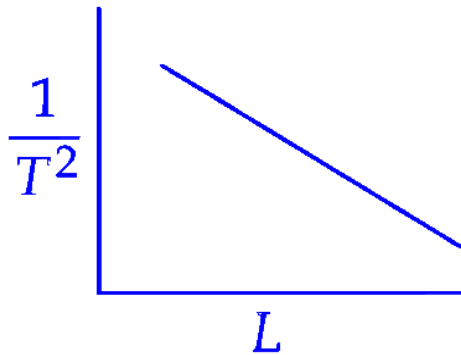
Options:



C.



D.



Answer: A

Solution:

The time period (T) of a simple pendulum of length L is given by the standard formula :

$$T = 2\pi\sqrt{L/g}$$

Where g is the acceleration due to gravity.

The graphs in the options plot $\frac{1}{T^2}$ versus L .

On rearranging the formula for the time period,

$$T^2 = \frac{4\pi^2 L}{g}$$

$$\Rightarrow \frac{1}{T^2} = \frac{g}{4\pi^2 L}$$

$$\Rightarrow \frac{1}{T^2} = \left(\frac{g}{4\pi^2}\right) \frac{1}{L}$$

Since g and $4\pi^2$ are constants, the term $\frac{g}{4\pi^2}$ is just a constant value. Let $\frac{g}{4\pi^2} = k$, where k is any positive constant.

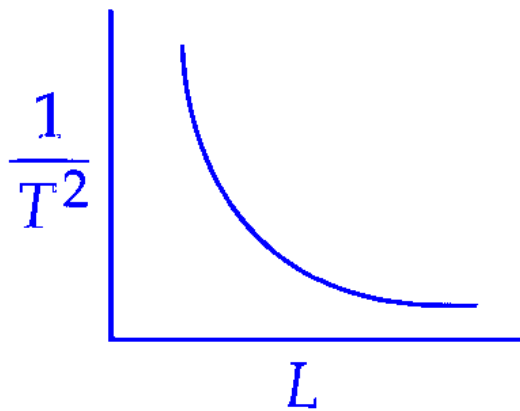
Therefore, the relationship is :

$$y = kx$$

where $y = \frac{1}{T^2}$ and $x = L$.

The mathematical equation $y = kx$ (or $xy = \text{constant}$) represents a rectangular hyperbola, where $\frac{1}{T^2} \rightarrow 0$ if $L \rightarrow \infty$, i.e., x-axis is horizontal asymptote. Also, $\frac{1}{T^2} \rightarrow \infty$ if $L \rightarrow 0$, i.e., y-axis is horizontal asymptote

So, the resulting graph is,



Therefore, the correct option is (A).

Question5

A simple pendulum of string length 30 cm performs 20 oscillations in 10 s . The length of the string required for the pendulum to perform 40 oscillations in the same time duration is

_____ cm . [Assume that the mass of the pendulum remains same.]

JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

0.75

B.

7.5

C.

15

D.

120

Answer: B**Solution:**

The time period (T) of a simple pendulum is the time taken to complete one full oscillation.

For a simple pendulum of the length of the string (L) and the acceleration due to gravity (g) the formula for the time period is,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

So, the time period is directly proportional to the square root of the length ($T \propto \sqrt{L}$).

$$T^2 \propto L \Rightarrow T^2 \propto L$$

Case - 1

The initial length of the string is $L_1 = 30$ cm

$N_1 = 20$ is the number of oscillations completed in total time $t_1 = 10$ s

So, the initial time period of oscillation is

$$T_1 = \frac{\text{Total Time}}{\text{Number of Oscillations}} = \frac{10}{20} = 0.5 \text{ s}$$

Case - 2

$N_2 = 40$ is the number of oscillations completed in the same time $t_2 = t_1 = 10$ s

So, the final time period of oscillation is,

$$T_2 = \frac{\text{Total Time}}{\text{Number of Oscillations}} = \frac{10}{40} = 0.25 \text{ s}$$

We need to find the required length (L_2) corresponding to the final length of the string, Using the proportionality $L \propto T^2$

$$\frac{L_2}{L_1} = \left(\frac{T_2}{T_1} \right)^2$$

$$\Rightarrow \frac{L_2}{30} = \left(\frac{0.25}{0.50} \right)^2 = \frac{1}{4}$$

$$\Rightarrow \frac{L_2}{30} = \frac{1}{4} \Rightarrow L_2 = \frac{30}{4} \text{ cm} = 7.5 \text{ cm}$$

Therefore, the required length of the string is 7.5 cm .

Hence, the correct option is (B).

Question6

A spring of force constant 15 N/m is cut into two pieces. If the ratio of their length is 1 : 3, then the force constant of smaller piece is _____ N/m.

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Options:

A.

20

B.

45

C.

60

D.

15

Answer: C

Solution:

The spring constant k of a spring is inversely proportional to the length L of the spring.

$$k \propto \frac{1}{L}$$

This means that the product of the spring constant and the length is a constant for a given spring:

$$k \cdot L = \text{Constant}$$

The force constant of original spring is $k_o = 15 \text{ N/m}$.

Let the original length be L .

The spring is cut into two pieces with a length ratio of 1 : 3.

$$x + 3x = L \Rightarrow x = \frac{L}{4}$$

Hence, the length of smaller spring is $L_1 = x = \frac{L}{4}$ and the length of larger spring is $L_2 = 3x = \frac{3L}{4}$.

Let k_1 be the force constant of the smaller piece.

Using the inverse proportionality relationship:

$$k_o L_o = k_1 L_1$$

$$\Rightarrow 15 \times L = k_1 \times \left(\frac{L}{4}\right) \Rightarrow k_1 = 60 \text{ N/m}$$

Therefore, the force constant of smaller piece is 60 N/m.

Hence, the correct option is (C).

Question7

A cylindrical block of mass M and area of cross section A is floating in a liquid of density ρ and with its axis vertical. When depressed a little and released the block starts oscillating. The period of oscillation is _____

JEE Main 2026 (Online) 24th January Morning Shift

Options:

A.

$$2\pi\sqrt{\frac{\rho A}{Mg}}$$

B.

$$\pi\sqrt{\frac{2M}{\rho Ag}}$$

C.

$$2\pi\sqrt{\frac{M}{\rho Ag}}$$

D.

$$\pi\sqrt{\frac{\rho A}{Mg}}$$

Answer: C

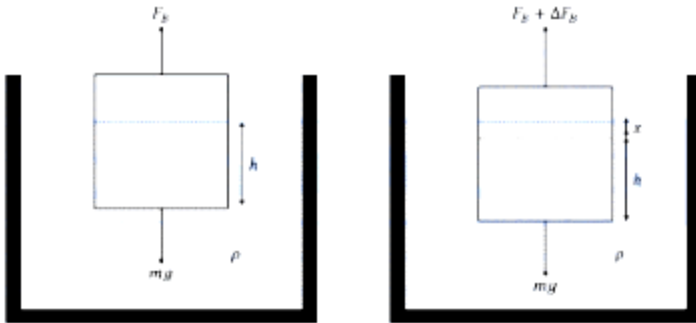
Solution:

Initially, the block is floating. The weight of the block is balanced by the buoyant force (F_B).

Let h be the height of the submerged part of the block.

$$Mg = F_B = V_{\text{submerged}} \times \rho_{\text{liquid}} \times g$$

$$\Rightarrow Mg = (A \times h)\rho g$$



When the block is depressed by a small distance x downwards, the volume of the displaced liquid increases by $A \cdot x$. This creates an extra buoyant force which acts as the restoring force.

The additional buoyant force is :

$$F_{\text{restoring}} = -(\text{Additional Volume}) \times \rho \times g$$

$$F_{\text{restoring}} = -(Ax)\rho g$$

Since the force is proportional to displacement x and acts opposite to it ($F \propto -x$), the motion is Simple Harmonic Motion (SHM).

Using Newton's Second Law ($F = Ma$) :

$$Ma = -(\rho Ag)x$$

$$\Rightarrow a = -(\rho AgM)x.$$

Standard equation for SHM acceleration is $a = -\omega^2 x$

Comparing the two equations:

$$\omega^2 = \frac{\rho Ag}{M} \Rightarrow \omega = \sqrt{\frac{\rho Ag}{M}}$$

The time period T is given by $T = \frac{2\pi}{\omega}$.

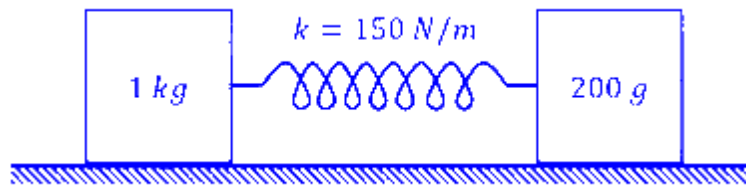
$$\Rightarrow T = 2\pi \sqrt{\frac{M}{\rho Ag}}$$

Hence, the correct option is (C).

Question8

As shown in the figure, a spring is kept in a stretched position with some extension by holding the masses 1 kg and 0.2 kg with a

separation more than spring natural length and are released. Assuming the horizontal surface to be frictionless, the angular frequency (in SI unit) of the system is :



JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

20

B.

5

C.

30

D.

27

Answer: C

Solution:

For a two-body system, the angular frequency (ω) is given by :

$$\omega = \sqrt{\frac{k}{\mu}}$$

Where :

- k is the spring constant.
- μ is the reduced mass of the system.

The reduced mass is calculated using the formula :

$$\mu = \frac{m_1 \cdot m_2}{m_1 + m_2}$$

Given values of masses are :

- $m_1 = 1 \text{ kg}$
- $m_2 = 200 \text{ g} = 0.2 \text{ kg}$

$$\mu = \frac{1 \times 0.2}{1 + 0.2} = \frac{0.2}{1.2} = \frac{1}{6} \text{ kg}$$

So the angular frequency is,

$$\omega = \sqrt{\frac{150}{1/6}}$$

$$\Rightarrow \omega = \sqrt{900} = 30 \text{ rad/s}$$

Therefore, the angular frequency of the system in SI units is 30 . Hence, the correct option is (C).

Question9

The equation of motion of a particle is given by $x = a \sin(50t + \pi/3)$ cm. The particle will come to rest at time t_1 and it will have zero acceleration at time t_2 . The t_1 and t_2 respectively are _____.

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

$$\frac{\pi}{300} \text{ s}, \frac{\pi}{75} \text{ s}$$

B.

$$\frac{\pi}{75} \text{ s}, \frac{\pi}{300} \text{ s}$$

C.

$$\frac{\pi}{300} \text{ s}, \frac{\pi}{25} \text{ s}$$

D.

$$\frac{\pi}{50} \text{ s}, \frac{\pi}{100} \text{ s}$$

Answer: A

Solution:

The velocity v is the first derivative of displacement x with respect to time t .

$$v = \frac{dx}{dt} = \frac{d}{dt} \left[a \sin \left(50t + \frac{\pi}{3} \right) \right]$$

$$v = 50a \cos \left(50t + \frac{\pi}{3} \right)$$

The particle comes to rest when $v = 0$.

$$50a \cos \left(50t_1 + \frac{\pi}{3} \right) = 0 \Rightarrow \cos \left(50t_1 + \frac{\pi}{3} \right) = 0$$

The first positive time occurs when the phase is $\frac{\pi}{2}$:

$$50t_1 + \frac{\pi}{3} = \frac{\pi}{2}$$

$$50t_1 = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$$

$$t_1 = \frac{\pi}{300} \text{ s}$$

The acceleration a is the derivative of velocity v .

$$a = \frac{dv}{dt} = \frac{d}{dt} \left[50a \cos \left(50t + \frac{\pi}{3} \right) \right]$$

$$a = -2500a \sin \left(50t + \frac{\pi}{3} \right)$$

Acceleration is zero when $\sin \left(50t_2 + \frac{\pi}{3} \right) = 0$.

The first positive time (other than the initial state if it is zero) occurs when the phase is π :

$$50t_2 + \frac{\pi}{3} = \pi$$

$$50t_2 = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

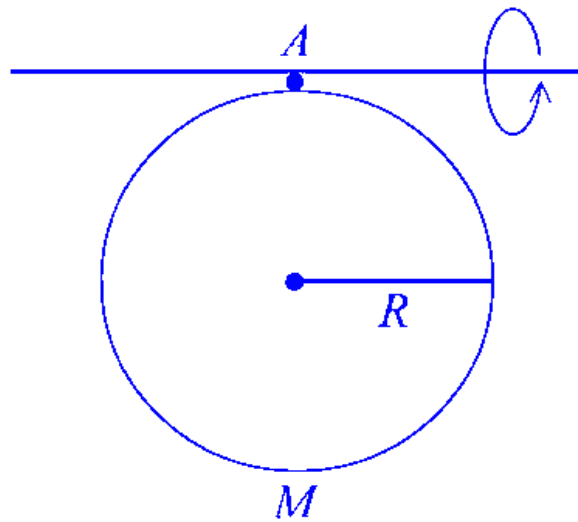
$$t_2 = \frac{2\pi}{150} = \frac{\pi}{75} \text{ s}$$

Therefore, the particle will come to rest time $\frac{\pi}{300}$ s and it will have zero acceleration at time $t_2 \frac{\pi}{75}$ s. Thus, the correct option is (A).

Question10

A uniform disc of radius R and mass M is free to oscillate about the axis A as shown in the figure. For small oscillations the time period is _____ .

(g is acceleration due to gravity)



JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$$2\pi\sqrt{\frac{5R}{4g}}$$

B.

$$2\pi\sqrt{\frac{2R}{3g}}$$

C.

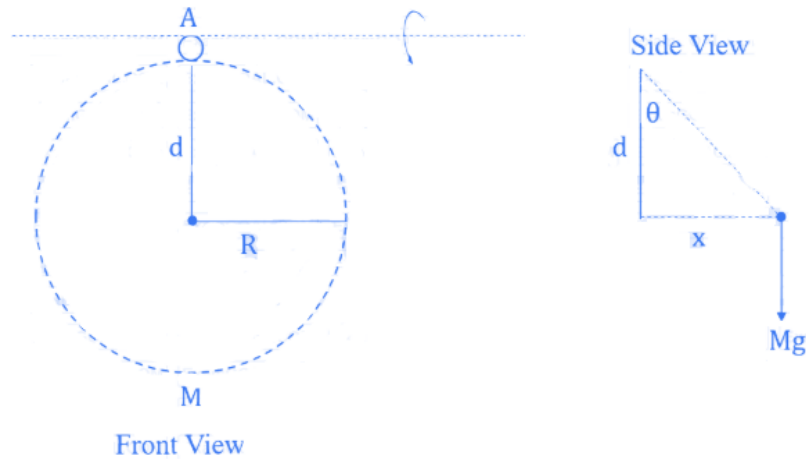
$$2\pi\sqrt{\frac{3R}{2g}}$$

D.

$$2\pi\sqrt{\frac{3R}{g}}$$

Answer: A

Solution:



For a physical pendulum that oscillates about a horizontal axis. For a small angular displacement θ , the restoring torque τ is:

$$\tau = -Mgd \sin \theta$$

For small angular displacement $\sin \theta \approx \theta$

$$\Rightarrow \tau \approx -Mgd \theta$$

Where d is the distance from the pivot A to the centre of mass.

Since $\tau = I\alpha = I \frac{d^2\theta}{dt^2}$, where I is the moment of inertia about the pivot:

$$I \frac{d^2\theta}{dt^2} + Mgd \theta = 0 \Rightarrow \frac{d^2\theta}{dt^2} + \left(\frac{Mgd}{I} \right) \theta = 0$$

This is the equation for SHM with $\omega^2 = \frac{Mgd}{I}$.

So, the time period T of oscillation is:

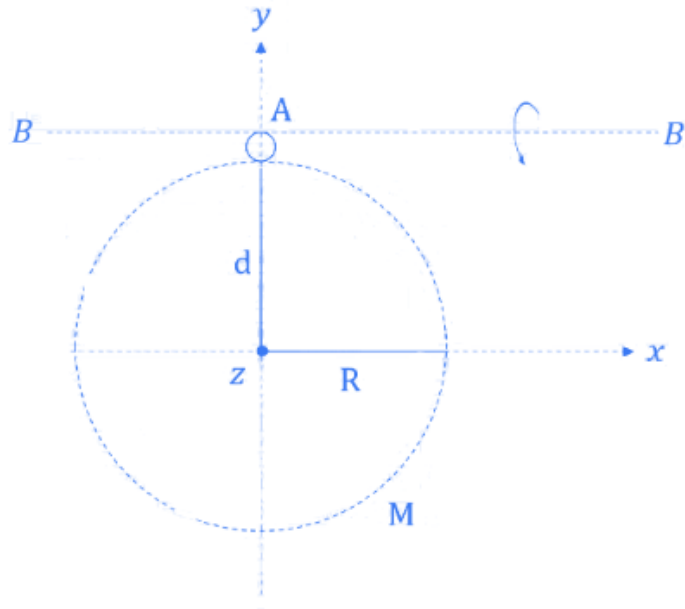
$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{Mgd}}$$

The axis of rotation A is at the edge of the disc.

The moment of inertia of a uniform disc about its centre of mass (axis perpendicular to the plane) is,

$$I_{\text{cm}} = \frac{1}{2}MR^2$$

Using perpendicular axis theorem, if the moment of inertia about diameter of ring is I_d then



$$I_{\text{cm}} = I_z = I_x + I_y = I_d + I_d \Rightarrow I_d = \frac{I_z}{2}$$

$$\Rightarrow I_d = \frac{I_{\text{cm}}}{2} = \frac{\frac{1}{2}MR^2}{2} = \frac{1}{4}MR^2$$

Now, the given axis is parallel to the diameter. So, using the parallel axis theorem $I = I_{\text{cm}} + Md^2$. Since the pivot A is at a distance $d = R$ from the centre :

$$I_A = \frac{1}{4}MR^2 + M(R)^2$$

$$\Rightarrow I_A = \frac{5}{4}MR^2$$

The distance from the pivot A to the centre of mass is $d = R$.

Substituting I_A and d into the time period formula :

$$T = 2\pi\sqrt{\frac{\frac{5}{4}MR^2}{MgR}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{5R}{4g}}$$

Therefore, the time period for small oscillations is $2\pi\sqrt{\frac{5R}{4g}}$.

Thus, the correct option is (A).

Question11

Match List - I with List - II.

List - I		List - II	
A.	$\sin^2 \omega t$	I.	Periodic with time period $T = \frac{\pi}{\omega}$ but not simple harmonic motion (SHM)
B.	$\sin^3(2\omega t)$	II.	Periodic with time period $T = \frac{2\pi}{\omega}$ but Not SHM
C.	$\sin(\omega t) + \cos(\pi\omega t)$	III.	Periodic with time period $T = \frac{\pi}{\omega}$ and SHM
D.	$\cos \omega t + \cos 2\omega t$	IV.	Non-periodic

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

A-III, B-I, C-IV, D-II

B.

A-II, B-I, C-III, D-IV

C.

A-III, B-II, C-IV, D-I

D.

A-II, B-I, C-IV, D-III

Answer: A

Solution:

For the displacement equation $y = \sin^2 \omega t$

Using the trigonometric identity $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$. So, $y = \frac{1}{2} - \frac{1}{2} \cos(2\omega t)$.

For SHM, the acceleration must be proportional to displacement $\left(\frac{d^2 y}{dt^2} = -\omega^2 y \right)$.

From the given equation, the function is shifted by $\frac{1}{2}$ and amplitude is $\frac{1}{2}$, so it oscillates between 0 and 1 .

While the oscillatory part is harmonic, the function represents a periodic motion and simple harmonic motion

The angular frequency is 2ω .

$$T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}.$$

Therefore, it is periodic with $T = \frac{\pi}{\omega}$ and SHM, i.e., A $\rightarrow$ III

For the displacement equation $y = \sin^3(2\omega t)$

Using the identity

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta \Rightarrow \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$$

Here, $\theta = 2\omega t$.

$$\text{So, } y = \frac{3 \sin(2\omega t) - \sin(6\omega t)}{4} = \frac{3}{4} \sin(2\omega t) - \frac{1}{4} \sin(6\omega t).$$

It is a superposition of two periodic functions $y_1 = \frac{3}{4} \sin(2\omega t)$ and $y_2 = -\frac{1}{4} \sin(6\omega t)$.

The time period is the LCM of time periods of both the periodic functions.

The period of first function is $T_1 = 2\pi/2\omega = \frac{\pi}{\omega}$.

The period of second function is $T_2 = \frac{2\pi}{6\omega} = \frac{\pi}{3\omega}$.

So, the time period of function is $T = \frac{\pi}{\omega}$.

As it contains multiple frequencies, so it is not SHM.

Therefore, it is a periodic function with time period $T = \frac{\pi}{\omega}$ but not SHM. (B $\rightarrow$ I)

For the displacement equation $y = \sin(\omega t) + \cos(\pi\omega t)$

It is a sum of two periodic functions $y_1 = \sin(\omega t)$ and $y_2 = \cos(\pi\omega t)$

It will be periodic only if the ratio of their time periods is a rational number.

$$\omega_1 = \omega \Rightarrow T_1 = \frac{2\pi}{\omega}$$

$$\omega_2 = \pi\omega \Rightarrow T_2 = \frac{2\pi}{\pi\omega} = \frac{2}{\omega}$$

$$\text{Ratio } \frac{T_1}{T_2} = \frac{2\pi/\omega}{2/\omega} = \pi$$

Since π is an irrational number, the function is non-periodic. (C $\rightarrow$ IV)

For the displacement equation $y = \cos \omega t + \cos 2\omega t$

It is also the sum of two periodic function $y_1 = \cos \omega t$ and $y_2 = \cos 2\omega t$

$$\omega_1 = \omega \Rightarrow T_1 = \frac{2\pi}{\omega} \text{ and } \omega_2 = 2\omega \Rightarrow T_2 = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}.$$

The ratio of time periods is, $\frac{T_1}{T_2} = 2$

The ratio is rational. The fundamental period is $T = \frac{2\pi}{\omega}$.

So, it is a periodic with time period $T = \frac{\pi}{\omega}$ (D $\rightarrow$ III)

Therefore, the correct match is A - II, B - I, C - IV, D - III.

Hence, the correct option is (D).

Question12

A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec . The time required by it to move from $x = A$ to $x = \frac{A}{\sqrt{2}}$ is _____ sec.

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

1/4

B.

5/4

C.

5/8

D.

3/8

Answer: C

Solution:

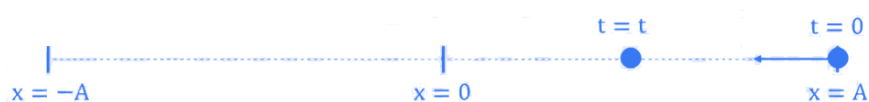
For a particle starting from the extreme position ($x = A$) at $t = 0$, the displacement x at any time t is given by:

$$x = A \cos(\omega t)$$

Where ω is the angular frequency, defined as $\omega = \frac{2\pi}{T}$.

The given time period is $T = 5\text{sec}$:

$$\omega = \frac{2\pi}{5} \text{ rad/s}$$



The particle moves to $x = \frac{A}{\sqrt{2}}$. Substituting into the displacement equation:

$$\frac{A}{\sqrt{2}} = A \cos\left(\frac{2\pi}{5}t\right)$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \cos\left(\frac{2\pi}{5}t\right)$$

Since $\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$:

$$\frac{\pi}{4} = \frac{2\pi}{5}t$$

$$\Rightarrow t = \frac{5}{8}\text{sec}$$

Therefore, the correct option is (C).

Question13

A spring stretches by 2 mm when it is loaded with a mass of 200 g . From equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and maximum energy in the spring are _____ Hz and _____ J, respectively.

(Take $g = 10 \text{ m/s}^2$)

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Options:

A.

$$\frac{5\sqrt{50}}{\pi} \text{ and } 8 \times 10^{-3}$$

B.

$$\frac{5\sqrt{50}}{\pi} \text{ and } 8$$

C.

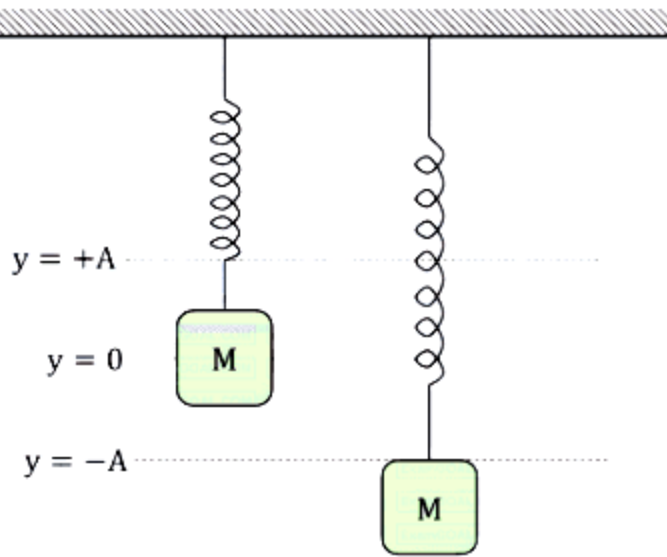
$$10\sqrt{50} \text{ and } 2 \times 10^{-3}$$

D.

$$\frac{5\sqrt{50}}{\pi} \text{ and } 16 \times 10^{-3}$$

Answer: A

Solution:



When the mass is at rest (equilibrium), the downward gravitational force is balanced by the upward spring force.

$$mg = kx_0$$

Where, mass of the suspended load is $m = 200 \text{ g} = 0.2 \text{ kg}$,

The acceleration due to gravity is $g = 10 \text{ m/s}^2$,

The extension in spring at equilibrium position is $x_0 = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

Putting the values,

$$k = \frac{mg}{x_0} = \frac{0.2 \times 10}{2 \times 10^{-3}} = \frac{2}{2} \times 10^{-3} = 1000 \text{ N/m}$$

The angular frequency (ω) of a mass-spring system is derived from the restoration force $F = -kx$.

Since $F = ma$, we have $ma + kx = 0$,

$$\omega = \sqrt{\frac{k}{m}}$$

The linear frequency is $f = \frac{\omega}{2\pi}$:

$$f = \frac{1}{2\pi} \sqrt{\frac{1000}{0.2}} = \frac{1}{2\pi} \sqrt{5000}$$

$$\Rightarrow f = \frac{1}{2\pi} \sqrt{100 \times 50} = \frac{10\sqrt{50}}{2\pi} = \frac{5\sqrt{50}}{\pi} \text{ Hz}$$

The mass is pulled down a further 2 mm from equilibrium and released. This additional displacement is the amplitude (A).

$$A = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$$

The maximum energy in the spring is the at the point of maximum extension, i.e.,

$$x_{\max} = 4 \text{ mm} = 4 \times 10^{-3} \text{ m}$$

$$E = \frac{1}{2} k x_{\max}^2$$

$$\Rightarrow E = \frac{1}{2} \times 1000 \times (4 \times 10^{-3})^2$$

$$\Rightarrow E = 500 \times (16 \times 10^{-6}) = 8000 \times 10^{-6} = 8 \times 10^{-3} \text{ J}$$

Therefore, the frequency associated with the system is $\frac{5\sqrt{50}}{\pi}$ Hz and the maximum energy in the spring is 8×10^{-3} J.

Thus, the correct option is (A).

Question14

The frequency of oscillation of a mass m suspended by a spring is v_1 . If the length of the spring is cut to half, the same mass oscillates with frequency v_2 . The value of v_2/v_1 is ____ .

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

1

B.

2

C.

$\sqrt{2}$

D.

$\sqrt{3}$

Answer: C

Solution:

When a mass m is attached to a spring with spring constant k , the restoring force is given by Hooke's law

$$F = -kx$$

According to Newton's Second Law,

$$m \frac{d^2x}{dt^2} = -kx \Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Comparing this to the standard SHM equation $\frac{d^2x}{dt^2} + \omega^2x = 0$, we find the angular frequency as,

$$\omega = \sqrt{\frac{k}{m}}$$

Since frequency $v = \frac{\omega}{2\pi}$, the frequency of oscillation is:

$$v = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$\Rightarrow v \propto \sqrt{k}$$

A spring constant k is inversely proportional to the natural length L of the spring $\left(k \propto \frac{1}{L}\right)$.

$$k \cdot L = \text{constant}$$

Let the initial length be L_1 and initial spring constant be k_1 .

The new length is $L_2 = \frac{L_1}{2}$.

Let the new spring constant is k_2 .

$$k_1 L_1 = k_2 L_2 \Rightarrow k_1 L_1 = k_2 \left(\frac{L_1}{2}\right)$$

$$\Rightarrow k_2 = 2k_1$$

Using the relation between spring constant and frequency of oscillation,

$$\frac{v_2}{v_1} = \frac{\frac{1}{2\pi} \sqrt{\frac{k_2}{m}}}{\frac{1}{2\pi} \sqrt{\frac{k_1}{m}}}$$

$$\Rightarrow \frac{v_2}{v_1} = \sqrt{\frac{k_2}{k_1}}$$

Substituting $k_2 = 2k_1$:

$$\frac{v_2}{v_1} = \sqrt{\frac{2k_1}{k_1}}$$

$$\Rightarrow \frac{v_2}{v_1} = \sqrt{2}$$

Therefore, the value of the ratio $\frac{v_2}{v_1}$ is $\sqrt{2}$.

Hence, the correct option is (C).

Question15

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : A simple pendulum is taken to a planet of mass and radius, 4 times and 2 times, respectively, than the Earth. The time period of the pendulum remains same on earth and the planet.

Reason (R) : The mass of the pendulum remains unchanged at Earth and the other planet.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. (A) is false but (R) is true
- C. (A) is true but (R) is false
- D. Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Answer: D

Solution:

Assertion (A): A simple pendulum transported to a planet where the mass and radius are 4 times and 2 times that of the Earth, respectively, has the same time period as it does on Earth.

Reason (R): The mass of the pendulum remains unchanged whether on Earth or the other planet.

Explanation:

The acceleration due to gravity g on a planet is given by the formula:

$$g = \frac{GM}{R^2}$$

where G is the gravitational constant, M is the mass of the planet, and R is the radius of the planet.

For the new planet, the gravitational acceleration g' is:

$$g' = \frac{G(4M)}{(2R)^2} = \frac{4GM}{4R^2} = \frac{GM}{R^2} = g$$

Thus, the gravitational acceleration on this new planet is the same as on Earth, g .

The time period T of a simple pendulum is determined by:

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

where ℓ is the length of the pendulum, which indicates that the time period T is independent of the mass of the pendulum.

Therefore, while Assertion (A) is true and Reason (R) is true, Reason (R) does not correctly explain Assertion (A) because the time period of the pendulum is also independent of the pendulum's mass, and the key factor is the unchanged gravitational acceleration.

Question16

A light hollow cube of side length 10 cm and mass 10 g , is floating in water. It is pushed down and released to execute simple harmonic oscillations. The time period of oscillations is $y\pi \times 10^{-2}$ s, where the value of y is(Acceleration due to gravity, $g = 10 \text{ m/s}^2$, density of water = 10^3 kg/m^3)

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

- A. 2
- B. 4
- C. 1
- D. 6

Answer: A

Solution:

We can determine the time period of the oscillations by considering that when the cube is depressed by a small displacement, the additional buoyant force provided by the displaced water acts as a restoring force. Here's a step-by-step explanation:

Define the given values:

Side length of the cube, $L = 10 \text{ cm} = 0.1 \text{ m}$.

Mass of the cube, $m = 10 \text{ g} = 0.01 \text{ kg}$.

Density of water, $\rho = 10^3 \text{ kg/m}^3$.

Acceleration due to gravity, $g = 10 \text{ m/s}^2$.

Cross-sectional area of the cube (face area),

$$A = L^2 = (0.1)^2 = 0.01 \text{ m}^2.$$

When the cube is depressed by a small distance x , the additional volume of water displaced is

$$\Delta V = A x.$$

Thus, the additional buoyant force is:

$$F_b = \rho g \Delta V = \rho g A x.$$

This force acts in the upward (restoring) direction. Notice that the force is proportional to the displacement x , which is the hallmark of simple harmonic motion (SHM).

The effective spring constant, k , associated with this SHM is:

$$k = \rho g A.$$

The period of oscillation for SHM is given by:

$$T = 2\pi \sqrt{\frac{m}{k}},$$

which, upon substituting for k , becomes:

$$T = 2\pi \sqrt{\frac{m}{\rho g A}}.$$

Substitute the given values into the expression:

$$T = 2\pi \sqrt{\frac{0.01}{1000 \times 10 \times 0.01}}.$$

Calculate the denominator:

$$1000 \times 10 \times 0.01 = 100,$$

so,

$$T = 2\pi \sqrt{\frac{0.01}{100}} = 2\pi \sqrt{0.0001}.$$

Since,

$$\sqrt{0.0001} = 0.01,$$

we have:

$$T = 2\pi \times 0.01 = 0.02\pi \text{ s.}$$

The problem states that the time period is given by:

$$y\pi \times 10^{-2} \text{ s.}$$

Writing our result in the same form:

$$0.02\pi \text{ s} = 2\pi \times 10^{-2} \text{ s.}$$

We see that $y = 2$.

Therefore, the correct answer is:

Option A: 2.

Question17

A particle is executing simple harmonic motion with time period 2 s and amplitude 1 cm . If D and d are the total distance and displacement covered by the particle in 12.5 s , then $\frac{D}{d}$ is

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. 10

B. $\frac{16}{5}$

C. 25

D. $\frac{15}{4}$

Answer: C

Solution:

$$x(t) = \cos(\pi t)$$

The particle executes simple harmonic motion (SHM) with amplitude $A = 1$ cm and period $T = 2$ s. In one complete cycle (2 s), the motion is as follows:

It starts at $x = 1$ cm.

It moves to $x = -1$ cm (covering a distance of 2 cm).

It returns to $x = 1$ cm (covering another 2 cm).

Thus, the total distance traveled in one complete cycle is:

$$D_{\text{cycle}} = 2 + 2 = 4 \text{ cm.}$$

In 12.5 s, the number of cycles is:

$$\frac{12.5}{2} = 6.25 \text{ cycles.}$$

For the 6 complete cycles (12 s), the distance traveled is:

$$D_{\text{complete}} = 6 \times 4 = 24 \text{ cm.}$$

For the remaining 0.5 s, determine the displacement. At $t = 12$ s, the particle is at:

$$x(12) = \cos(12\pi) = 1 \text{ cm.}$$

After an extra 0.5 s (i.e., at $t = 12.5$ s), the position becomes:

$$x(12.5) = \cos(12.5\pi) = \cos\left(12\pi + \frac{\pi}{2}\right) = \cos\frac{\pi}{2} = 0 \text{ cm.}$$

The distance covered in this interval is:

$$D_{\text{extra}} = |1 - 0| = 1 \text{ cm.}$$

So, the total distance traveled is:

$$D = D_{\text{complete}} + D_{\text{extra}} = 24 + 1 = 25 \text{ cm.}$$

The net displacement, which is the difference between the final and initial positions, is calculated as follows:

Initial position at $t = 0$:

$$x(0) = \cos 0 = 1 \text{ cm,}$$

Final position at $t = 12.5$ s:

$$x(12.5) = 0 \text{ cm.}$$

Thus, the displacement is:

$$d = |0 - 1| = 1 \text{ cm.}$$

The ratio of the total distance traveled to the displacement is:

$$\frac{D}{d} = \frac{25}{1} = 25.$$

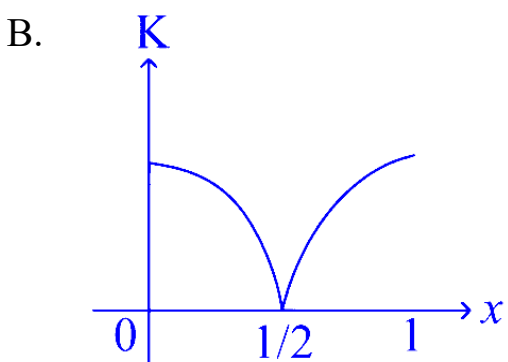
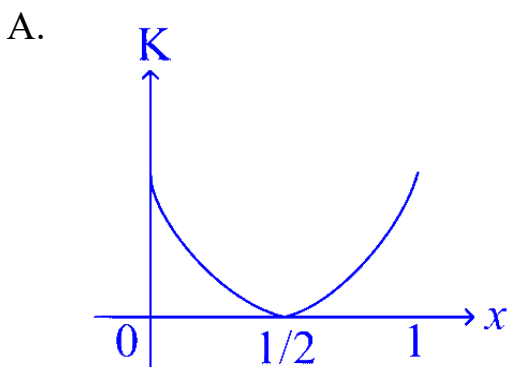
Therefore, the correct answer is 25.

Question18

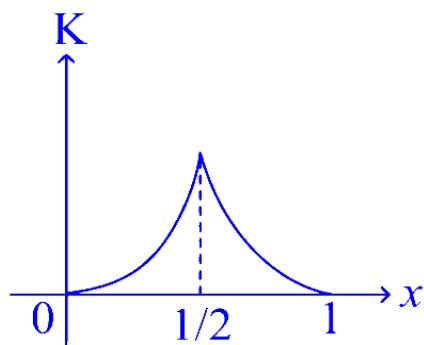
A particle oscillates along the x -axis according to the law, $x(t) = x_0 \sin^2\left(\frac{t}{2}\right)$ where $x_0 = 1$ m. The kinetic energy (K) of the particle as a function of x is correctly represented by the graph

JEE Main 2025 (Online) 24th January Evening Shift

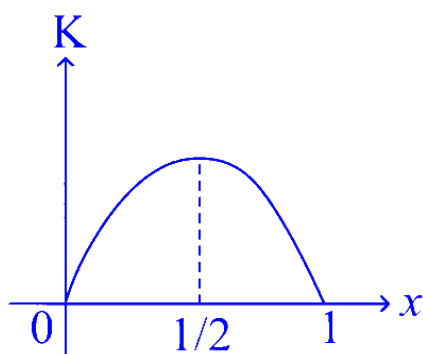
Options:



C.



D.



Answer: D

Solution:

$$x = x_0 \sin^2 \frac{t}{2} = x_0 \left(\frac{1 - \cos t}{2} \right)$$

$$x - \frac{x_0}{2} = \frac{-\cos t}{2}$$

where $x_0 = 1$

$$x - \frac{1}{2} = \frac{-\cos t}{2}$$

Particle is oscillating between $x = 0$ to $x = 1$

Question19

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Knowing initial position x_0 and initial momentum p_0 is enough to determine the position and momentum at any time t for a simple harmonic motion with a given angular frequency ω .

Reason (R) : The amplitude and phase can be expressed in terms of X_0 and p_0 .

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

(A) is true but (R) is false

B.

Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

C.

(A) is false but (R) is true

D.

Both (A) and (R) are true and (R) is the correct explanation of (A)

Answer: D

Solution:

$$x = A \sin(\omega t + \phi)$$

$$x_0 = A \sin \phi \quad \dots (1)$$

$$p = mA\omega \cos(\omega t + \phi)$$

$$p_0 = mA\omega \cos \phi \quad \dots (2)$$

$$(2)/(1) \Rightarrow \tan \phi = \left(\frac{x_0}{p_0} \right) m\omega$$

$$\sin \phi = \frac{x_0 m\omega}{\sqrt{(m\omega x_0)^2 + p_0^2}}$$

$$\text{From (1), } A = \frac{x_0}{\sin \phi} = \frac{\sqrt{(m\omega x_0)^2 + p_0^2}}{m\omega}$$

This means we can explain assertion with the given reason.

Question20

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Time period of a simple pendulum is longer at the top of a mountain than that at the base of the mountain.

Reason (R) : Time period of a simple pendulum decreases with increasing value of acceleration due to gravity and vice-versa.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

Both (A) and (R) are true but (R) is not the correct explanation of (A).

B.

(A) is true but (R) is false.

C.

Both (A) and (R) are true and (R) is the correct explanation of (A).

D.

(A) is false but (R) is true.

Answer: C

Solution:

As h increases, g decreases, T increases

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$
$$g = \frac{g_0 R^2}{(R + h)^2}$$

Question21

Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2 , respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

$$\sqrt{\frac{k_2}{k_1}}$$

B.

$$\sqrt{\frac{k_1}{k_2}}$$

C.

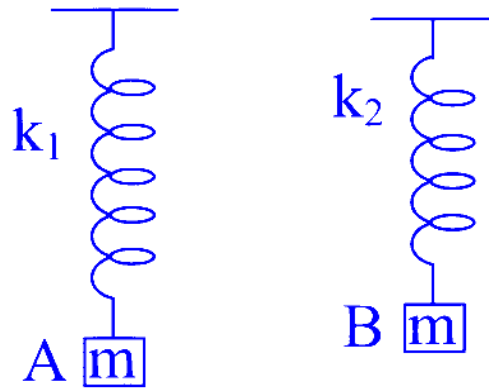
$$\frac{k_2}{k_1}$$

D.

$$\frac{k_1}{k_2}$$

Answer: B

Solution:



We know, for an oscillation,

$$V_{\max} = wA \dots (1)$$

where, A = amplitude

w = frequency of oscillation

$$\text{and } w = \sqrt{\frac{k}{m}} \dots (2)$$

$$\text{from (1) and (2), } V_{\max} = \sqrt{\frac{k}{m}} A$$

$$\Rightarrow V_{\max} \propto \sqrt{k} \text{ (As m and A are same (or constant))}$$

$$\Rightarrow \frac{(V_{\max})_A}{(V_{\max})_B} = \sqrt{\frac{k_1}{k_2}}$$

Hence, option 2 is correct.

Question22

A particle is subjected to two simple harmonic motions as :

$x_1 = \sqrt{7} \sin 5t$ cm and $x_2 = 2\sqrt{7} \sin \left(5t + \frac{\pi}{3}\right)$ cm where x is displacement and t is time in seconds. The maximum acceleration of the particle is $x \times 10^{-2} \text{ ms}^{-2}$. The value of x is :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. $5\sqrt{7}$

B. 125

C. $25\sqrt{7}$

D. 175

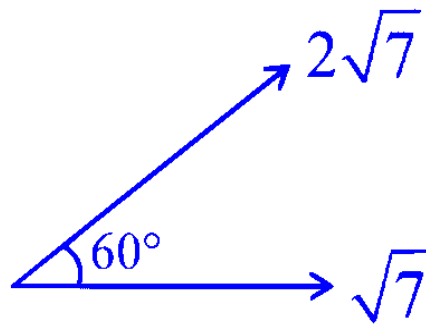
Answer: D

Solution:

$$x_1 = \sqrt{7} \sin 5t$$

$$x_2 = 2\sqrt{7} \sin \left(5t + \frac{\pi}{3} \right)$$

From phasor,



$\therefore$ Amplitude of resultant SHM = 7

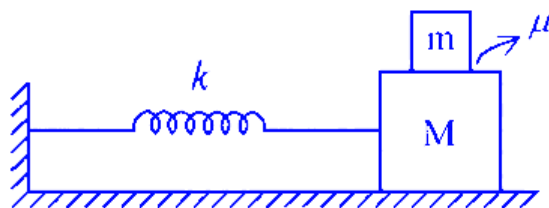
$$\phi = \tan^{-1} \frac{2\sqrt{7} \times \sqrt{3}/2}{\sqrt{7} + 2\sqrt{7} \times \frac{1}{2}} = \tan^{-1} \frac{\sqrt{21}}{2\sqrt{7}} = \tan^{-1} \frac{\sqrt{3}}{2}$$

$$\therefore X_R = 7 \sin(5t + \phi)$$

$$a_R = -7 \times 25 \sin(5t + \phi)$$

$$\therefore a_{\max} = 175 \text{ cm/sec} = 175 \times 10^{-2} \text{ m/sec}$$

Question23



Two blocks of masses m and M , ($M > m$), are placed on a frictionless table as shown in figure. A massless spring with spring constant k is attached with the lower block. If the system is slightly displaced and released, then (μ = coefficient of friction between the two blocks)

A. The time period of small oscillation of the two blocks is

$$T = 2\pi\sqrt{\frac{(m+M)}{k}}$$

B. The acceleration of the blocks is $a = -\frac{kx}{M+m}$ (x = displacement of the blocks from the mean position)

C. The magnitude of the frictional force on the upper block is $\frac{m\mu|x|}{M+m}$

D. The maximum amplitude of the upper block, if it does not slip, is $\frac{\mu(M+m)g}{k}$

E. Maximum frictional force can be $\mu(M + m)g$.

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. B, C, D Only

B. C, D, E Only

C. A, B, D Only

D. A, B, C Only

Answer: C

Solution:

(A) As both blocks moving together so

$$\text{Time period} = 2\pi\sqrt{\frac{m}{K}}; \text{ where } m = M + m$$

$$T = 2\pi\sqrt{\frac{M + m}{K}}$$

(B) Let block is displaced by x in (+ve) direction so force on block will be in(-ve) direction

$$F = -Kx$$

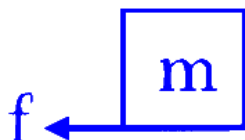
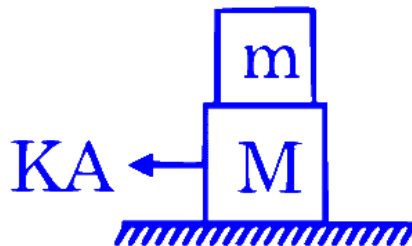
$$(M + m)a = -Kx$$

$$a = -\frac{Kx}{(M+m)}$$

(C) As upper block is moving due to friction thus

$$f = ma = \frac{mKx}{(M+m)}$$

(D) This option is like two block problem in friction for maximum amplitude, force on block is also maximum, for which both blocks are moving together.



$$KA = (M + m)a$$

$$a = \frac{KA}{(M + m)}$$

$$f = ma = \frac{mKA}{(M+m)}$$

$$f_{\max} = f_L = \mu mg$$

$$f = \mu mg$$

$$\frac{mKA}{(M+m)} = \mu mg$$

$$A = \frac{\mu(M+m)g}{K}$$

(E) Maximum friction can be μmg as force is acting between blocks & normal force here is mg .

Question24

Two simple pendulums having lengths l_1 and l_2 with negligible string mass undergo angular displacements θ_1 and θ_2 , from their mean positions, respectively. If the angular accelerations of both pendulums are same, then which expression is correct?

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $\theta_1 l_2 = \theta_2 l_1$

B. $\theta_1 l_1 = \theta_2 l_2$

C. $\theta_1 l_2^2 = \theta_2 l_1^2$

D. $\theta_1 l_1^2 = \theta_2 l_2^2$

Answer: A

Solution:

Angular Frequency: The angular frequency (ω) of a simple pendulum is given by:

$$\omega = \sqrt{\frac{g}{\ell}}$$

where g is the acceleration due to gravity and ℓ is the pendulum length.

Angular Acceleration: The angular acceleration (α) can be expressed in terms of angular displacement (θ) and angular frequency:

$$\alpha = -\omega^2\theta$$

Equating Angular Accelerations: Since the angular accelerations of the two pendulums are equal, we equate them:

$$\frac{g}{\ell_1}\theta_1 = \frac{g}{\ell_2}\theta_2$$

Simplifying the Expression: By canceling out g on both sides, we derive:

$$\theta_1\ell_2 = \theta_2\ell_1$$

Thus, the correct expression that relates the displacements and lengths of the pendulums is $\theta_1\ell_2 = \theta_2\ell_1$.

Question25

A block of mass 2 kg is attached to one end of a massless spring whose other end is fixed at a wall. The spring-mass system moves on a frictionless horizontal table. The spring's natural length is 2 m and spring constant is 200 N/m. The block is pushed such that the length of the spring becomes 1 m and then released. At distance x m ($x < 2$) from the wall, the speed of the block will be

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

$$10[1 - (2 - x)^2]^{\frac{1}{2}} \text{ m/s}$$

B.

$$10[1 - (2 - x)^2]^{\frac{3}{2}} \text{ m/s}$$

C.

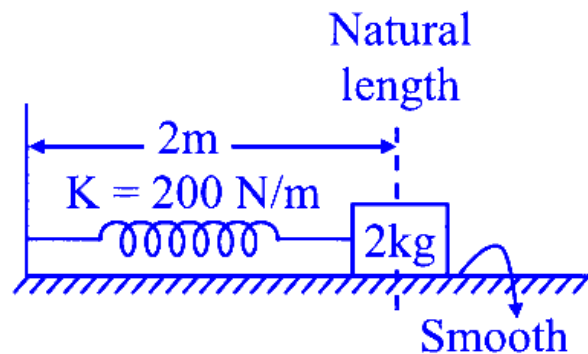
$$10[1 - (2 - x)^2] \text{ m/s}$$

D.

$$10[1 - (2 - x)^2]^2 \text{ m/s}$$

Answer: A

Solution:



Given:

Natural length of the spring: 2 meters

Initial compression of the spring (x_i): 1 meter

Final compression of the spring when the block is at distance x (x_f): $(2 - x)$ meters

Solution:

To find the block's speed, we'll use energy conservation. Initially, the spring's potential energy is fully converted to kinetic energy and spring potential energy at the new position x .

Conservation of Energy:

Initial kinetic energy (K_i) is 0 since the block starts from rest.

Initial potential energy (U_i) due to spring compression is $\frac{1}{2} K x_i^2$.

At a distance x , the kinetic energy (K_f) is $\frac{1}{2} m v^2$ and the potential energy (U_f) is $\frac{1}{2} K x_f^2$.

$$K_i + U_i = K_f + U_f$$

$$0 + \frac{1}{2} K x_i^2 = \frac{1}{2} m v^2 + \frac{1}{2} K x_f^2$$

Rearrange to find v^2 :

$$\frac{1}{2} m v^2 = \frac{1}{2} K (x_i^2 - x_f^2)$$

Substitute known values:

$$\frac{1}{2} \times 2 \times v^2 = \frac{1}{2} \times 200 \times (1^2 - (2 - x)^2)$$

Simplify and solve for v^2 :

$$v^2 = 100 [1 - (2 - x)^2]$$

Finally, take the square root to find v :

$$v = 10[1 - (2 - x)^2]^{1/2}$$

Therefore, the speed of the block at distance x from the wall is given by the expression above.

Question26

A particle executes simple harmonic motion with an amplitude of 4cm. At the mean position, velocity of the particle is 10cm/s. The distance of the particle from the mean position when its speed becomes 5cm/s is $\sqrt{\alpha}$ cm, where $\alpha = \underline{\hspace{1cm}}$

[27-Jan-2024 Shift 1]

Answer: 12

Solution:

$$V_{\text{at mean position}} = A\omega \Rightarrow 10 = 4\omega$$

$$\omega = \frac{5}{2}$$

$$V = \omega \sqrt{A^2 - x^2}$$

$$5 = \frac{5}{2} \sqrt{4^2 - x^2} \Rightarrow x^2 = 16 - 4$$

$$x = \sqrt{12} \text{ cm}$$

Question27

A ball suspended by a thread swings in a vertical plane so that its magnitude of acceleration in the extreme position and lowest position are equal. The angle (θ) of thread deflection in the extreme position will be :

[27-Jan-2024 Shift 2]

Options:

A.

$$\tan^{-1}(\sqrt{2})$$

B.

$$2\tan^{-1}(1/2)$$

C.

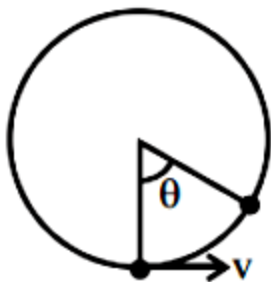
$$\tan^{-1}(1/2)$$

D.

$$2\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

Answer: B

Solution:



Loss in kinetic energy = Gain in potential energy

$$\Rightarrow \frac{1}{2}mv^2 = mg\ell(1 - \cos \theta)$$

$$\Rightarrow \frac{v^2}{\ell} = 2g(1 - \cos \theta)$$

$$\text{Acceleration at lowest point} = -\frac{v^2}{\ell}$$

$$\text{Acceleration at extreme point} = g\sin \theta$$

$$\text{Hence, } \frac{v^2}{\ell} = g\sin \theta$$

$$\therefore \sin \theta = 2(1 - \cos \theta)$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{1}{2} \Rightarrow \theta = 2\tan^{-1}\left(\frac{1}{2}\right)$$

Question28

When the displacement of a simple harmonic oscillator is one third of its amplitude, the ratio of total energy to the kinetic energy is $x/8$, where $x = \underline{\hspace{2cm}}$

[29-Jan-2024 Shift 1]

Answer: 9

Solution:

$$\text{Let total energy} = E = \frac{1}{2}KA^2$$

$$U = \frac{1}{2}K\left(\frac{A}{3}\right)^2 = \frac{KA^2}{2 \times 9} = \frac{E}{9}$$

$$KE = E - \frac{E}{9} = \frac{8E}{9}$$

$$\text{Ratio } \frac{\text{Total}}{\text{KE}} = \frac{E}{\frac{8E}{9}} = \frac{9}{8}$$

$$X = 9$$

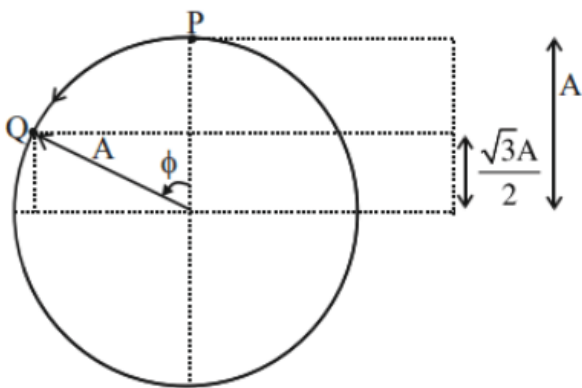
Question29

A simple harmonic oscillator has an amplitude A and time period 6π second. Assuming the oscillation starts from its mean position, the time required by it to travel from $x = A$ to $x = \sqrt{3}/2 A$ will be π/x s, where $x = \underline{\hspace{2cm}}$:

[29-Jan-2024 Shift 2]

Answer: 2

Solution:



From phasor diagram particle has to move from P to Q in a circle of radius equal to amplitude of SHM.

$$\cos \phi = \frac{\frac{\sqrt{3}A}{2}}{A} = \frac{\sqrt{3}}{2}$$

$$\phi = \frac{\pi}{6}$$

Now, $\frac{\pi}{6} = \omega t$

$$\frac{\pi}{6} = \frac{2\pi}{T}t$$

$$\frac{\pi}{6} = \frac{2\pi}{6\pi}t$$

$$t = \frac{\pi}{2}$$

So, $x = 2$

Question30

A simple pendulum is placed at a place where its distance from the earth's surface is equal to the radius of the earth. If the length of the string is 4m, then the time period of small oscillations will be _____.
[. take $g = \pi^2 \text{ ms}^{-2}$]

[30-Jan-2024 Shift 2]

Answer: 8

Solution:

Acceleration due to gravity $g' = \frac{g}{4}$

$$T = 2\pi \sqrt{\frac{4\ell}{g}}$$

$$T = 2\pi \sqrt{\frac{4 \times 4}{g}}$$

$$T = 2\pi \frac{4}{\pi} = 8 \text{ s}$$

Question31

A particle performs simple harmonic motion with amplitude A . Its speed is increased to three times at an instant when its displacement is $2A/3$. The new amplitude of motion is $nA/3$. The value of n is ____

[31-Jan-2024 Shift 1]

Answer: 7

Solution:

$$v = \omega \sqrt{A^2 - x^2}$$

$$\text{at } x = \frac{2A}{3}$$

$$v = \omega \sqrt{A^2 - \left(\frac{2A}{3}\right)^2} = \frac{\sqrt{5}A\omega}{3}$$

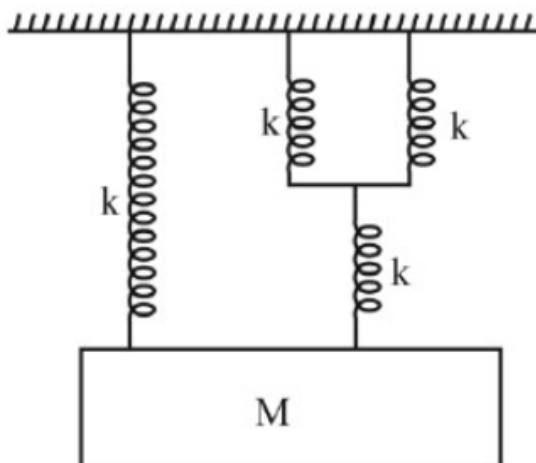
New amplitude = A'

$$v' = 3v = \sqrt{5}A\omega = \omega \sqrt{(A')^2 - \left(\frac{2A}{3}\right)^2}$$

$$A' = \frac{7A}{3}$$

Question32

The time period of simple harmonic motion of mass M in the given figure is $\pi \sqrt{\frac{\alpha M}{5K}}$, where the value of α is ____



[31-Jan-2024 Shift 2]

Answer: 12

Solution:

$$k_{eq} = \frac{2k \cdot k}{3k} + k = \frac{5k}{3}$$

$$\text{Angular frequency of oscillation } (\omega) = \sqrt{\frac{k_{eq}}{m}}$$

$$(\omega) = \sqrt{\frac{5k}{3m}}$$

$$\text{Period of oscillation } (\tau) = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{3m}{5k}}$$

$$= \pi \sqrt{\frac{12m}{5k}}$$

Question33

A mass m is suspended from a spring of negligible mass and the system oscillates with a frequency f_1 . The frequency of oscillations if a mass $9m$ is suspended from the same spring is f_2 . The value of f_1/f_2 is:

[1-Feb-2024 Shift 2]

Answer: 3

Solution:

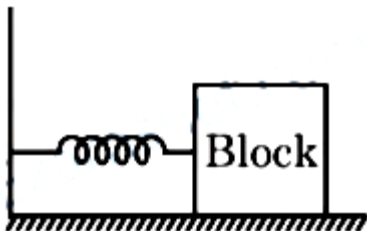
$$f_1 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$f_2 = \frac{1}{2\pi} \sqrt{\frac{k}{9m}}$$

$$\frac{f_1}{f_2} = \sqrt{\frac{9}{1}} = \frac{3}{1}$$

Question34

For a simple harmonic motion in a mass spring system shown, the surface is frictionless. When the mass of the block is 1 kg, the angular frequency is ω_1 . When the mass block is 2 kg the angular frequency is ω_2 . The ratio ω_2/ω_1 is :



[30-Jan-2023 Shift 2]

Options:

A. $\sqrt{2}$

B. $\frac{1}{\sqrt{2}}$

C. 2

D. $\frac{1}{2}$

Answer: B

Solution:

$$\omega = \sqrt{\frac{k}{m}}$$

$$\frac{\omega_2}{\omega_1} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{2}}$$

Question35

The velocity of a particle executing SHM varies with displacement (x) as $4v^2 = 50 - x^2$. The time period of oscillations is $\frac{x}{7}$ s. The value of x is _____.

(Take $\pi = \frac{22}{7}$)

[30-Jan-2023 Shift 2]

Answer: 88

Solution:

$$4v^2 = 50 - x^2$$

$$\Rightarrow v = \frac{1}{2}\sqrt{50 - x^2}$$

$$\omega = \frac{1}{2}$$

$$T = \frac{2\pi}{\omega} = 4\pi = \frac{88}{7}$$

$$x = 88$$

Question36

The maximum potential energy of a block executing simple harmonic motion is 25 J. A is amplitude of oscillation. At A/2, the kinetic energy of the block is
[31-Jan-2023 Shift 1]

Options:

A. 37.5J

B. 9.75J

C. 18.75J

D. 12.5J

Answer: C

Solution:

$$u_{\max} = \frac{1}{2}m\omega^2 A^2 = 25J$$

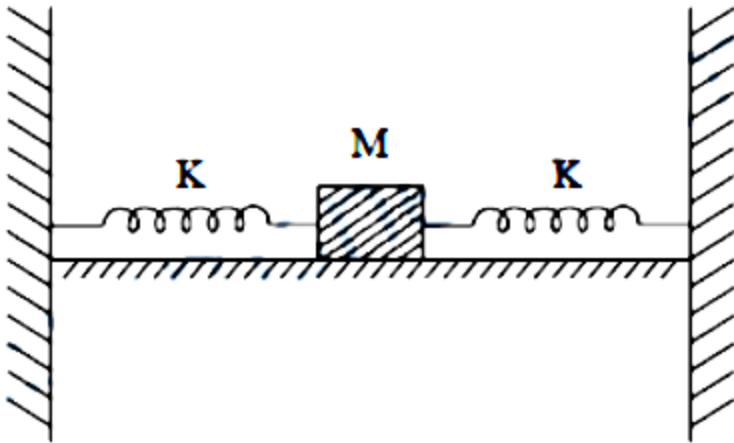
$$KE \text{ at } \frac{A}{2} = \frac{1}{2}mv_1^2 = \frac{1}{2}m\omega^2 \left(A^2 - \frac{A^2}{4} \right)$$

$$KE = \frac{1}{2}m\omega^2 \frac{3A^2}{4} = \frac{3}{4} \left(\frac{1}{2}m\omega^2 A^2 \right)$$

$$KE = \frac{3}{4} \times 25 = 18.75J$$

Question37

In the figure given below. a block of mass $M = 490g$ placed on a frictionless table is connected with two springs having same spring constant ($K = 2Nm^{-1}$). If the block is horizontally displaced through 'X' m then the number of complete oscillations it will make in 14π seconds will be _____



[31-Jan-2023 Shift 1]

Answer: None

Solution:



$K_{eff} = K + K$ as both springs are in use in parallel

$$= 2k$$

$$= 2 \times 2 = 4\text{N/m} \quad m = 490 \text{ gm}$$

$$= 0.49 \text{ kg}$$

$$T = 2\pi \sqrt{\frac{m}{K_{eff}}} = 2\pi \sqrt{\frac{0.49 \text{ kg}}{4}}$$

$$= 2\pi \sqrt{\frac{49}{400}} = 2\pi \frac{7}{20} = \frac{7\pi}{10}$$

No. of oscillation in the 14π is

$$N = \frac{\text{time}}{T} = \frac{14\pi}{7\pi/10} = 20$$

Ans in 20 .

Question38

A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If time taken by particle to go from $x = 0$ to $\frac{A}{2}$ is 2 s; then

time taken by particle in going from $x = \frac{A}{2}$ to A is :

[25-Jan-2023 Shift 2]

Options:

A. 3 s

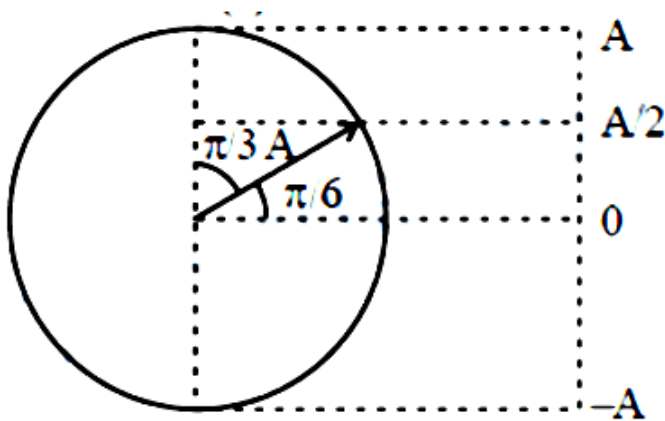
B. 2 s

C. 1.5 s

D. 4 s

Answer: D

Solution:



Let time from 0 to $A/2$ is t_1

& from $A/2$ to A is t_2

then $\omega t_1 = \pi/6$

$\omega t_2 = \pi/3$

$$\frac{t_1}{t_2} = \frac{1}{2}$$

$$t_2 = 2t_1 = 2 \times 2 = 4 \text{ sec}$$

Question39

A particle of mass 250g executes a simple harmonic motion under a periodic force $F = (-25 x)$ N. The particle attains a maximum speed of 4m/ s during its oscillation. The amplitude of the motion is

_____ cm.
[29-Jan-2023 Shift 2]

Answer: 40

Solution:

$$\begin{aligned}\frac{1}{4}a &= -25x; a = -100x \\ \omega^2 &= 100 \quad \omega = 10, \\ \omega A &= 4 \quad A = \frac{4}{10} = 0.4\text{m} \\ A &= 40 \text{ cm}\end{aligned}$$

Question40

The general displacement of a simple harmonic oscillator is $x = A \sin \omega t$. Let T be its time period. The slope of its potential energy (U) - time (t) curve will be maximum when $t = \frac{T}{\beta}$. The value of β is _____.

[30-Jan-2023 Shift 1]

Answer: 8

Solution:

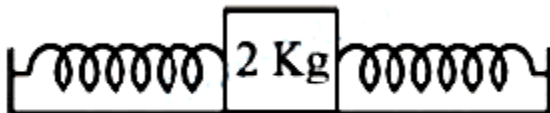
$$\begin{aligned}x &= A \sin (\omega t) \\ U_{(x)} &= \frac{1}{2} k x^2 \\ \frac{dU}{dt} &= \frac{1}{2} k 2x \frac{dx}{dt} \\ &= k A^2 \omega \sin \omega t \cos \omega t \times \frac{2}{2}\end{aligned}$$

$$\left(\frac{dU}{dt} \right)_{\max} = \frac{kA^2\omega}{2} (\sin 2\omega t)_{\max}$$

$$2\omega t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{4\omega} = \frac{T}{8} \Rightarrow \beta = 8$$

Question41

A block of mass 2 kg is attached with two identical springs of spring constant 20N/m each. The block is placed on a frictionless surface and the ends of the springs are attached to rigid supports (see figure). When the mass is displaced from its equilibrium position, it executes a simple harmonic motion. The time period of oscillation is $\frac{\pi}{\sqrt{x}}$ in SI unit. The value of x is _____



[24-Jan-2023 Shift 1]

Answer: 5

Solution:

$$F = -2kx, a = -\frac{2kx}{m}, \omega = \sqrt{\frac{2k}{m}} = \sqrt{\frac{2 \times 20}{2}}$$

$$= \sqrt{20} \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{20}} = \frac{\pi}{\sqrt{5}}$$

$$x = 5$$

Question42

A body of mass 200g is tied to a spring of spring constant 12.5N/m, while the other end of spring is fixed at point O. If the body moves about O in a circular path on a smooth horizontal surface with constant angular speed 5 rad/s, then the ratio of extension in the

spring to its natural length will be :
[24-Jan-2023 Shift 2]

Options:

A. 1 : 2

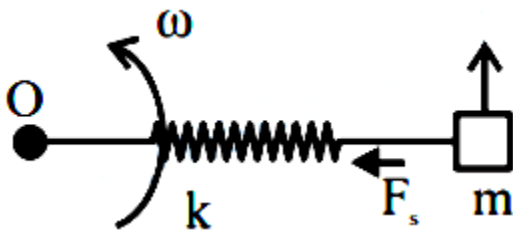
B. 1 : 1

C. 2 : 3

D. 2 : 5

Answer: C

Solution:



Natural length = L_0

Extension = x

$$kx = m(L_0 + x)\omega^2$$

$$\Rightarrow 12.5x = \frac{1}{5}(L_0 + x)25 \Rightarrow 1.5x = L_0$$

$$\Rightarrow \frac{x}{L_0} = \frac{2}{3}$$

Question43

A mass m attached to free end of a spring executes SHM with a period of 1 s. If the mass is increased by 3 kg the period of oscillation increases by one second, the value of mass m is ____ kg.
[24-Jan-2023 Shift 2]

Answer: 1

Solution:

$$T = 2\pi \sqrt{\frac{m}{k}} = 1$$

$$T' = 2\pi \sqrt{\frac{m+3}{k}} = 2$$

$$\frac{T}{T'} = \sqrt{\frac{m}{m+3}} = \frac{1}{2}$$

$$\Rightarrow \frac{m}{m+3} = \frac{1}{4}$$

$$m = 1$$

Question44

The general displacement of a simple harmonic oscillator is $x = A \sin \omega t$. Let T be its time period. The slope of its potential energy (U) - time (t) curve will be maximum when $t = \frac{T}{\beta}$. The value of β is _____.

[30-Jan-2023 Shift 1]

Answer: 8

Solution:

$$x = A \sin (\omega t)$$

$$U_{(x)} = \frac{1}{2} k x^2$$

$$\frac{dU}{dt} = \frac{1}{2} k 2x \frac{dx}{dt}$$

$$= k A^2 \omega \sin \omega t \cos \omega t \times \frac{2}{2}$$

$$\left(\frac{dU}{dt} \right)_{\max} = \frac{k A^2 \omega}{2} (\sin 2\omega t)_{\max}$$

$$2\omega t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{4\omega} = \frac{T}{8} \Rightarrow \beta = 8$$

Question45

The amplitude of a particle executing SHM is 3 cm. The displacement at which its kinetic energy will be 25% more than the potential energy is: _____ cm.
[1-Feb-2023 Shift 1]

Answer: 2

Solution:

$$K E = P E + \frac{P E}{4}$$

$$K E = \frac{5}{4} P E$$

$$\frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{5}{4} \times \frac{1}{2} m \omega^2 x^2$$

$$[v = \omega \sqrt{A^2 - x^2}]$$

$$A^2 - x^2 = \frac{5}{4} x^2$$

$$\frac{9x^2}{4} = A^2$$

$$x = \frac{2}{3} A$$

$$\therefore x = \frac{2}{3} \times 3 \text{ cm}$$

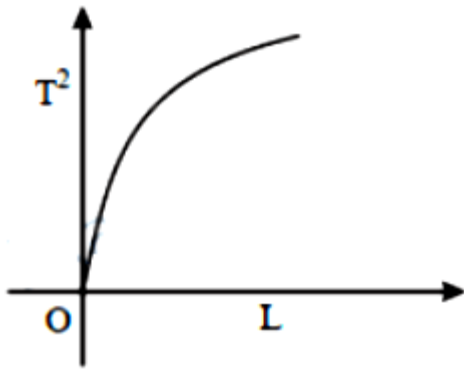
$$x = 2 \text{ cm}$$

Question46

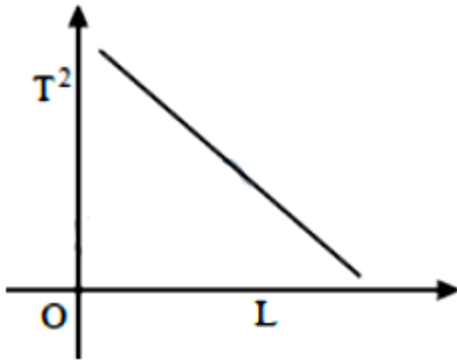
Choose the correct length (L) versus square of time period (T^2) graph for a simple pendulum executing simple harmonic motion.
[1-Feb-2023 Shift 2]

Options:

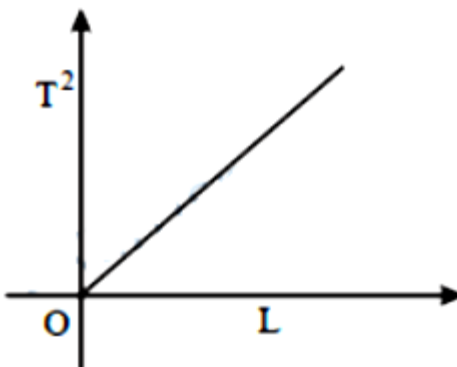
A.



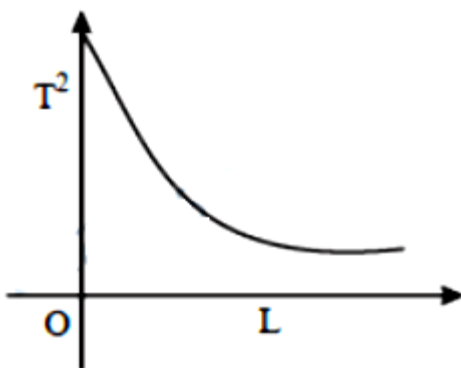
B.



C.



D.



Answer: C

Solution:

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$T^2 = \frac{4\pi^2}{g} \times \ell$$

$$T^2 \propto \ell$$

Question47

A simple pendulum with length 100 cm and bob of mass 250g is executing S.H.M. of amplitude 10 cm. The maximum tension in the string is found to be $\frac{x}{40}$ N. The value of x is _____.

[6-Apr-2023 shift 2]

Answer: 99

Solution:

For pendulum

$$T_{\max} = mg + \frac{mv^2}{L} \dots (1)$$

$$\text{Given } m = \frac{1}{4} \text{ kg, } L = 1 \text{ m, } g = 9.8 \text{ m/s}^2$$

$$\text{and amplitude } A = \frac{1}{10} \text{ m}$$

$$\text{For SHM, } KE_{\max} = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 A^2$$

$$\text{using } \omega = \sqrt{\frac{g}{L}}$$

$$mv^2 = m \left(\sqrt{\frac{g}{L}} \right)^2 A^2 = \frac{mgA}{L} \dots (2)$$

using (2) in (1)

$$T_{\max} = 2mg + \frac{mgA^2}{L^2}$$

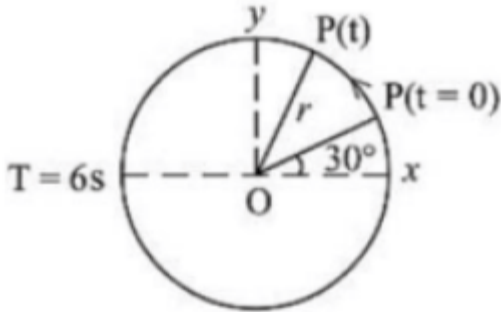
$$= mg \left[1 + \frac{1}{10^2} \right] = \frac{1}{4} \times 9.8 \times \frac{101}{100}$$

$$\text{or } T_{\max} = \frac{98.98}{40}$$

Therefore x = 99

Question48

For particle P revolving round the centre O with radius of circular path r and angular velocity ω , as shown in below figure, the projection of OP on the x-axis at time t is



[8-Apr-2023 shift 2]

Options:

A. $x(t) = r \cos \left(\omega t + \frac{\pi}{6} \right)$

B. $x(t) = r \cos \left(\omega t - \frac{\pi}{6} \omega \right)$

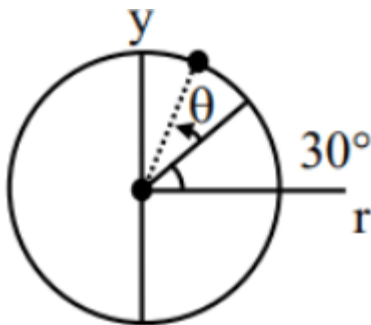
C. $x(t) = r \cos(\omega t)$

D. $x(t) = r \sin \left(\omega t + \frac{\pi}{6} \right)$

Answer: A

Solution:

$$\theta = \omega t$$



Angle from x axis $= \omega t + \frac{\pi}{6}$

Projection of OP on x axis $= r \cos \left(\omega t + \frac{\pi}{6} \right)$

Question49

A particle executes S.H.M. of amplitude A along x -axis. At $t = 0$, the position of the particle is $x = \frac{A}{2}$ and it moves along positive x -axis. The displacement of particle in time t is $x = A \sin(\omega t + \delta)$, then the value δ will be
[10-Apr-2023 shift 1]

Options:

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{6}$

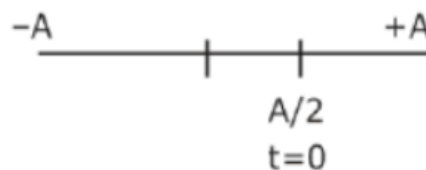
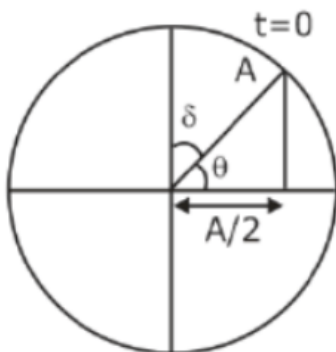
Answer: D

Solution:

$$\cos \theta = \frac{A}{2 \times A} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

$$\delta = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$$



Question50

A rectangular block of mass 5 kg attached to a horizontal spiral spring executes simple harmonic motion of amplitude 1m and time period 3.14 s. The maximum force exerted by spring on block is _____ N.

[10-Apr-2023 shift 2]

Answer: 20

Solution:

When an object executes S.H.M, its maximum acceleration is given by $a_{\max} = \omega^2 A$

Where $\omega = \frac{2\pi}{T}$

Therefore, $a_{\max} = \frac{4\pi^2 A}{T^2}$

(Max force) $F_{\max} = ma_{\max} = 5 \times \frac{4 \times 3.14 \times 3.14}{3.14 \times 3.14} \times (1)$
 $= 20\text{N}$

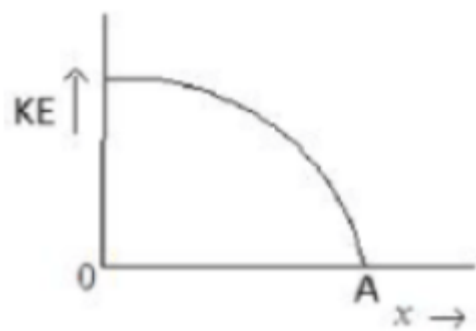
Question51

The variation of kinetic energy (KE) of a particle executing simple harmonic motion with the displacement (x) starting from mean position to extreme position (A) is given by

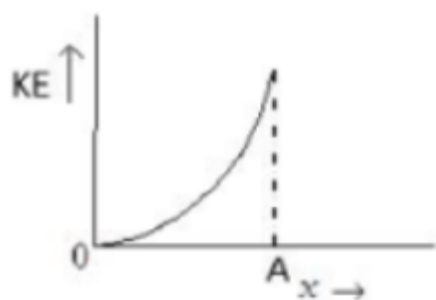
[11-Apr-2023 shift 1]

Options:

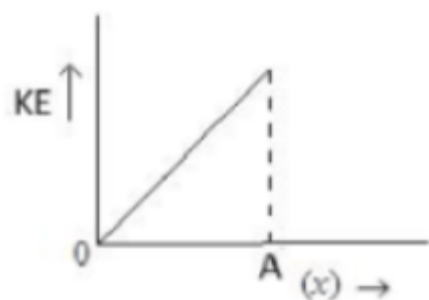
A.



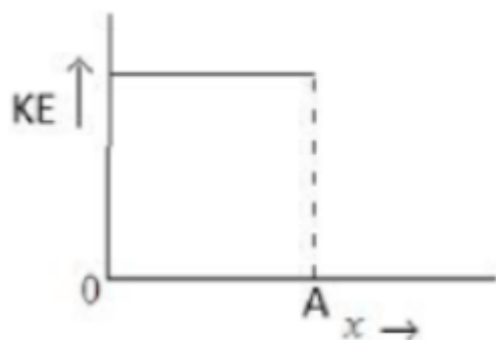
B.



C.



D.



Answer: A

Solution:

$$K \cdot E = T \cdot E - P \cdot E$$

$$K \cdot E = \frac{1}{2}KA^2 - \frac{1}{2}Kx^2$$

Graph b/w $K \cdot E$ and x will be parabola
Option $\rightarrow$ (1)

Question 52

A particle is executing simple harmonic motion (SHM). The ratio of potential energy and kinetic energy of the particle when its displacement is half of its amplitude will be [12-Apr-2023 shift 1]

Options:

A. 1 : 1

B. 1 : 3

C. 1 : 4

D. 2 : 1

Answer: B

Solution:

Ratio of PE and KE = ?

When $x = \frac{A}{2}$
x \ disp.

$$KE = \frac{1}{2}m\omega^2[A^2 - x^2]$$

$$= \frac{1}{2}m\omega^2\left[A^2 - \frac{A^2}{4}\right]$$

$$KE = \frac{1}{2}m\omega^2\left[\frac{3A^2}{4}\right] \dots (1)$$

$$\text{And } PE = \frac{1}{2}m\omega^2x^2$$

$$= \frac{1}{2}m\omega^2\frac{A^2}{4}$$

$$\frac{PE}{KE} = \frac{m\omega^2A^2 \times 8}{8m\omega^2A^2 \times 3} = \frac{1}{3}$$

$$\Rightarrow 1 : 3$$

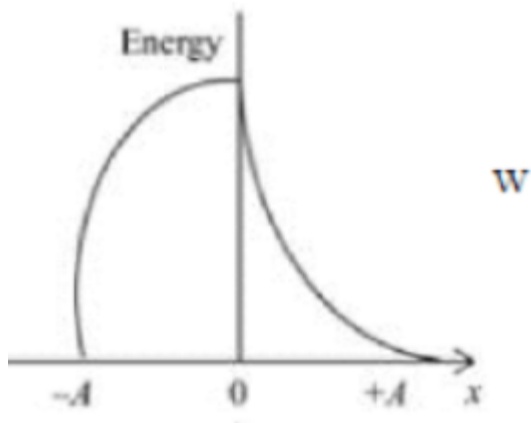
Question53

Which graph represents the difference between total energy and potential energy of a particle executing SHM vs it's distance from mean position?

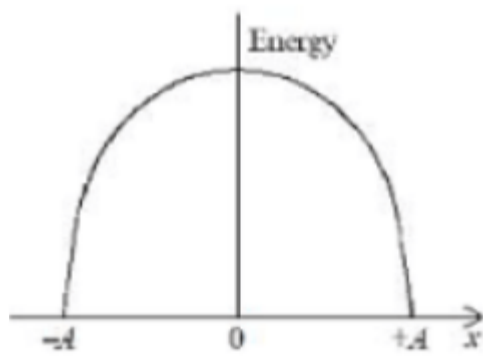
[13-Apr-2023 shift 1]

Options:

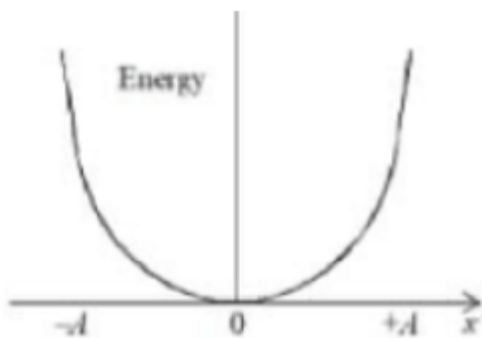
A.



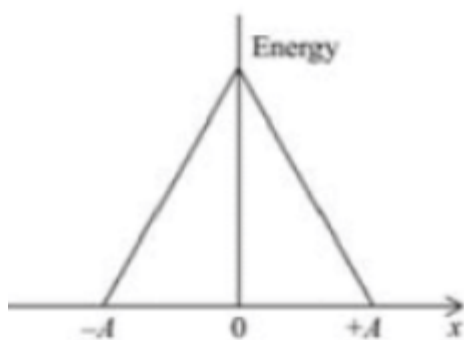
B.



C.



D.



Answer: B

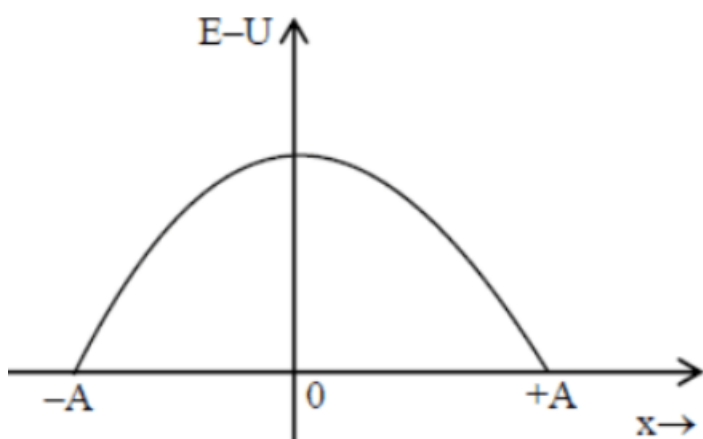
Solution:

Total energy in SHM = E

$$E = K + U$$

$$E - U = K$$

$$E - U = \frac{1}{2}m\omega^2(A^2 - x^2)$$



Question 54

At a given point of time the value of displacement of a simple harmonic oscillator is given as $y = A \cos(30^\circ)$. If amplitude is 40 cm and kinetic energy at that time is 200J, the value of force constant is $1.0 \times 10^x \text{ Nm}^{-1}$. The value of x is _____
[13-Apr-2023 shift 1]

Answer: 4

Solution:

$$v = \omega \sqrt{A^2 - x^2}$$

$$y = A \times \frac{\sqrt{3}}{2}$$

$$v = \omega \sqrt{A^2 - \frac{3A^2}{4}} = \frac{\omega A}{2}$$

Given, KE = 200J

$$\frac{1}{2} m \frac{\omega^2 A^2}{4} = 200$$

$$KA^2 = 1600 \quad (K = m\omega^2)$$

$$K = \frac{1600}{(40 \times 10^{-2})^2}$$

$$K = 10^4 \text{ N/m}$$

$$x = 4$$

Question55

A particle executes SHM of amplitude A. The distance from the mean position when its's kinetic energy becomes equal to its potential energy is :

[13-Apr-2023 shift 2]

Options:

A. $\sqrt{2}A$

B. $\frac{1}{2}A$

C. $\frac{1}{\sqrt{2}}A$

D. $2A$

Answer: C

Solution:

$$K.E = P.E$$

$$\frac{1}{2}M\omega^2(A^2 - x^2) = \frac{1}{2}M\omega^2x^2$$

$$(A^2 - x^2) = x^2$$

$$2x^2 = A^2$$

$$x = \pm \frac{A}{\sqrt{2}}$$

Question 56

In a linear Simple Harmonic Motion (SHM)

(A) Restoring force is directly proportional to the displacement.

(B) The acceleration and displacement are opposite in direction.

(C) The velocity is maximum at mean position.

(D) The acceleration is minimum at extreme points.

Choose the correct answer from the options given below :

[15-Apr-2023 shift 1]

Options:

A. (C) and (D) only

B. (A), (C) and (D) only

C. (A), (B) and (C) only

D. (A), (B) and (D) only

Answer: C

Solution:

In SHM,

$$F \propto -x \rightarrow A$$

$$a \propto -x \rightarrow B$$

$$V_{\text{mean}} \rightarrow \text{maximum} \rightarrow C$$

$$a_{\text{extreme}} \rightarrow \text{maximum}$$

Hence, (A), (B) and (C) are true.

Question 57

Two massless springs with spring constants $2k$ and $9k$, carry $50g$ and $100g$ masses at their free ends. These two masses oscillate vertically such that their maximum velocities are equal. Then, the ratio of their respective amplitudes will be:

[24-Jun-2022-Shift-2]

Options:

A. $1 : 2$

B. $3 : 2$

C. $3 : 1$

D. $2 : 3$

Answer: B

Solution:

$$\begin{aligned}\omega_1 A_1 &= \omega_2 A_2 \\ \Rightarrow \frac{A_1}{A_2} &= \frac{\omega_2}{\omega_1} \\ &= \sqrt{\frac{k_2}{m_2}} \times \sqrt{\frac{m_1}{k_1}} = \sqrt{\frac{9k}{100} \times \frac{50}{2k}} = \frac{3}{2}\end{aligned}$$

Question58

The displacement of simple harmonic oscillator after 3 seconds starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is :

[27-Jun-2022-Shift-1]

Options:

A. 6s

B. 8s

C. 12s

D. 36s

Answer: D

Solution:

Time taken by the harmonic oscillator to move from mean position to half of amplitude is $\frac{T}{12}$

$$\text{So, } \frac{T}{12} = 3$$

$$T = 36 \text{ sec.}$$

Question59

The equation of a particle executing simple harmonic motion is given by $x = \sin \pi \left(t + \frac{1}{3} \right)$ m. At $t = 1$ s, the speed of particle will be
(Given : $\pi = 3.14$)
[27-Jun-2022-Shift-2]

Options:

A. 0 cm s^{-1}

B. 157 cm s^{-1}

C. 272 cm s^{-1}

D. 314 cm s^{-1}

Answer: B

Solution:

$$x = \sin \left(\pi t + \frac{\pi}{3} \right) \text{ m}$$

$$\Rightarrow \frac{dx}{dt} = \pi \cos \left(\pi t + \frac{\pi}{3} \right)$$

$$= \pi \cos \left(\pi + \frac{\pi}{3} \right) \text{ at } t = 1 \text{ s}$$

$$= -\frac{\pi}{2} \text{ m/s}$$

$$\text{or } \left| \frac{dx}{dt} \right| = 157 \text{ cm/s}$$

Question60

A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is 6s. The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is s.

[27-Jun-2022-Shift-2]

Answer: 1

Solution:

$$A = 8 \text{ cm}$$

$$T = 6 \text{ s}$$

$$A \cos\left(\frac{2\pi t}{T}\right) = \frac{A}{2}$$

$$\Rightarrow \frac{2\pi t}{T} = \frac{\pi}{3}$$

$$\text{or } t = \frac{T}{6} = 1 \text{ s}$$

Question61

A body is performing simple harmonic with an amplitude of 10 cm. The velocity of the body was tripled by air jet when it is at 5 cm from its mean position. The new amplitude of vibration is $\sqrt{x}$ cm.

The value of x is _____

[29-Jun-2022-Shift-1]

Answer: 700

Solution:

$$\begin{aligned}v &= \omega \sqrt{A^2 - y^2} \\ \Rightarrow 3\omega \sqrt{10^2 - 5^2} &= \omega \sqrt{(A')^2 - 5^2} \\ \Rightarrow 9 \times 75 &= (A')^2 - 25 \\ \Rightarrow A' &= \sqrt{28 \times 25} \text{ cm} \\ \Rightarrow x &= 700\end{aligned}$$

Question62

The motion of a simple pendulum executing S.H.M. is represented by the following equation.

$y = A \sin(\pi t + \phi)$, where time is measured in second. The length of pendulum is

[29-Jun-2022-Shift-2]

Options:

A. 97.23 cm

B. 25.3 cm

C. 99.4 cm

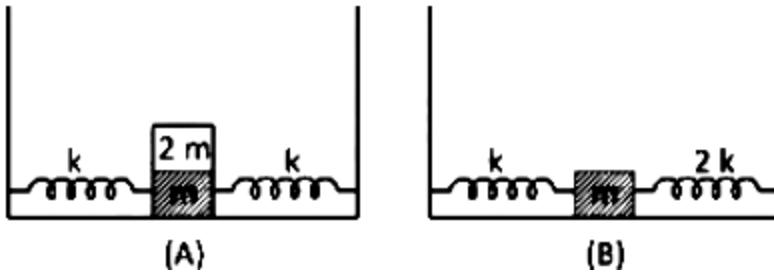
D. 406.1 cm

Answer: C

Solution:

$$\begin{aligned}\omega &= \pi = \sqrt{\frac{g}{l}} \\ \text{So, } l &= \frac{g}{\pi^2} \\ &\simeq 99.4 \text{ cm (Nearest value)}\end{aligned}$$

Question63



In figure (A), mass ' $2m$ ' is fixed on mass ' m ' which is attached to two springs of spring constant k . In figure (B), mass ' m ' is attached to two springs of spring constant ' k ' and ' $2k$ '. If mass ' m ' in (A) and in (B) are displaced by distance ' x ' horizontally and then released, then time period T_1 and T_2 corresponding to (A) and (B) respectively follow the relation.

[25-Jul-2022-Shift-1]

Options:

A. $\frac{T_1}{T_2} = \frac{3}{\sqrt{2}}$

B. $\frac{T_1}{T_2} = \sqrt{\frac{3}{2}}$

C. $\frac{T_1}{T_2} = \sqrt{\frac{2}{3}}$

D. $\frac{T_1}{T_2} = \frac{\sqrt{2}}{3}$

Answer: A

Solution:

Both the springs are in parallel combination in both the diagrams so

$$T_1 = 2\pi \sqrt{\frac{3m}{2k}}$$

$$\text{and } T_2 = 2\pi \sqrt{\frac{m}{3k}}$$

$$\text{So, } \frac{T_1}{T_2} = \frac{3}{\sqrt{2}}$$

The length of a seconds pendulum at a height $h = 2R$ from earth surface will be:

(Given $R =$ Radius of earth and acceleration due to gravity at the surface of earth, $g = \pi^2 \text{ms}^{-2}$)

[25-Jul-2022-Shift-2]

Options:

A. $\frac{2}{9}m$

B. $\frac{4}{9}m$

C. $\frac{8}{9}m$

D. $\frac{1}{9}m$

Answer: D

Solution:

$$g = \frac{GM}{(R+h)^2} = \frac{GM}{9R^2} = \frac{g_0}{9}$$
$$\Rightarrow T = 2\pi \sqrt{\frac{1}{g}} = 2\pi \sqrt{\frac{1}{\frac{g_0}{9}}}$$

$$\Rightarrow 2 = 2\pi \sqrt{\frac{9l}{g_0}}$$

$$\Rightarrow 1 = \frac{g_0}{9\pi^2} = \frac{1}{9}m$$

Question65

Two waves executing simple harmonic motions travelling in the same direction with same amplitude and frequency are superimposed. The resultant amplitude is equal to the $\sqrt{3}$ times of amplitude of individual motions. The phase difference between the two motions is ____ (degree).

[25-Jul-2022-Shift-2]

Answer: 60

Solution:

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \varphi}$$

$$\sqrt{3}A = \sqrt{A^2 + A^2 + 2A^2 \cos \varphi}$$

$$3A^2 = 2A^2 + 2A^2 \cos \varphi$$

$$\cos \varphi = \frac{1}{2}$$

$$\varphi = 60^\circ$$

Question66

**When a particle executes Simple Harmonic Motion, the nature of graph of velocity as a function of displacement will be :
[26-Jul-2022-Shift-1]**

Options:

A. Circular

B. Elliptical

C. Sinusoidal

D. Straight line

Answer: B

Solution:

$$\text{Let } x = A \sin \omega t$$

$$\Rightarrow v = A\omega \cos \omega t$$

$$\Rightarrow v = \pm \omega \sqrt{A^2 - x^2}$$

$$\Rightarrow \frac{v^2}{\omega^2} + x^2 = A^2$$

$$\Rightarrow \text{Ellipse}$$

Question 67

As per given figures, two springs of spring constants k and $2k$ are connected to mass m . If the period of oscillation in figure (a) is $3s$, then the period of oscillation in figure (b) will be $\sqrt{x}s$. The value of x is _____.

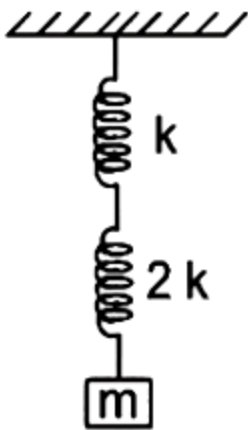


figure (a)

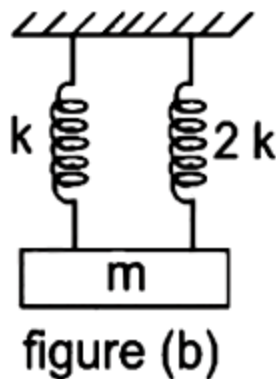


figure (b)

[26-Jul-2022-Shift-2]

Answer: 2

Solution:

For figure (a) :

$$K_{eq} = \frac{K \times 2K}{K + 2K} = \frac{2K}{3}$$

$$T = 2\pi \sqrt{\frac{m}{K_{eq}}} = 2\pi \sqrt{\frac{m}{2K/3}} = 2\pi \sqrt{\frac{3m}{2K}}$$

For figure (b):

$$K_{eq} = 3K, T' = 2\pi \sqrt{\frac{m}{3K}}$$

$$\frac{T'}{T} = \sqrt{\frac{m \times 2K}{3K \times 3m}} = \frac{\sqrt{2}}{3}$$

$$T' = \sqrt{2}$$

$$x = 2$$

Question68

Two identical positive charges Q each are fixed at a distance of ' $2a$ ' apart from each other. Another point charge q_0 with mass ' m ' is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 will be :
[27-Jul-2022-Shift-1]

Options:

A. $\sqrt{\frac{4\pi^3 \epsilon_0 m a^3}{q_0 Q}}$

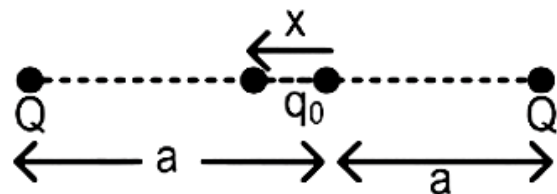
B. $\sqrt{\frac{q_0 Q}{4\pi^3 \epsilon_0 m a^3}}$

C. $\sqrt{\frac{2\pi^2 \epsilon_0 m a^3}{q_0 Q}}$

D. $\sqrt{\frac{8\pi^3 \epsilon_0 m a^3}{q_0 Q}}$

Answer: A

Solution:



($x < a$) (α is acceleration)

$$F_{\text{net}} = - \left(\frac{kq_0 Q}{(a-x)^2} - \frac{kQq_0}{(a+x)^2} \right)$$

$$m\alpha = - \frac{kq_0 Q}{a^4} 4ax$$

$$\Rightarrow \alpha = - \frac{4kq_0Q}{ma^3}x$$

$$\text{So, } T = 2\pi \sqrt{\frac{4\pi\epsilon_0 ma^3}{4q_0Q}}$$

$$\text{or } T = \sqrt{\frac{4\pi^3\epsilon_0 ma^3}{q_0Q}}$$

Question69

Two bar magnets oscillate in a horizontal plane in earth's magnetic field with time periods of 3s and 4s respectively. If their moments of inertia are in the ratio of 3 : 2, then the ratio of their magnetic moments will be:

[27-Jul-2022-Shift-1]

Options:

A. 2 : 1

B. 8 : 3

C. 1 : 3

D. 27 : 16

Answer: B

Solution:

$$T = 2\pi \sqrt{\frac{I}{MB_H}}$$

$$\frac{T_1}{T_2} = \frac{2\pi \sqrt{\frac{I_1}{M_1 B_H}}}{2\pi \sqrt{\frac{I_2}{M_2 B_H}}} = \frac{3}{4}$$

$$\sqrt{\frac{I_1}{I_2} \times \frac{M_2}{M_1}} = \frac{3}{4}$$

$$\sqrt{\frac{I_1}{I_2}} \times \sqrt{\frac{M_2}{M_1}} = \frac{3}{4}$$

$$\sqrt{\frac{3}{2}} \times \sqrt{\frac{M_2}{M_1}} = \frac{3}{4}$$

$$\frac{3}{2} \times \frac{M_2}{M_1} = \frac{9}{16}$$

$$\frac{M_1}{M_2} = \frac{8}{3}$$

Question70

A mass 0.9 kg, attached to a horizontal spring, executes SHM with an amplitude A_1 . When this mass passes through its mean position, then a smaller mass of 124g is placed over it and both masses move together with amplitude A_2 . If the ratio $\frac{A_1}{A_2}$ is $\frac{\alpha}{\alpha-1}$, then the value of α will be _____.

[27-Jul-2022-Shift-1]

Answer: 16

Solution:

$$\begin{aligned} \frac{1}{2}kA^2 &= \frac{p^2}{2m} \\ \Rightarrow \left(\frac{A_1}{A_2} \right)^2 &= \frac{m_2}{m_1} = \frac{1024}{900} \\ \Rightarrow \frac{A_1}{A_2} &= \frac{32}{30} = \frac{16}{15} = \frac{16}{16-1} \\ \therefore \alpha &= 16 \end{aligned}$$

Question71

A compass needle of oscillation magnetometer oscillates 20 times per minute at a place P of dip 30° . The number of oscillations per minute become 10 at another place Q of 60° dip. The ratio of the total

magnetic field at the two places ($B_Q : B_P$) is :

[27-Jul-2022-Shift-2]

Options:

A. $\sqrt{3} : 4$

B. $4 : \sqrt{3}$

C. $\sqrt{3} : 2$

D. $2 : \sqrt{3}$

Answer: A

Solution:

$$T = 2\pi \sqrt{\frac{I}{B_H M}}$$

$$T_1 = 3 \text{ sec} = 2\pi \sqrt{\frac{I}{(B_P \cos 30^\circ)M}}$$

$$T_2 = 6 \text{ sec} = 2\pi \sqrt{\frac{I}{(B_Q \cos 60^\circ)M}}$$

$$\frac{3}{6} = \sqrt{\frac{I}{\left(B_P \frac{\sqrt{3}}{2}\right) \times (B_Q/2)}}$$

$$\frac{3}{6} = \sqrt{\left(\frac{B_Q}{\sqrt{3}B_P}\right)}$$

$$\frac{\sqrt{3}}{4} = \frac{B_Q}{B_P}$$

$$B_Q : B_P = \sqrt{3} : 4$$

Question72

The potential energy of a particle of mass 4 kg in motion along the x-axis is given by $U = 4(1 - \cos 4x)$ J. The time period of the particle for small oscillation ($\sin\theta \simeq \theta$) is $\left(\frac{\pi}{K}\right)$ s. The value of K is _____.

[28-Jul-2022-Shift-2]

Answer: 2

Solution:

$$U = 4(1 - \cos 4x)$$

$$F = -\frac{dU}{dx} = -4(+\sin 4x)4 = -16 \sin(4x)$$

For small θ

$$\sin \theta \approx \theta$$

$$F = -64x$$

$$a = -64x/m = -16x$$

$$\omega^2 = 16$$

$$T = \frac{2\pi}{\omega} = \frac{\pi}{2}$$

Question 73

The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α , is given by :
[29-Jul-2022-Shift-1]

Options:

A. $2\pi\sqrt{L/(g \cos \alpha)}$

B. $2\pi\sqrt{L/(g \sin \alpha)}$

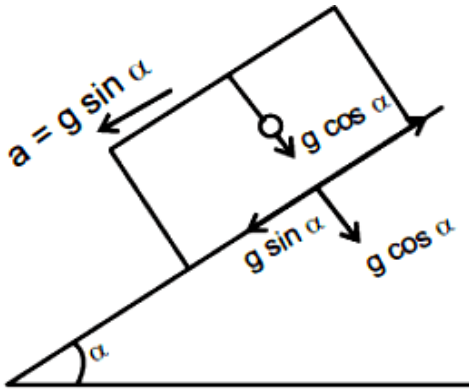
C. $2\pi\sqrt{L/g}$

D. $2\pi\sqrt{L/(g \tan \alpha)}$

Answer: A

Solution:

$$g_{\text{eff}} = g \cos \alpha$$



$$T = 2\pi \sqrt{\frac{L}{g \cos \alpha}}$$

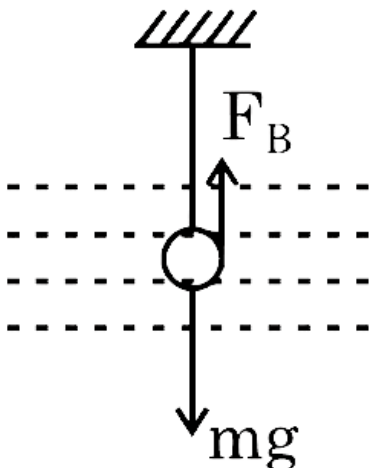
Question74

The metallic bob of simple pendulum has the relative density 5 . The time period of this pendulum is 10s. If the metallic bob is immersed in water, then the new time period becomes $5\sqrt{x}$ s. The value of x will be _____.

[29-Jul-2022-Shift-2]

Answer: 5

Solution:



$$\begin{aligned}
 mg' &= mg - F_B \\
 g' &= \frac{mg - F_B}{m} \\
 &= \frac{\rho_B Vg - \rho_w Vg}{\rho_B V} \\
 &= \left(\frac{\rho_B - \rho_w}{\rho_B} \right) g \\
 &= \frac{5-1}{5} \times g \\
 &= \frac{4}{5}g
 \end{aligned}$$

We know, $T = 2\pi \sqrt{\frac{\ell}{g}}$

$$\frac{T'}{T} = \sqrt{\frac{g}{g'}} = \sqrt{\frac{g}{\frac{4}{5}g}} = \sqrt{\frac{5}{4}}$$

$$T' = T \sqrt{\frac{5}{4}} = \frac{10}{2}\sqrt{5}$$

$$T' = 5\sqrt{5}$$

Question75

When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement is
[24 Feb 2021 Shift 2]

Options:

- A. circular
- B. elliptical
- C. parabolic
- D. straight line

Answer: B

Solution:

Since, the particle is executing SHM. Therefore, displacement equation of wave will be

$$y = A \sin \omega t$$

$$\Rightarrow y/A = \sin \omega t$$

and wave velocity equation will be

$$v_y = \frac{dy}{dt} = A\omega \cos \omega t$$

$$\Rightarrow v_y/A\omega = \cos \omega t$$

$$\text{Now, } \sin^2 \omega t + \cos^2 \omega t = 1$$

$$\therefore (y/A)^2 + (v_y/A\omega)^2 = 1$$

This equation is similar to the equation of ellipse.

Question76

Given below are two statements:

Statement I A second's pendulum has a time period of 1s.

Statement II It takes precisely one second to move between the two extreme positions.

In the light of the above statements, choose the correct answer from the options given below.

[26 Feb 2021 Shift 2]

Options:

A. Both Statement I and Statement II are false.

B. Statement I is false but Statement II is true.

C. Statement I is true but Statement II is false.

D. Both Statement I and Statement II are true.

Answer: B

Solution:

Statement I is false because time period of second's pendulum is always 2s.
Therefore, time taken to move between two extreme positions will be $T/2 = 2/2 = 1\text{s}$
Hence, option (b) is the correct.

Question 77

Time period of a simple pendulum is T . The time taken to complete $5/8$ oscillations starting from mean position is $\frac{\alpha}{\beta}T$. The value of α is

[26 Feb 2021 Shift 2]

Answer: 7

Solution:

Given, angular displacement to complete $\frac{5}{8} \left(= \frac{1}{2} + \frac{1}{8} \right)$ rev

$$= \left(\pi + \frac{\pi}{6} \right) \text{rad}$$

$$= \left(\frac{7\pi}{6} \right) \text{rad}$$

$$\text{Since, } \omega = \frac{2\pi}{T} = \frac{\theta}{t}$$

$$\Rightarrow \theta = \frac{2\pi}{T} \cdot t \Rightarrow \frac{7\pi}{6} = \frac{2\pi}{T} t$$

$$\Rightarrow t = \frac{7T}{12} = \frac{\alpha T}{\beta}$$

Hence, $\alpha = 7$

Question 78

If two similar springs each of spring constant K_1 are joined in series, the new spring constant and time period would be changed by a

factor

[26 Feb 2021 Shift 1]

Options:

A. $\frac{1}{2}, \sqrt{2}$

B. $\frac{1}{4}, \sqrt{2}$

C. $\frac{1}{4}, 2\sqrt{2}$

D. $\frac{1}{2}, 2\sqrt{2}$

Answer: A

Solution:

Let series equivalent of spring constant $= k_{eq}$ and T be the time period.

In series arrangement, $\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$

$$\Rightarrow \frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_1} = \frac{2}{k_1} \Rightarrow k_{eq} = \frac{k_1}{2}$$

$$\text{As, } T = 2\pi \sqrt{\frac{m}{k_1}}$$

where, m is mass of body connected with spring.

$$\Rightarrow T \propto \sqrt{\frac{1}{k_1}}$$

$$\text{and } T' \propto \sqrt{\frac{2}{k_1}} \Rightarrow T' = \sqrt{2}T$$

Question79

Assume that a tunnel is dug along a chord of the earth, at a

perpendicular distance $\left(\frac{R}{2}\right)$ from the earth's centre, where R is the radius of the earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period?

[26 Feb 2021 Shift 1]

Options:

A. $\frac{2\pi R}{g}$

B. $\frac{g}{2\pi R}$

C. $\frac{1}{2\pi} \sqrt{\frac{g}{R}}$

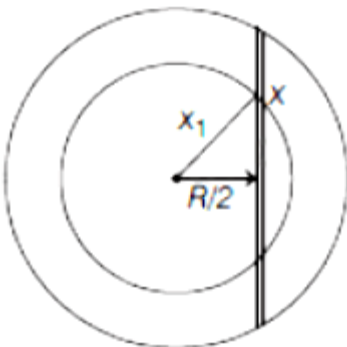
D. $2\pi \sqrt{\frac{R}{g}}$

Answer: D

Solution:

Given, radius of earth = R

Distance of chord from centre of earth = $\frac{R}{2}$



Let x_1 be the radius of inner circle and M be the mass of earth.

$$\therefore m' (\text{effective mass of earth}) = \frac{M}{\frac{4}{3}\pi R^3} \cdot \frac{4}{3}\pi x_1^3$$

$$\Rightarrow m' = \frac{M}{R^3} x_1^3$$

If F is the gravitational force exerted by earth on particle at position x and ω be the angular velocity in time period T, then

$$F = \frac{Gm'm}{x_1^2} = \frac{Gm}{x_1^2} \cdot \frac{M}{R^3} x_1^3$$

$$\Rightarrow m\omega^2 x_1 = \frac{GMmx_1}{R^3}$$

$$\Rightarrow \omega = \sqrt{\frac{GM}{R^3}}$$

$$F = \frac{Gm'm}{x_1^2} = \frac{Gm}{x_1^2} \cdot R^3 x_1^3$$

$$\Rightarrow m\omega^2 x_1 = \frac{GMmx_1}{R^3}$$

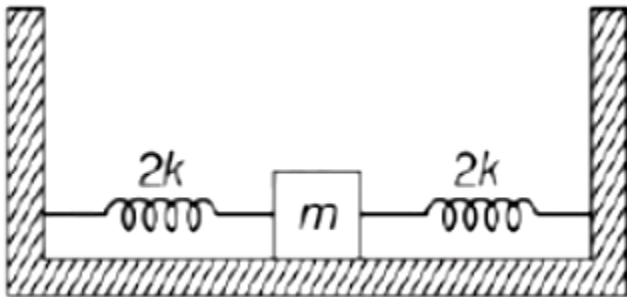
$$\Rightarrow \omega = \sqrt{\frac{GM}{R^3}}$$

Since, $\omega = \frac{2\pi}{T}$ and $GM = gR^2$

Substituting the above values in Eq. (i), we get $1 \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{gR^2}{R^3}} \Rightarrow T = 2\pi \sqrt{\frac{R}{g}}$

Question80

Two identical springs of spring constant $2k$ are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. The time period of oscillations of this system is



[25 Feb 2021 Shift 2]

Options:

A. $2\pi \sqrt{\frac{m}{2k}}$

B. $2\pi \sqrt{\frac{m}{k}}$

C. $\pi \sqrt{\frac{m}{k}}$

D. $\pi \sqrt{\frac{m}{2k}}$

Answer: C

Solution:

Let spring constants of two springs be k_1 and k_2 . Since, two springs are connected in parallel connection and parallel equivalent spring constant, $k_{eq} = k_1 + k_2$

$$\Rightarrow k_{eq} = 2k + 2k = 4k$$

$$\text{As, time period, } T = 2\pi \sqrt{\frac{m}{k_{eq}}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{4k}} = \frac{2\pi}{2} \sqrt{\frac{m}{k}} = \pi \sqrt{\frac{m}{k}}$$

Question81

**If the time period of a 2m long simple pendulum is 2s, the acceleration due to gravity at the place, where pendulum is executing SHM is
[25 Feb 2021 Shift 1]**

Options:

A. $\pi^2 \text{ms}^{-2}$

B. 9.8ms^{-2}

C. $2\pi^2 \text{ms}^{-2}$

D. 16ms^{-2}

Answer: C

Solution:

Given, length of simple pendulum, $l = 2\text{m}$

Time period, $T = 2\text{s}$

Let g_{eff} be the acceleration due to gravity.

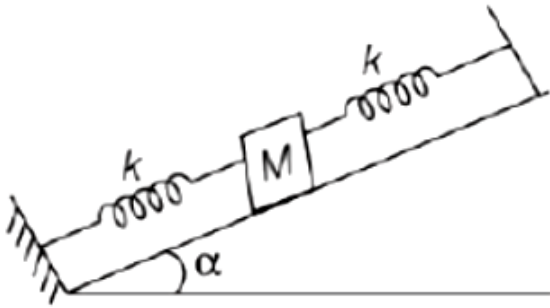
$$\therefore \text{Time period, } T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}}$$

$$\Rightarrow g_{\text{eff}} = 4\pi^2 \frac{l}{T^2}$$

$$= 4\pi^2 \cdot \frac{2}{4} = 2\pi^2 \text{ms}^{-2}$$

Question82

In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k , then the frequency of oscillation of given body is



[24 Feb 2021 Shift 2]

Options:

- A. $\frac{1}{2\pi} \sqrt{\frac{k}{2M}}$
- B. $\frac{1}{2\pi} \sqrt{\frac{2k}{M g \sin \alpha}}$
- C. $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$
- D. $\frac{1}{2\pi} \sqrt{\frac{k}{M g \sin \alpha}}$

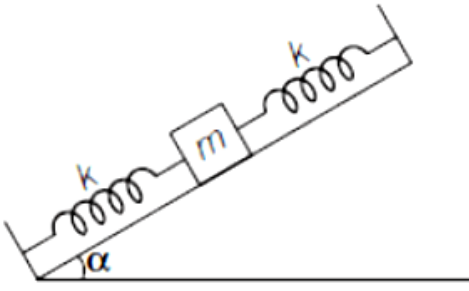
Answer: C

Solution:

Let T be the time period of oscillation, then

$$T = 2\pi \sqrt{\frac{M}{k_{eq}}}$$

$$\therefore T = 2\pi \sqrt{\frac{M}{2k}} \quad [\because k_{eq} = k + k]$$



and frequency (f) = $\frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{2k}{M}}$

Question83

A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin, i.e. $ma = -\alpha x^2$. The distance at which the particle stops is
[24 Feb 2021 Shift 2]

Options:

A. $\left(\frac{3mv_0^2}{2\alpha} \right)^{\frac{1}{2}}$

B. $\left(\frac{2mv_0}{3\alpha} \right)^{\frac{1}{3}}$

C. $\left(\frac{2mv_0^2}{3\alpha} \right)^{\frac{1}{2}}$

D. $\left(\frac{3mv_0^2}{2\alpha} \right)^{\frac{1}{3}}$

Answer: D

Solution:

Given, speed of projection = v_0

Damping force, $F = ma = -\alpha x^2$

$$\Rightarrow a = -\alpha x^2/m$$

$$\text{Also, } \Rightarrow v dv = a dx = -\frac{\alpha}{m} x^2 dx$$

Integrating both sides, we get

$$\int_{v_0}^v v dv = \int_0^x -\frac{\alpha}{m} x^2 dx$$

$$\Rightarrow \left(\frac{v^2}{2} \right)_{v_0}^0 = -\frac{\alpha}{m} \left(\frac{x^3}{3} \right)_0^x$$

$$\Rightarrow 0 - \frac{v_0^2}{2} = -\frac{\alpha}{m} \frac{x^3}{3} \Rightarrow x = \left(\frac{3m}{2} \frac{v_0^2}{\alpha} \right)^{1/3}$$

Question84

A student is performing the experiment of resonance column. The diameter of the column tube is 6cm. The frequency of the tuning fork is 504H z. Speed of the sound at the given temperature is 336m/s. The zero of the meter scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is
[25 Feb 2021 Shift 1]

Options:

- A. 13cm
- B. 16.6cm
- C. 8.4cm
- D. 14.8cm

Answer: D

Solution:

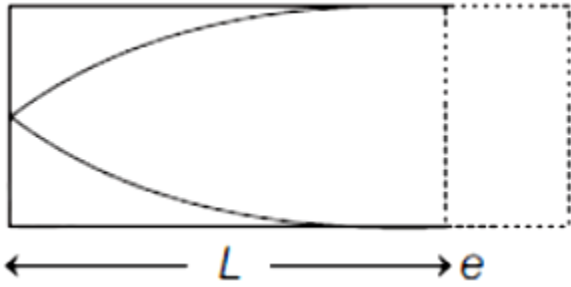
Given, diameter of the column tube,

$$d = 6\text{cm} = 6 \times 10^{-2}\text{m}$$

Frequency of tuning fork, $f = 504\text{H z}$ Speed of sound at given temperature, $v = 336\text{ms}^{-1}$

As, this is a closed organ pipe.

Let L be the length of tube and λ be the wavelength, then



$$\text{and } L + e = \lambda/4 = v/4f \quad \left(\because \lambda = \frac{v}{f} \right)$$

$$e = 0.6 \times d / 2$$

$$= \frac{6}{10} \times \frac{6 \times 10^{-2}}{2} = 0.018$$

$$\Rightarrow L + e = \frac{v}{4f} = \frac{336}{4 \times 504}$$

$$\Rightarrow L = \frac{336}{2016} - 0.018 = 0.1667 - 0.018$$

$$= 0.1487\text{m} = 14.87\text{cm}$$

$$\sim \text{eq } 14.8\text{cm}$$

Question 85

A particle executes SHM, the graph of velocity as a function of displacement is

[26 Feb 2021 Shift 2]

Options:

- A. a circle
- B. a parabola
- C. an ellipse
- D. a helix

Answer: C

Solution:

Velocity of particle in SHM in terms of displacement x is given as

$$v = \omega \sqrt{A^2 - x^2}$$

$$\Rightarrow v = A\omega \sqrt{1 - x^2/A^2}$$

$$\Rightarrow \frac{v}{A\omega} = \sqrt{1 - x^2/A^2}$$

$$\Rightarrow \frac{v^2}{A^2 \omega^2} = 1 - \frac{x^2}{A^2}$$

$$\Rightarrow \frac{v^2}{A^2 \omega^2} + \frac{x^2}{A^2} = 1 \dots (i)$$

As equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Eq. (i) is similar to the equation of ellipse. So, the graph between velocity and displacement is an ellipse.

Question 86

A particle executes SHM with amplitude a and time period T . The displacement of the particle when its speed is half of maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is
[26 Feb 2021 Shift 2]

Answer: 3

Solution:

Given, amplitude of particle in SHM = a

Maximum velocity, $v = A\omega$

Let displacement be y when speed, $v_2 = v/2 = \frac{A\omega}{2}$

By using equation of wave velocity,

$$v_2^2 = \omega^2(A^2 - y^2)$$

$$\therefore \left(\frac{\omega A}{2} \right)^2 = \omega^2(A^2 - y^2)$$

$$\Rightarrow \frac{\omega^2 A^2}{4} = \omega^2(A^2 - y^2)$$

$$\Rightarrow \frac{A^2}{4} = A^2 - y^2$$

$$\Rightarrow y = \frac{\sqrt{3}}{2}A$$

Hence, $x = 3$

Question 87

$Y = A \sin(\omega t + \phi_0)$ is the time-displacement equation of SHM. At $t = 0$, the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x-direction. Then, the initial phase angle ϕ_0 will be
[25 Feb 2021 Shift 2]

Options:

A. $\frac{\pi}{3}$

B. $\frac{5\pi}{6}$

C. $\frac{\pi}{6}$

D. $\frac{2\pi}{3}$

Answer: C

Solution:

Given, displacement-time equation,

$$Y = A \sin(\omega t + \phi_0)$$

Here, A is amplitude, ω is angular frequency, t is time taken and ϕ_0 is the phase constant.

At $t = 0$, $Y = A/2$

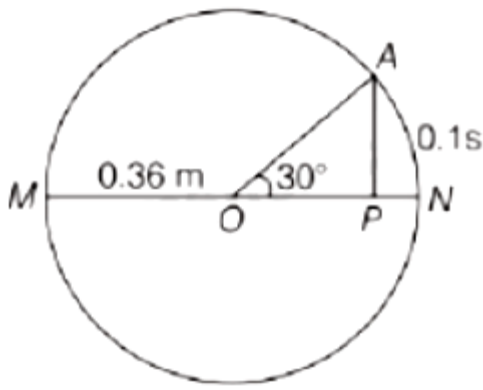
$$\therefore Y = A/2 = A \sin(0 + \phi_0) = A \sin \phi_0$$

$$\Rightarrow \sin \phi_0 = 1/2$$

$$\Rightarrow \phi_0 = \pi/6$$

Question88

The point A moves with a uniform speed along the circumference of a circle of radius 0.36m and covers 30° in 0.1s. The perpendicular projection P from A on the diameter MN represents the simple harmonic motion of P. The restoration force per unit mass when P touches M will be



[25 Feb 2021 Shift 2]

Options:

- A. 100 N
- B. 9.87 N
- C. 50 N
- D. 0.49 N

Answer: B

Solution:

Given, radius of circle, $R = 0.36\text{m}$

Angular distance, $\theta = 30^\circ = \pi/6\text{rad}$

Let l be the arc length.

$$\therefore l = R\theta$$

$$\Rightarrow l = \frac{36}{100} \times \frac{\pi}{6} = \frac{6\pi}{100}\text{m}$$

As, speed on circular track (v) = Arc length(l)/Time(t)

$$\Rightarrow v = \frac{6\pi}{100 \times 0.1} = \frac{6\pi}{10}\text{ms}^{-1}$$

If F be the restoration force and a_r be the radial acceleration ($= v^2/R$), then

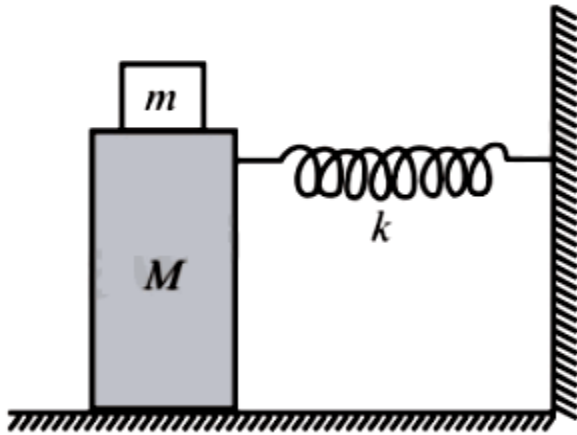
$$F = ma_r$$

$$\begin{aligned} a_r &= \frac{F}{m} = \frac{v^2}{R} = \left(\frac{6\pi}{10}\right)^2 \times \frac{100}{36} \\ &= \frac{36 \times 9.87}{100} \times \frac{100}{36} = 9.87\text{N} \end{aligned}$$

Question89

In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of

the spring is k . The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillation will be :



[24feb2021shift1]

Options:

A. $A \sqrt{\frac{M-m}{M}}$

B. $A \sqrt{\frac{M}{M+m}}$

C. $A \sqrt{\frac{M+m}{M}}$

D. $A \sqrt{\frac{M}{M-m}}$

Answer: B

Solution:

Let the initial velocity of M is V . On putting m on M , let velocity becomes V' .
Momentum of system remains conserved.

$$\therefore p_i = p_f \Rightarrow M V = (m + M) V'$$

$$\Rightarrow M A \omega = (m + M) A' \omega (\because V = A \omega)$$

$$\Rightarrow M A \sqrt{\frac{k}{M}} = (m + M) A' \sqrt{\frac{k}{m + M}}$$

$$\Rightarrow A' = A \sqrt{\frac{M}{M + m}}$$

Question90

Two particles A and B of equal masses are suspended from two massless springs of spring constants k_1 and k_2 , respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A and B is
[17 Mar 2021 Shift 2]

Options:

A. $\frac{k_2}{k_1}$

B. $\frac{k_1}{k_2}$

C. $\sqrt{\frac{k_1}{k_2}}$

D. $\sqrt{\frac{k_2}{k_1}}$

Answer: D

Solution:

We know that, the expression of maximum velocity during oscillation,

$$V_{\max} = A\omega$$

$$\text{Given, } V_{\max}(A) = V_{\max}(B) \Rightarrow A_1\omega_1 = A_2\omega_2$$

Now, the ratio of the amplitude during oscillation,

$$\frac{A_1}{A_2} = \frac{\omega_2}{\omega_1}$$

We know that, $\omega = \sqrt{k/m}$

where, k is the spring constant,

m is the mass of the object.

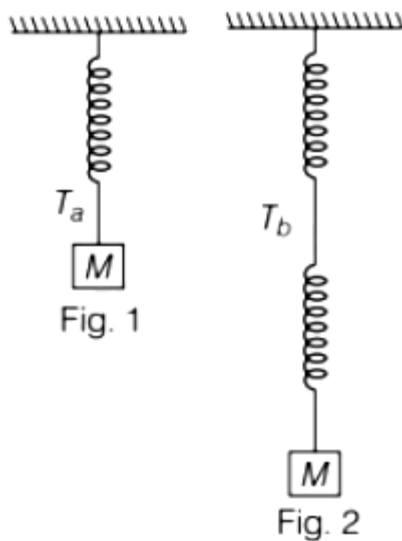
$$\text{Substituting the value of } \omega \text{ in Eq. (i), we get } \frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}}$$

Question91

Consider two identical springs each of spring constant k and negligible mass compared to the mass M as shown. Fig.1 shows one of them and Fig.2 shows their series combination. The ratios of time period of oscillation of the two SHM is $\frac{T_b}{T_a} = \sqrt{x}$, where value of x is

.....

(Round off to the nearest integer)



[17 Mar 2021 Shift 1]

Answer: 2

Solution:

Time period of Fig. 1 can be given as

$$T_a = 2\pi \sqrt{\frac{M}{k}}$$

where, M is mass of the suspended object and k is the force constant.

In Fig. 2, both the springs are in series combination, therefore its time period can be given as

$$T_b = 2\pi \sqrt{\frac{M}{k_{eq}}} = 2\pi \sqrt{\frac{M}{k/2}} \quad \left(\because k_{eq} = \frac{k \times k}{k + k} \right)$$

$$\text{Now, } \frac{T_b}{T_a} = \frac{2\pi \sqrt{\frac{M}{k/2}}}{2\pi \sqrt{\frac{M}{k}}}$$

$$\Rightarrow \frac{T_b}{T_a} = \sqrt{2} \dots (i)$$

According to question, the ratio of time period of oscillation of two SHM is $T_b/T_a = \sqrt{x}$, so on comparing it with Eq. (i) we can say, $x = 2$

Question92

Time period of a simple pendulum is T inside a lift, when the lift is stationary. If the lift moves upwards with an acceleration $g/2$, then the time period of pendulum will be
[16 Mar 2021 Shift 1]

Options:

A. $\sqrt{3}T$

B. $\frac{T}{\sqrt{3}}$

C. $\sqrt{\frac{3}{2}}T$

D. $\sqrt{\frac{2}{3}}T$

Answer: D

Solution:

Time period of a simple pendulum can be given as

$$T = 2\pi \sqrt{\frac{L}{g}} \dots (i)$$

When the lift moves upwards, then effective acceleration is $\Rightarrow$

$$\Rightarrow g_{\text{eff}} = g + a = g + \frac{g}{2} = \frac{3g}{2}$$

$$\therefore \text{New time period, } T_1 = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{2L}{3g}}$$

$$\Rightarrow T_1 = \sqrt{\frac{2}{3}}T \quad [\text{From Eq. (i)}]$$

Question93

A block of mass 1kg attached to a spring is made to oscillate with an initial amplitude of 12cm. After 2min, the amplitude decreases to 6cm. Determine the value of the damping constant for this motion. (Take, $\ln 2 = 0.693$)
[17 Mar 2021 Shift 2]

Options:

- A. $0.69 \times 10^2 \text{ kg/s}$
- B. $3.3 \times 10^2 \text{ kg/s}$
- C. $1.16 \times 10^{-2} \text{ kg/s}$
- D. $5.7 \times 10^{-3} \text{ kg/s}$

Answer: C

Solution:

Given, mass block, $m = 1\text{kg}$
 Initial amplitude, $A_0 = 12\text{cm}$
 Final amplitude, $A = 6\text{cm}$
 The time taken to reduce the amplitude, $t = 2\text{min} = 120\text{s}$
 Using the expression of damped oscillation,

$$A = A_0 e^{-\frac{b}{2m}t}$$

Substituting the values in the above equation, we get

$$6 = 12 e^{-\frac{b(120)}{2(1)}}$$

$$\Rightarrow e^{60b} = 2 \text{ or } 60b = 10 \log 2$$

$$\Rightarrow b = \frac{0.693}{60} = 1.16 \times 10^{-2} \text{ kg/s}$$

Question94

The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is

[18 Mar 2021 Shift 2]

Options:

A. $\sin(\omega t) = \cos(\omega t)$

B. $\cos(\omega t) + \cos(2\omega t) + \cos(3\omega t)$

C. $\sin^2(\omega t)$

D. $3 \cos\left(\frac{\pi}{4} - 2\omega t\right)$

Answer: D

Solution:

Time period, $T = \frac{2\pi}{\omega'}$

Given, time period, $T = \frac{\pi}{\omega}$

$$\frac{2\pi}{\omega'} = \frac{\pi}{\omega} \Rightarrow \omega' = 2\omega$$

Options (a) and (b) are incorrect.

Option (c),

$$\sin^2 \omega t = \frac{1}{2}(2\sin^2 \omega t) = \frac{1}{2}(1 - \cos 2\omega t)$$

Hence, the angular frequency is 2ω .

Option (d),

$$3 \cos\left(\frac{\pi}{4} - 2\omega t\right)$$

Angular frequency of SHM is 2ω .

Option (d) is the correct answer.

Question 95

A particle performs simple harmonic motion with a period of 2s. The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}$ s. The value of a to the nearest integer is

[18 Mar 2021 Shift 1]

Answer: 6

Solution:

Given, the time period of the simple harmonic motion, $T = 2\text{s}$

Displacement covered by the particle, $x = A/2$

Here, A is the amplitude of the particle.

The general equation of the simple harmonic motion,

$$x = A \sin \omega t$$

$$\Rightarrow \frac{A}{2} = A \sin \left(\frac{2\pi}{T}t \right) \Rightarrow \frac{1}{2} = \sin \left(\frac{2\pi}{2}t \right)$$

$$\Rightarrow \sin \frac{\pi}{6} = \sin(\pi t) \Rightarrow \frac{\pi}{6} = \pi t$$

$$\Rightarrow t = \frac{1}{6}$$

The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{6}\text{s}$.

Comparing with $t = \frac{1}{a}$

Hence, the value of the a is 6 .

Question96

For what value of displacement the kinetic energy and potential energy of a simple harmonic oscillation become equal?

[17 Mar 2021 Shift 1]

Options:

A. $x = 0$

B. $x = \pm A$

C. $x = \pm \frac{A}{\sqrt{2}}$

D. $x = \frac{A}{2}$

Answer: C

Solution:

$$\text{Potential energy of SHM} = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2$$

where m = mass of particle,

ω = angular velocity

and x = displacement.

$$\text{Kinetic energy of SHM} = \frac{1}{2}m(\omega\sqrt{A^2 - x^2})^2$$

Here, A is the amplitude of SHM.

According to question,

Potential energy of SHM = Kinetic energy of SHM

$$\frac{1}{2}m\omega^2x^2 = \frac{1}{2}m(\omega\sqrt{A^2 - x^2})^2$$

$$\Rightarrow \omega^2x^2 = \omega^2(\sqrt{A^2 - x^2})^2 \Rightarrow \omega^2x^2 = \omega^2(A^2 - x^2)$$

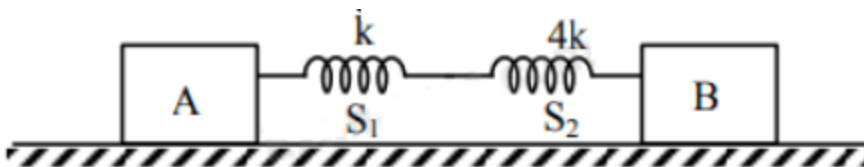
$$\Rightarrow \omega^2x^2 = \omega^2A^2 - \omega^2x^2 \Rightarrow 2\omega^2x^2 = \omega^2A^2$$

$$\Rightarrow 2x^2 = A^2 \Rightarrow x^2 = \frac{A^2}{2}$$

$$\Rightarrow x = \pm \sqrt{\frac{A^2}{2}} \Rightarrow x = \pm \frac{A}{\sqrt{2}}$$

Question 97

In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be _____ rad/s when $k = 20 \text{ N/m}$.



[25 Jul 2021 Shift 1]

Answer: 10

Solution:

$$\omega = \sqrt{\frac{k_{eq}}{\mu}}$$

μ = reduced mass

springs are in series connection

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

$$k_{eq} = \frac{k \times 4k}{k + 4k} = \frac{4k}{5}$$

$$k_{eq} = \frac{4 \times 20}{5} \text{ N/m} = 16 \text{ N/m}$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{0.2 \times 0.8}{0.2 + 0.8} = 0.16 \text{ kg}$$

$$\omega = \sqrt{\frac{16}{0.16}} = \sqrt{100} = 10$$

Question98

T_0 is the time period of a simple pendulum at a place. If the length of the pendulum is reduced to $\frac{1}{16}$ times of its initial value, the modified timeperiod is :
[22 Jul 2021 Shift 2]

Options:

A. T_0

B. $8\pi T_0$

C. $4T_0$

D. $\frac{1}{4}T_0$

Answer: C

Solution:

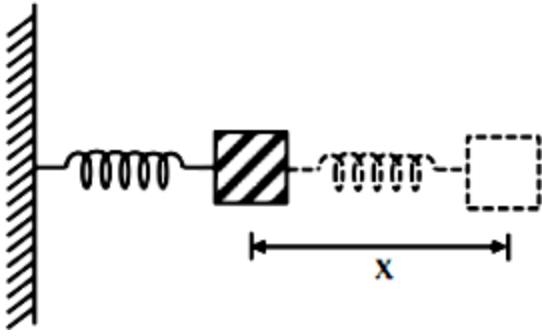
$$T_0 = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{New time period } T = 2\pi \sqrt{\frac{l/16}{g}} = \frac{2\pi}{4} \sqrt{\frac{l}{g}}$$

$$T = \frac{T_0}{4}$$

Question99

The motion of a mass on a spring, with spring constant K is as shown in figure.



The equation of motion is given by $x(t) = A \sin \omega t + B \cos \omega t$ with $\omega = \sqrt{\frac{K}{m}}$. Suppose that at time $t = 0$, the position of mass is $x(0)$ and velocity $v(0)$, then its displacement can also be represented as $x(t) = C \cos(\omega t - \phi)$, where C and ϕ are :

[22 Jul 2021 Shift 2]

Options:

A. $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$

B. $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{x(0)\omega}{2v(0)} \right)$

C. $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{x(0)\omega}{v(0)} \right)$

D. $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$

Answer: D

Solution:

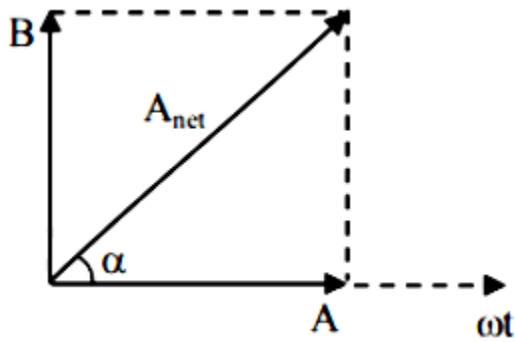
$$x = A \sin \omega t + B \cos \omega t$$

$$v = \frac{dx}{dt} = A\omega \cos \omega t - B\omega \sin \omega t$$

$$\text{At } t = 0, x(0) = B$$

$$v(0) = A\omega$$

$$x = A \sin \omega t + B \sin(\omega t + 90^\circ)$$



$$A_{\text{net}} = \sqrt{A^2 + B^2}$$

$$\tan \alpha = \frac{B}{A} \Rightarrow \cot \alpha = \frac{A}{B}$$

$$\Rightarrow x = \sqrt{A^2 + B^2} \sin(\omega t + \alpha)$$

$$\Rightarrow x = \sqrt{A^2 + B^2} \cos(\omega t - (90 - \alpha))$$

$$x = C \cos(\omega t - \phi)$$

$$\Rightarrow C = \sqrt{A^2 + B^2}$$

$$C = \sqrt{\frac{[v(0)]^2}{\omega^2} + [x(0)]^2}$$

$$\phi = 90 - \alpha$$

$$\tan \alpha = \cot \alpha = \frac{A}{B}$$

$$\Rightarrow \tan \phi = \frac{v(0)}{x(0) \cdot \omega}$$

$$\phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$$

Question100

A particle executes simple harmonic motion represented by displacement function as

$$x(t) = A \sin(\omega t + \phi)$$

If the position and velocity of the particle at $t = 0$ s are 2cm and $2\omega \text{ cm s}^{-1}$ respectively, then its amplitude is $x\sqrt{2}$ cm where the value of x is ____.

[27 Jul 2021 Shift 2]

Answer: 2

Solution:

$$x(t) = A \sin(\omega t + \phi)$$

$$v(t) = A\omega \cos(\omega t + \phi)$$

$$2 = A \sin \phi \dots\dots(1)$$

$$2\omega = A\omega \cos \phi \dots\dots(2)$$

From (1) and (2) $\tan \phi = 1$

$$\phi = 45^\circ$$

Putting value of ϕ in equation (1)

$$2 = A \left\{ \frac{1}{\sqrt{2}} \right\}$$

$$A = 2\sqrt{2}$$

$$x = 2$$

Question101

A particle starts executing simple harmonic motion (SHM) of amplitude 'a' and total energy E . At any instant, its kinetic energy is $\frac{3E}{4}$ then its displacement 'y' is given by:

[27 Jul 2021 Shift 1]

Options:

A. $y = a$

B. $y = \frac{a}{\sqrt{2}}$

C. $y = \frac{a\sqrt{3}}{2}$

D. $y = \frac{a}{2}$

Answer: D

Solution:

$$E = \frac{1}{2}K a^2$$

$$\frac{3E}{4} = \frac{1}{2}K (a^2 - y^2)$$

$$\frac{3}{4} \times \frac{1}{2}K a^2 = \frac{1}{2}K (a^2 - y^2)$$

$$y^2 = a^2 - \frac{3a^2}{4}$$

$$y = \frac{a}{2}$$

Question102

In a simple harmonic oscillation, what fraction of total mechanical energy is in the form of kinetic energy, when the particle is midway between mean and extreme position.

[25 Jul 2021 Shift 2]

Options:

A. $\frac{1}{2}$

B. $\frac{3}{4}$

C. $\frac{1}{3}$

D. $\frac{1}{4}$

Answer: B

Solution:

$$K = \frac{1}{2}m\omega^2(A^2 - x^2)$$

$$= \frac{1}{2}m\omega^2\left(A^2 - \frac{A^2}{4}\right)$$

$$= \frac{1}{2}m\omega^2\left(\frac{3A^2}{4}\right)$$

$$K = \frac{3}{4}\left(\frac{1}{2}m\omega^2A^2\right)$$

Question103

A particle is making simple harmonic motion along the X-axis. If at a distances x_1 and x_2 from the mean position the velocities of the

particle are v_1 and v_2 respectively. The time period of its oscillation is given as:

[20 Jul 2021 Shift 2]

Options:

$$A. T = 2\pi \sqrt{\frac{x_2^2 + x_1^2}{v_1^2 - v_2^2}}$$

$$B. T = 2\pi \sqrt{\frac{x_2^2 + x_1^2}{v_1^2 + v_2^2}}$$

$$C. T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 + v_2^2}}$$

$$D. T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

Answer: D

Solution:

$$v^2 = \omega^2(A^2 - x^2)$$

$$A^2 = x_1^2 + \frac{v_1^2}{\omega^2} = x_2^2 + \frac{v_2^2}{\omega^2}$$

$$\omega^2 = \frac{v_2^2 - v_1^2}{x_1^2 - x_2^2}$$

$$T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

Question104

A particle of mass 1 kg is hanging from a spring of force constant 100Nm^{-1} . The mass is pulled slightly downward and released, so that it executes free simple harmonic motion with time period T. The

time when the kinetic energy and potential energy of the system will become equal, is $\frac{T}{x}$. The value of x is.

[31 Aug 2021 Shift 1]

Answer: 8

Solution:

Given, mass of particle, $m = 1\text{kg}$

Spring constant, $k = 100\text{Nm}^{-1}$

Let time period is T .

Kinetic energy (KE) = Potential energy (PE) = E

Amplitude = A

Angular velocity = ω

Wave displacement = x

KE in SHM = PE in SHM

$$\Rightarrow \frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{2}kx^2$$

$$\Rightarrow m\omega^2(A^2 - x^2) = kx^2 \dots(i)$$

Since, force, $F = m\omega^2x = kx$

$$\Rightarrow k = m\omega^2$$

Substituting the value in Eq. (i), we get

$$\Rightarrow m\omega^2(A^2 - x^2) = m\omega^2x^2$$

$$\Rightarrow A^2 - x^2 = x^2$$

$$\Rightarrow A^2 = 2x^2$$

$$\Rightarrow x = \frac{A}{\sqrt{2}}$$

Since, wave displacement, $x = A \sin \omega t$

$$\Rightarrow \frac{A}{\sqrt{2}} = A \sin \omega t$$

$$\Rightarrow \sin\left(\frac{\pi}{4}\right) = \sin \omega t$$

$$\Rightarrow \omega t = \frac{\pi}{4}$$

$$\Rightarrow \frac{2\pi}{T} \cdot t = \frac{\pi}{4}$$

$$\Rightarrow t = \frac{T}{8}$$

Comparing the given value in the question i.e. $\frac{T}{x}$, the value of $x = 8$

Question105

A bob of mass m suspended by a thread of length l undergoes simple harmonic oscillations with time period T . If the bob is immersed in a liquid that has density $\frac{1}{4}$ times that of the bob and the length of the thread is increased by $\frac{1}{3}$ rd of the original length, then the time period of the simple harmonic oscillations will be
[31 Aug 2021 Shift 2]

Options:

A. T

B. $\frac{3}{2}T$

C. $\frac{3}{4}T$

D. $\frac{4}{3}T$

Answer: D

Solution:

Given that, a bob of mass m suspended by a thread of length l undergoes SHM with time period,

$$T_1 = T = 2\pi \sqrt{\frac{l}{g}} \dots (i)$$

In liquid, effective gravity,

$$g_{\text{eff}} = g \left(\frac{\sigma - \rho}{\sigma} \right)$$

where, ρ = density of liquid
 and σ = density of body (bob).

$$\text{Given, } \rho = \frac{\sigma}{4}$$

$$\Rightarrow \sigma = 4\rho$$

Substituting this value, we get

$$g_{\text{eff}} = g \left(\frac{4\rho - \rho}{4\rho} \right) = \frac{3g}{4}$$

Now, the new time period of pendulum in liquid is

$$T' = 2\pi \sqrt{\frac{l'}{g_{\text{eff}}}} \dots (ii)$$

$$\text{Here, } l' = l + \frac{1}{3}l = \frac{4l}{3}$$

Substituting the values in Eq. (ii), we get

$$T' = 2\pi \sqrt{\frac{\frac{4l}{3}}{\frac{3g}{4}}} = 2\pi \sqrt{\frac{4l}{3} \times \frac{4}{3g}}$$

$$= \left(2\pi \sqrt{\frac{L}{g}} \right) \frac{4}{3}$$

$$\Rightarrow T' = \frac{4T}{3} \text{ [from Eq. (i)]}$$

Question 106

The acceleration due to gravity is found upto an accuracy of 4% on a planet. The energy supplied to a simple pendulum to known mass m to undertake oscillations of time period T is being estimated. If time period is measured to an accuracy of 3%, the accuracy to which E is known as%.

[26 Aug 2021 Shift 2]

Answer: 14

Solution:

Given, accuracy of acceleration due to gravity = 4%

Accuracy of time period = 3%

Energy stored in pendulum at any instant is given as

$$E = \frac{mgL\theta^2}{2} \dots (i)$$

Time period of pendulum is given by expression

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Rearrange the above expression for L , we get

$$L = g \left(\frac{T}{2\pi} \right)^2 \dots (ii)$$

Substituting the value of L in Eq. (i), we get

$$E = \frac{mg^2\theta^2}{2} (T2\pi)^2 = \frac{mg^2\theta^2 T^2}{8\pi^2}$$

Now, the accuracy in measurement of energy can be calculated as

$$\frac{\Delta E}{E} \times 100 = 2 \frac{\Delta g}{g} \times 100 + 2 \frac{\Delta T}{T} \times 100 \quad (\because m, \theta \text{ and } \pi \text{ are constant})$$

$$\Rightarrow \frac{\Delta E}{E} \times 100 = 2 \times 4\% + 2 \times 3\% = 14\%$$

Thus, the accuracy to which E is known is 14%.

Question107

For a body executing SHM

A. potential energy is always equal to its kinetic energy.

B. average potential and kinetic energy over any given time interval are always equal.

C. sum of the kinetic and potential energy at any point of time is constant.

D. average kinetic energy in one time period is equal to average potential energy in one time period.

Choose the most appropriate option from the options given below.

[31 Aug 2021 Shift 2]

Options:

A. (C) and (D)

B. Only (C)

C. (B) and (C)

D. Only (B)

Answer: A

Solution:

As we know that, for a body executing SHM the average kinetic energy is equal to average potential energy over a complete time period

$$\text{i.e. } KE_{av} = PE_{av} = \frac{1}{4}m\omega^2a^2$$

$$\therefore \text{Total mechanical energy at any time} = \text{Sum of kinetic and potential energy} = \frac{1}{2}m\omega^2a^2 = \text{constant.}$$

where, m = mass of body executing SHM,

ω = angular frequency

and a = amplitude of SHM.

Hence, (c) and (d) are true.

Question108

Two simple harmonic motion, are represented by the equations

$y_1 = 10 \sin \left(3\pi t + \frac{\pi}{3} \right)$ and $y_2 = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)$ Ratio of amplitude of y_1 to $y_2 = x : 1$. The value of x is

[27 Aug 2021 Shift 2]

Answer: 1

Solution:

Given, $y_1 = 10 \sin \left(3\pi t + \frac{\pi}{3} \right)$

$A_1 = 10$

$y_2 = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)$

$= 5 \sin 3\pi t + 5\sqrt{3} \cos 3\pi t$

Amplitude of SHM₂ is

$A_2 = \sqrt{5^2 + (5\sqrt{3})^2}$

$= \sqrt{25 + 75}$

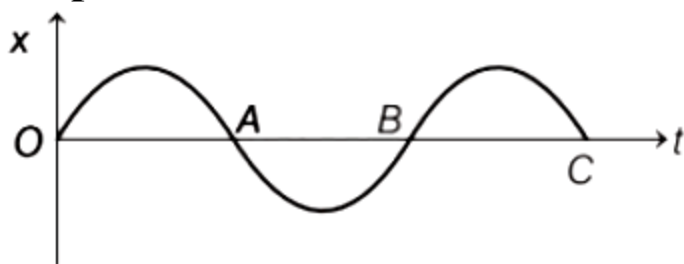
$= \sqrt{100} = 10$

$\therefore$ Ratio of amplitudes, $\frac{A_1}{A_2} = \frac{10}{10} = 1 : 1$

$x = 1$

Question109

The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure.

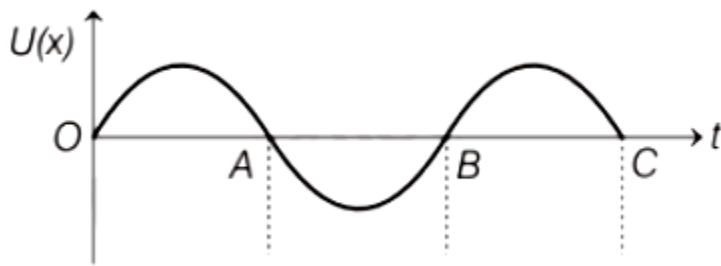


The potential energy $U(x)$ versus time (t) plot of the particle is correctly shown in figure :

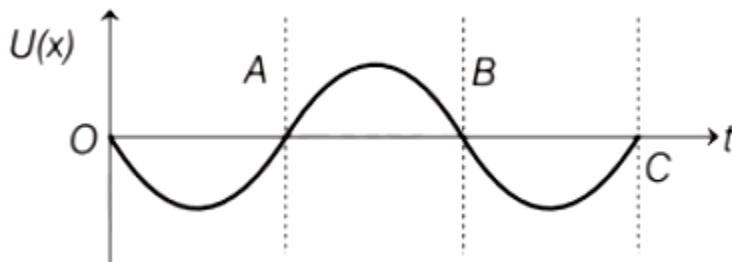
[27 Aug 2021 Shift 1]

Options:

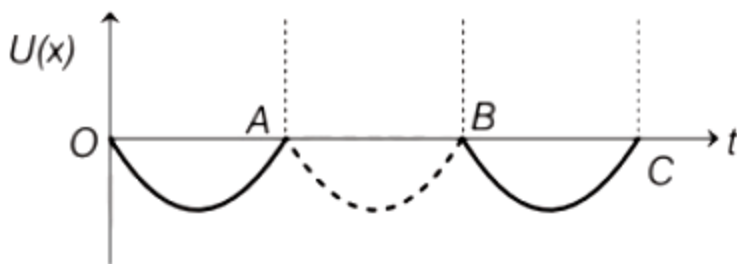
A.



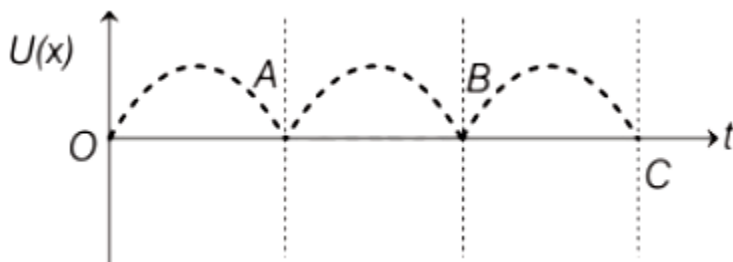
B.



C.



D.



Answer: D

Solution:

It is given that motion is free simple harmonic motion.

Consider a spring mass system performing free simple harmonic motion.

Displacement of mass is given as

$$x = x_0 \sin \omega t$$

The potential energy stored in spring for displacement x can be given as

$$PE = \frac{1}{2}kx^2$$

Substituting the value of x , we get

$$PE = \frac{1}{2}kx_0^2 \sin^2 \omega t$$

Here, k is spring constant and x_0 is also constant.

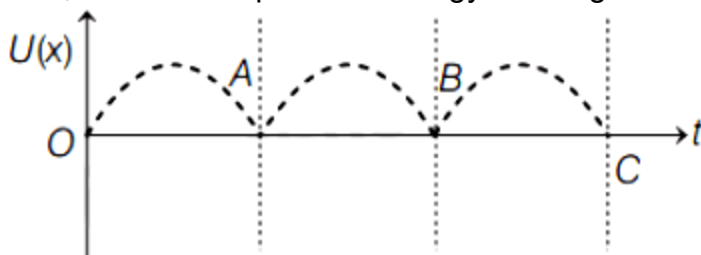
$$PE = C \sin^2 \omega t \left(\because C = \frac{1}{2}kx_0^2 \right)$$

Simplify the value of $\sin^2 \omega t$, we get

$$PE = C \left[\frac{1 - \cos 2 \omega t}{2} \right]$$

The value of function $\cos 2 \omega t$ will always lies between -1 and 1 .

Thus, the value of potential energy will be greater than or equal to zero.



$$0 \leq PE$$

This condition is only satisfies by graph (d).

Question110

Two simple harmonic motions are represented by the equations

$$x_1 = 5 \sin \left(2\pi t + \frac{\pi}{4} \right) \text{ and}$$

$$x_2 = 5\sqrt{2}(\sin 2 \pi t + \cos 2 \pi t)$$

The amplitude of second motion istimes the amplitude in first motion.

[26 Aug 2021 Shift 2]

Answer: 2

Solution:

Equation of first simple harmonic motion,

$$x_1 = 5 \sin \left(2\pi t + \frac{\pi}{4} \right).$$

Equation of second simple harmonic motion,

$$x_2 = 5\sqrt{2}(\sin 2\pi t + \cos 2\pi t)$$

Simplifying the second equation of simple harmonic motion as follows

$$x_2 = 5\sqrt{2} \left(\sin 2\pi t \times \frac{1}{\sqrt{2}} + \cos 2\pi t \times \frac{1}{\sqrt{2}} \right) \sqrt{2}$$

$$= 10 \left(\sin 2\pi t \cos \frac{\pi}{4} + \cos 2\pi t \sin \frac{\pi}{4} \right)$$

$$= 10 \sin \left(2\pi t + \frac{\pi}{4} \right)$$

The ratio of amplitudes of two simple harmonic motions is

$$\frac{A_2}{A_1} = \frac{10}{5} = 2$$

Thus, amplitude of second motion is 2 times the amplitude in first motion.

Question111

A mass of 5 kg is connected to a spring. The potential energy curve of the simple harmonic motion executed by the system is shown in the figure. A simple pendulum of length 4m has the same period of oscillation as the spring system. What is the value of acceleration due to gravity on the planet where these experiments are performed?

[1 Sep 2021 Shift 2]

Options:

A. 10m/s^2

B. 5m/s^2

C. 4m/s^2

D. 9.8m/s^2

Answer: C

Solution:

From the potential energy curve,

$$U_{\max} = \frac{1}{2}kA^2$$

$$10 = \frac{1}{2}k(2)^2$$

$$\Rightarrow k = 5 \text{ N / m}$$

The length of the simple pendulum, $L = 4 \text{ m}$

Time period of spring,

$$T = 2\pi \sqrt{\frac{k}{m}}$$

Time period of simple pendulum,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

The time period of simple pendulum is same as the time period of the spring oscillation.

$$\Rightarrow 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{k}{m}}$$

Substituting the values in the above equations, we get

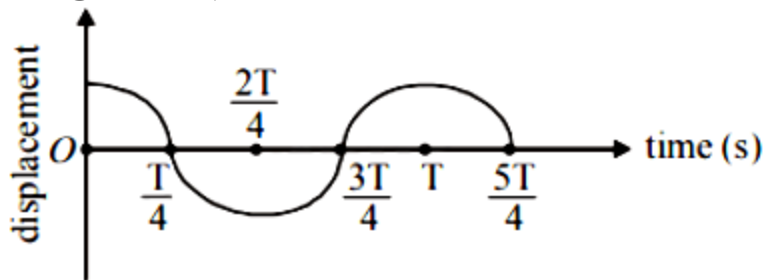
$$2\pi \sqrt{\frac{4}{g}} = 2\pi \sqrt{\frac{5}{5}}$$

$$g = 4 \text{ m/s}^2$$

$\therefore$ The acceleration due to the gravity on the planet is 4 m/s^2 .

Question 112

The displacement time graph of a particle executing S.H.M. is given in figure : (sketch is schematic and not to scale)



Which of the following statements is/are true for this motion?

- (1) The force is zero at $t = \frac{3T}{4}$
- (2) The acceleration is maximum at $t = T$
- (3) The speed is maximum at $t = \frac{T}{4}$
- (4) The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$

[Sep. 02, 2020 (II)]

Options:

- A. (1),(2) and (4)
- B. (2),(3) and (4)
- C. (1),(2) and (3)
- D. (1) and (4)

Answer: C

Solution:

From graph equation of SHM

$$X = A \cos \omega t$$

(1) At $\frac{3T}{4}$ particle is at mean position.

$\therefore$ Acceleration = 0, Force = 0

(2) At T particle again at extreme position so acceleration is maximum.

(3) At $t = \frac{T}{4}$, particle is at mean position so velocity is maximum.

Acceleration = 0

(4) When K E = P E

$$\Rightarrow \frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$$

Here, A = amplitude of SHM

x = displacement from mean position

$$\Rightarrow A^2 = 2x^2 \Rightarrow x = \frac{+A}{\sqrt{2}}$$

$$\Rightarrow \frac{A}{\sqrt{2}} = A \cos \omega t \Rightarrow t = \frac{T}{2}$$

$\therefore x = -A$ which is not possible

$\therefore$ 1, 2 and 3 are correct.

Question 113

An object of mass m is suspended at the end of a massless wire of length L and area of cross-section, A. Young modulus of the material of the wire is Y. If the mass is pulled down slightly its frequency of oscillation along the vertical direction is:

[Sep. 06, 2020 (I)]

Options:

A. $f = \frac{1}{2\pi} \sqrt{\frac{mL}{YA}}$

B. $f = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$

C. $f = \frac{1}{2\pi} \sqrt{\frac{mA}{YL}}$

$$D. f = \frac{1}{2\pi} \sqrt{\frac{Y L}{m A}}$$

Answer: B

Solution:

(b) An elastic wire can be treated as a spring and its spring constant.

$$k = \frac{Y A}{L} \left[\because Y = \frac{F}{A} / \frac{\Delta l}{l_0} \right]$$

Frequency of oscillation,

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{Y A}{m L}}$$

Question114

When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by $y(t) = y_0 \sin^2 \omega t$, where 'y' is measured from the lower end of unstretched spring.

Then ω is:

[Sep. 06, 2020 (II)]

Options:

$$A. \frac{1}{2} \sqrt{\frac{g}{y_0}}$$

$$B. \sqrt{\frac{g}{y_0}}$$

$$C. \sqrt{\frac{g}{2y_0}}$$

$$D. \sqrt{\frac{2g}{y_0}}$$

Answer: C

Solution:

$$y = y_0 \sin^2 \omega t$$

$$\Rightarrow y = \frac{y_0}{2}(1 - \cos 2\omega t) \quad \left(\because \sin^2 \omega t = \frac{1 - \cos 2\omega t}{2} \right)$$

$$\Rightarrow y - \frac{y_0}{2} = \frac{-y_0}{2} \cos 2\omega t$$

$$\Rightarrow y = A \cos 2\omega t$$

$$\therefore \text{Amplitude} = \frac{y_0}{2}$$

$$\text{Angular velocity} = 2\omega$$

$$\text{For equilibrium of mass, } \frac{ky_0}{2} = mg \Rightarrow \frac{k}{m} = \frac{2g}{y_0}$$

$$\text{Also, spring constant } k = m(2\omega)^2$$

$$\Rightarrow 2\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2g}{y_0}} \Rightarrow \omega = \frac{1}{2} \sqrt{\frac{2g}{y_0}} = \sqrt{\frac{g}{2y_0}}$$

Question115

A block of mass m attached to a massless spring is performing oscillatory motion of amplitude ' A ' on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system become fA . The value of f is :

[Sep. 03, 2020 (II)]

Options:

A. $\frac{1}{\sqrt{2}}$

B. 1

C. $\frac{1}{2}$

D. $\sqrt{2}$

Answer: A

Solution:

$$\text{Potential energy of spring} = \frac{1}{2}kx^2$$

Here, x = distance of block from mean position,

k = spring constant

At mean position, potential energy $= \frac{1}{2}kA^2$

At equilibrium position, half of the mass of block breaks off, so its potential energy becomes half.

Remaining energy $= \frac{1}{2} \left(\frac{1}{2}kA^2 \right) = \frac{1}{2}kA'^2$

Here, A' = New distance of block from mean position

$$\Rightarrow A' = \frac{A}{\sqrt{2}}$$

Question116

The position co-ordinates of a particle moving in a 3 – D coordinate system is given by

$$x = a \cos \omega t$$

$$y = a \sin \omega t$$

$$\text{and } z = a\omega t$$

The speed of the particle is:

[9 Jan 2019, II]

Options:

A. $\sqrt{2}a\omega$

B. $a\omega$

C. $\sqrt{3}a\omega$

D. $2a\omega$

Answer: A

Solution:

Here, $v_x = -a\omega \sin \omega t$, $v_y = a\omega \cos \omega t$ and $v_z = a\omega$

$$\Rightarrow v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\Rightarrow v = \sqrt{(-a\omega \sin \omega t)^2 + (a\omega \cos \omega t)^2 + (a\omega)^2}$$

$$v = \sqrt{2}a\omega$$

Question117

A particle undergoing simple harmonic motion has time dependent displacement given by $x(t) = A \sin \frac{\pi t}{90}$. The ratio of kinetic to potential energy of this particle at $t = 210$ s will be
[11 Jan 2019, I]

Options:

A. $\frac{1}{9}$

B. 1

C. 2

D. $\frac{1}{3}$

Answer: D

Solution:

$$\text{Kinetic energy, } k = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

$$\text{Potential energy, } U = \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t$$

$$\frac{k}{U} = \cot^2 \omega t = \cot^2 \frac{\pi}{90} (210) = \frac{1}{3}$$

Question118

A pendulum is executing simple harmonic motion and its maximum kinetic energy is K_1 . If the length of the pendulum is doubled and it performs simple harmonic motion with the same amplitude as in the first case, its maximum kinetic energy is K_2 .

[11 Jan 2019, II]

Options:

A. $K_2 = 2K_1$

B. $K_2 = \frac{K_1}{2}$

$$C. K_2 = \frac{K_1}{4}$$

$$D. K_2 = K_1$$

Answer: A

Solution:

$$K = \frac{1}{2}m\omega^2x^2$$

$$\Rightarrow K_{\max} = \frac{1}{2}m\omega^2A^2$$

$$A = L\theta$$

$$\omega = \sqrt{\frac{g}{L}}$$

$$\Rightarrow K = \frac{1}{2}m \cdot \frac{g}{L} \cdot L^2\theta^2$$

$$= \frac{1}{2}mgL\theta^2$$

$$\therefore \frac{K_1}{K_2} = \frac{L}{2L} = \frac{1}{2} \Rightarrow K_2 = 2K_1$$

$$\therefore K_1K_2 = \frac{L}{2L} = \frac{1}{2} \Rightarrow K_2 = 2K_1$$

Question119

**A particle is executing simple harmonic motion (SHM) of amplitude A, along the x -axis, about x = 0. When its potential Energy (PE) equals kinetic energy (KE), the position of the particle will be:
[9 Jan 2019, II]**

Options:

A. $\frac{A}{2}$

B. $\frac{A}{2\sqrt{2}}$

C. $\frac{A}{\sqrt{2}}$

D. A

Answer: C

Solution:

$$\text{Potential energy (U)} = \frac{1}{2}kx^2$$

$$\text{Kinetic energy (K)} = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$$

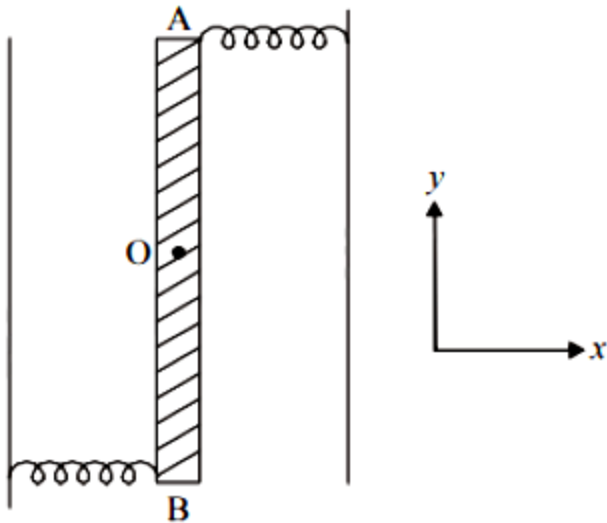
According to the question, $U = K$

$$\therefore \frac{1}{2}kx^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$$

$$\Rightarrow x^2 = A^2 \text{ or, } x = \pm \frac{A}{\sqrt{2}}$$

Question120

Two light identical springs of spring constant k are attached horizontally at the two ends of a uniform horizontal rod AB of length l and mass m . The rod is pivoted at its centre 'O' and can rotate freely in horizontal plane. The other ends of two springs are fixed to rigid supports as shown in figure. The rod is gently pushed through a small angle and released. The frequency of resulting oscillation is:



[12 Jan 2019, I]

Options:

A. $\frac{1}{2\pi} \sqrt{\frac{3k}{m}}$

B. $\frac{1}{2\pi} \sqrt{\frac{2k}{m}}$

C. $\frac{1}{2\pi} \sqrt{\frac{6k}{m}}$

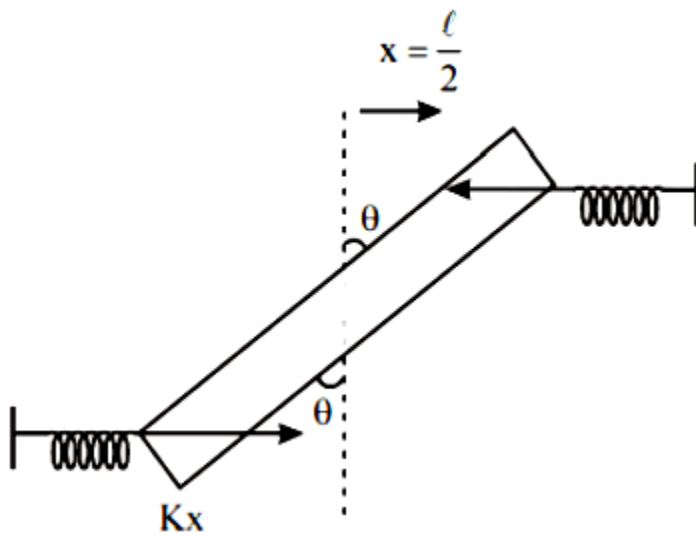
D. $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$

Answer: C

Solution:

Net torque due to spring force:

$$\tau = -2Kx \frac{l}{2} \cos \theta$$



$$\Rightarrow \tau = \left(\frac{Kl^2}{2} \right) \theta = -C\theta \quad \left[\text{let } C = \frac{Kl^2}{2} \right]$$

$\Rightarrow$ So, frequency of resulting oscillations

$$f = \frac{1}{2\pi} \sqrt{\frac{C}{I}} = \frac{1}{2\pi} \sqrt{\frac{\frac{Kl^2}{2}}{\frac{Ml^2}{12}}} = \frac{1}{2\pi} \sqrt{\frac{6K}{M}}$$

Question121

A simple pendulum, made of a string of length l and a bob of mass m , is released from a small angle θ_0 . It strikes a block of mass M , kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1 . The M is given by:

[12 Jan 2019, I]

Options:

A. $\frac{m}{2} \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$

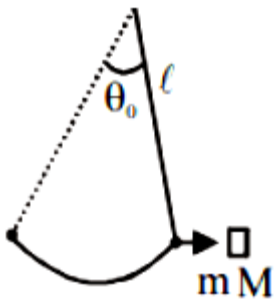
B. $m \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$

C. $m \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$

D. $\frac{m}{2} \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$

Answer: B

Solution:



Velocity before collision

$$v = \sqrt{2gl(1 - \cos \theta_0)}$$

Velocity after collision

$$v_1 = \sqrt{2gl(1 - \cos \theta_1)}$$

Using momentum conservation

$$mv = M V_m - mV_1$$

$$m\sqrt{2gl(1 - \cos \theta_0)} = M V_m - m\sqrt{2gl(1 - \cos \theta_1)}$$

$$\Rightarrow m\sqrt{2gl} \{ \sqrt{1 - \cos \theta_0} + \sqrt{1 - \cos \theta_1} \} = M V_m$$

$$\text{and } e = 1 = \frac{V_m + \sqrt{2gl(1 - \cos \theta_1)}}{\sqrt{2gl(1 - \cos \theta_0)}}$$

$$\sqrt{2gl} (\sqrt{1 - \cos \theta_0} - \sqrt{1 - \cos \theta_1}) = V_m \dots\dots(i)$$

$$m\sqrt{2gl} (\sqrt{1 - \cos \theta_0} + \sqrt{1 - \cos \theta_1}) = M V_m \dots\dots(ii)$$

Dividing (ii) by (i) we get

$$\frac{(\sqrt{1 - \cos \theta_0} + \sqrt{1 - \cos \theta_1})}{(\sqrt{1 - \cos \theta_0} - \sqrt{1 - \cos \theta_1})} = \frac{M}{m}$$

By componendo and dividendo rule

$$\frac{m - M}{m + M} = \frac{\sqrt{1 - \cos \theta_1}}{\sqrt{1 - \cos \theta_0}} = \frac{\sin\left(\frac{\theta_1}{2}\right)}{\sin\left(\frac{\theta_0}{2}\right)}$$

$$\Rightarrow \frac{M}{m} = \frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \Rightarrow M = m \frac{\theta_0 - \theta_1}{\theta_0 + \theta_1}$$

Question122

A simple harmonic motion is represented by

$$y = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t) \text{cm}$$

The amplitude and time period of the motion are :
[12 Jan 2019, II]

Options:

A. 10cm, $\frac{2}{3}$ s

B. 10cm, $\frac{3}{2}$ s

C. 5cm, $\frac{3}{2}$ s

D. 5cm, $\frac{2}{3}$ s

Answer: A

Solution:

Given: $y = 5[\sin(3\pi t) + \sqrt{3} \cos(3\pi t)]$

$$\Rightarrow y = 10 \sin\left(3\pi t + \frac{\pi}{3}\right)$$

$\therefore$ Amplitude = 10cm

Time period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{3\pi} = \frac{2}{3}$ s

Question123

A simple pendulum of length 1m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1rad /s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is best given by:

[11 Jan 2019, II]

Options:

A. 10^{-3} rad/s

B. 1 rad/s

C. 10^{-1} rad/s

D. 10^{-5} rad/s

Answer: A

Solution:

Angular frequency of pendulum $\omega = \sqrt{\frac{g}{l}}$

$\therefore$ relative change in angular frequency

$$\frac{\Delta\omega}{\omega} = \frac{1}{2} \frac{\Delta g}{g} \left[\text{as length remains constant} \right]$$

$$\Delta g = 2A\omega_s^2 \left[\omega_s = \text{angular frequency of support and, } A = \text{amplitude} \right]$$

$$\frac{\Delta\omega}{\omega} = \frac{1}{2} \times \frac{2A\omega_s^2}{g}$$

$$\Delta\omega = \frac{1}{2} \times \frac{2 \times 1^2 \times 10^{-2}}{10} = 10^{-3} \text{ rad/sec}$$

Question124

The mass and the diameter of a planet are three times the respective values for the Earth. The period of oscillation of a simple pendulum on the Earth is 2s. The period of oscillation of the same pendulum on the planet would be:

[11 Jan 2019, II]

Options:

A. $\frac{\sqrt{3}}{2}s$

B. $\frac{2}{\sqrt{3}}s$

C. $\frac{3}{2}s$

D. $2\sqrt{3}s$

Answer: D

Solution:

Acceleration due to gravity $g = \frac{GM}{R^2}$

$$\frac{g_p}{g_e} = M_p M_e \left(\frac{R_e}{R_p} \right)^2 = 3 \left(\frac{1}{3} \right)^2 = \frac{1}{3}$$

$$\text{Also } T \propto \frac{1}{\sqrt{g}} \Rightarrow \frac{T_p}{T_e} = \sqrt{\frac{g_e}{g_p}} = \sqrt{3}$$

$$\Rightarrow T_p = 2\sqrt{3}s$$

Question125

A particle executes simple harmonic motion with an amplitude of 5cm. When the particle is at 4cm from the mean position, the magnitude of its velocity in SI units is equal to that of its acceleration. Then, its periodic time in seconds is:

[10 Jan 2019, II]

Options:

A. $\frac{4\pi}{3}$

B. $\frac{3}{8}\pi$

C. $\frac{8\pi}{3}$

D. $\frac{7}{3}\pi$

Answer: C

Solution:

$$\text{Velocity, } v = \omega \sqrt{A^2 - x^2} \dots\dots(i)$$

$$\text{acceleration, } a = -\omega^2 x \dots\dots(ii)$$

and according to question,

$$|v| = |a|$$

$$\Rightarrow \omega \sqrt{A^2 - x^2} = \omega^2 x$$

$$\Rightarrow A^2 - x^2 = \omega^2 x^2$$

$$\Rightarrow 5^2 - 4^2 = \omega^2 (4^2)$$

$$3 = \omega \times 4 \Rightarrow \omega = \frac{3}{4}$$

$$\therefore T = 2\pi/\omega = \frac{2\pi}{3/4} = \frac{8\pi}{3}$$

Question126

A cylindrical plastic bottle of negligible mass is filled with 310ml of water and left floating in a pond with still water. If pressed downward slightly and released, it starts performing simple harmonic motion at angular frequency ω . If the radius of the bottle is 2.5cm then omega is close to: (density of water = 10^3 kg/m^3) [10 Jan 2019, II]

Options:

A. 3.75 rad s^{-1}

B. 1.25 rad s^{-1}

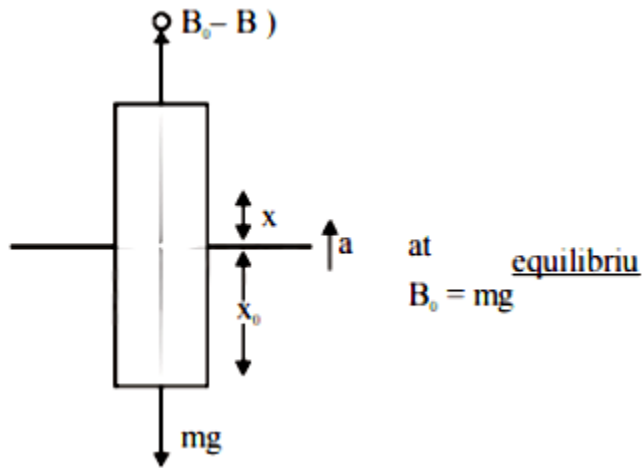
C. 2.50 rad s^{-1}

D. 5.00 rad s^{-1}

E. (Bonus)

Answer: E

Solution:



Extra boyant force $= \rho A x g$

$$B_0 + B = mg + ma$$

$$\therefore B = ma = \rho A x g = (\pi r^2 \rho g) x$$

$$a = \frac{(\pi r^2 \rho g) x}{m}$$

$$\text{using, } a = \omega^2 x$$

$$\Rightarrow \omega = \sqrt{\frac{\omega r^2 \rho g}{m}}$$

$$W \simeq 7.95 \text{ rad s}^{-1}$$

Question127

A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to :
 [9 Jan 2019, II]

Options:

A. 0.77

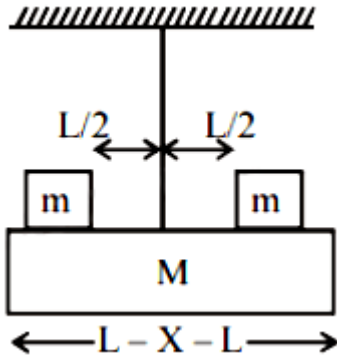
B. 0.57

C. 0.37

D. 0.17

Answer: C

Solution:



$$f_1 = \frac{1}{2\pi} \sqrt{\frac{C}{I}} \dots\dots (i)$$

$$= \frac{1}{2} \sqrt{3CML^2}$$

$$f_2 = \frac{1}{2\pi} \sqrt{CL^2 \left(\frac{M}{3} + \frac{M}{2} \right)} \dots\dots (ii)$$

As frequency reduces by 80%

$$\therefore f_2 = 0.8f_1 \Rightarrow \frac{f_2}{f_1} = 0.8 \dots\dots (iii)$$

Solving equations (i), (ii) & (iii)

$$\text{Ratio } \frac{m}{M} = 0.37$$

As frequency reduces by 80%

$$\therefore f_2 = 0.8f_1 \Rightarrow \frac{f_2}{f_1} = 0.8 \dots\dots (iii)$$

Solving equations (i), (ii) & (iii)

$$\text{Ratio } \frac{m}{M} = 0.37$$

Question128

The displacement of a damped harmonic oscillator is given by

$$x(t) = e^{0.1t} \cdot \cos(10\pi t + \phi). \text{ Here } t \text{ is in seconds.}$$

The time taken for its amplitude of vibration to drop to half of its initial value is close to :

[9 Jan 2019, II]

Options:

A. 4s

B. 7s

C. 13s

D. 27s

Answer: B

Solution:

Amplitude of vibration at time $t = 0$ is given by

$$A = A_0 e^{-0.1 \times 0} = 1 \times A_0 = A$$

$$\text{also at } t = t, \text{ if } A = \frac{A_0}{2}$$

$$\Rightarrow \frac{1}{2} = e^{-0.1t}$$

$$t = 10 \ln 2 \simeq 7\text{s}$$

Question129

A person of mass M is, sitting on a swing of length L and swinging with an angular amplitude θ_0 . If the person stands up when the swing passes through its lowest point, the work done by him, assuming that his centre of mass moves by a distance l ($l \ll L$), is close to :

[12 April 2019, II]

Options:

A. $mg l (1 - \theta_0^2)$

B. $mg l (1 + \theta_0^2 \text{ rht})$

C. $mg l$

D. $M g l \left(1 + \frac{\theta_0^2}{2} \right)$

Answer: B

Question130

**A simple pendulum oscillating in air has period T . The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$ th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is :
[9 April 2019 I]**

Options:

A. $2T \sqrt{\frac{1}{10}}$

B. $2T \sqrt{\frac{1}{14}}$

C. $4T \sqrt{\frac{1}{15}}$

D. $4T \sqrt{\frac{1}{14}}$

Answer: C

Solution:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

When immersed non viscous liquid

$$a_{\text{net}} = \left(g - \frac{g}{16} \right) = \frac{15g}{16}$$

$$\text{Now } T' = 2\pi \sqrt{\frac{l}{a_{\text{net}}}} = 2\pi \sqrt{\frac{l}{\frac{15g}{16}}} = \frac{4}{\sqrt{15}} T$$

Question131

A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original

amplitude is close to:
[8 April 2019, II]

Options:

A. 50s

B. 100s

C. 20s

D. 10s

Answer: C

Solution:

Time of half the amplitude is $= 2s$

Using, $A = A_0 e^{-kt}$

$$\frac{A_0}{2} = A_0 e^{-k \times 2} \dots\dots(i)$$

$$\text{and } \frac{A_0}{1000} = A_0 e^{-kt} \dots\dots(ii)$$

Dividing (i) by (ii) and solving, we get

$t = 20s$

Question132

Two simple harmonic motions, as shown, are at right angles. They are combined to form Lissajous figures.

$$x(t) = A \sin(at + \delta)$$

$$y(t) = B \sin(bt)$$

Identify the correct match below

[Online April 15, 2018]

Options:

A. Parameters: $A = B$, $a = 2b$; $\delta = \frac{\pi}{2}$; Curve: Circle

B. Parameters: $A = B$, $a = b$; $\delta = \frac{\pi}{2}$; Curve: Line

C. Parameters: $A \neq B$, $a = b$; $\delta = \frac{\pi}{2}$; Curve: Ellipse

D. Parameters: $A \neq B$, $a = b$; $\delta = 0$; Curve: Parabola

Answer: C

Solution:

From the two mutually perpendicular S.H.M.'s, the general equation of Lissajous figure,

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB} \cos \delta = \sin^2 \delta$$

$$x = A \sin(at + \delta)$$

$$y = B \sin(bt + r)$$

Clearly $A \neq B$ hence ellipse.

Question133

A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of 10^{12} /sec. What is the force constant of the bonds connecting one atom with the other? (Mole wt. of silver = 108 and Avagadro number = 6.02×10^{23} gmmol e^{-1}) [2018]

Options:

A. 6.4N /m

B. 7.1N /m

C. 2.2N /m

D. 5.5N /m

Answer: B

Solution:

$$\text{As we know, frequency in SHM } f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = 10^{12}$$

where m = mass of one atom

$$\text{Mass of one atom of silver,} = \frac{108}{(6.02 \times 10^{23})} \times 10^{-3} \text{kg}$$

$$\frac{1}{2\pi} \sqrt{\frac{k}{108 \times 10^{-3}}} \times 6.02 \times 10^{23} = 10^{12}$$

Solving we get, spring constant,
 $K = 7.1 \text{N /m}$

Question134

A particle executes simple harmonic motion and is located at $x = a$, b and c at t_0 , $2t_0$ and $3t_0$ respectively. The frequency of the oscillation is

[Online April 16, 2018]

Options:

A. $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+b}{2c} \right)$

B. $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+b}{3c} \right)$

C. $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{2a+3c}{b} \right)$

D. $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$

Answer: D

Solution:

Using $y = A \sin \omega t$

$$a = A \sin \omega t_0$$

$$b = A \sin 2 \omega t_0$$

$$c = A \sin 3 \omega t_0$$

$$a + c = A[\sin \omega t_0 + A \sin 3 \omega t_0] = 2A \sin 2 \omega t_0 \cos \omega t_0$$

$$\frac{a+c}{b} = 2 \cos \omega t_0$$

$$\Rightarrow \omega = \frac{1}{t_0} \cos^{-1} \left(\frac{a+c}{2b} \right) \Rightarrow f = \frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$$

Question135

An oscillator of mass M is at rest in its equilibrium position in a potential $V = \frac{1}{2}k(x - X)^2$. A particle of mass m comes from right with speed u and collides completely inelastically with M and sticks to it. This process repeats every time the oscillator crosses its equilibrium position. The amplitude of oscillations after 13 collisions is:

($M = 10, m = 5, u = 1, k = 1$),

[Online April 16, 2018]

Options:

A. $\frac{1}{2}$

B. $\frac{1}{\sqrt{3}}$

C. $\frac{2}{3}$

D. $\sqrt{\frac{3}{5}}$

Answer: B

Solution:

In first collision mu momentum will be imparted to system, in second collision when momentum of $(M + m)$ is in opposite direction mu momentum of particle will make its momentum zero.

On 13th collision, $[m] \rightarrow [M+12m]$; $[M + 13m] \rightarrow V$

$$mu = (M + 13m)v \Rightarrow v = \frac{mu}{M + 13m} = \frac{u}{15}$$

$$v = \omega A \Rightarrow \frac{u}{15} = \sqrt{\frac{K}{M + 13m}} \times A$$

$$\text{Putting value of } M, m, u \text{ and } K \text{ we get amplitude } A = \frac{1}{15} \sqrt{\frac{75}{1}} = \frac{1}{\sqrt{3}}$$

Question136

The ratio of maximum acceleration to maximum velocity in a simple harmonic motion is 10s^{-1} . At, $t = 0$ the displacement is 5m . What is the maximum acceleration ? The initial phase is $\frac{\pi}{4}$
[Online April 8,2017]

Options:

- A. 500 m/s^2
- B. $500\sqrt{2}\text{ m/s}^2$
- C. 750 m/s^2
- D. $750\sqrt{2}\text{ m/s}^2$

Answer: B

Solution:

Maximum velocity in SH M, $v_{\max} = a\omega$

Maximum acceleration in SH M, $A_{\max} = a\omega^2$

where a and ω are maximum amplitude and angular frequency.

Given that, $\frac{A_{\max}}{v_{\max}} = 10$

i.e., $\omega = 10\text{s}^{-1}$

Displacement is given by

$$x = a \sin(\omega t + \pi/4)$$

at $t = 0$, $x = 5$

$$5 = a \sin \pi/4$$

$$5 = a \sin 45^\circ \Rightarrow a = 5\sqrt{2}$$

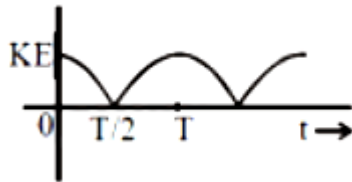
$$\text{Maximum acceleration } A_{\max} = a\omega^2 = 500\sqrt{2}\text{m/s}^2$$

Question137

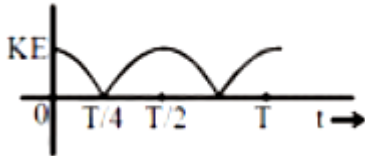
A particle is executing simple harmonic motion with a time period T . At time $t = 0$, it is at its position of equilibrium. The kinetic energy-time graph of the particle will look like:
[2017]

Options:

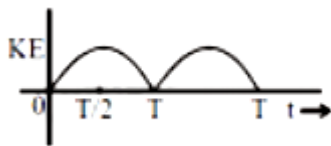
A.



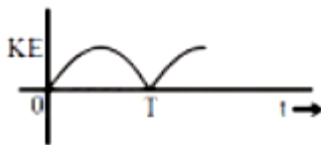
B.



C.



D.



Answer: B

Solution:

For a particle executing SHM

At mean position; $t = 0$, $\omega t = 0$, $y = 0$, $V = V_{\max} = a\omega$

$$\therefore K.E. = K E_{\max} = \frac{1}{2} m \omega^2 a^2$$

At extreme position: $t = \frac{T}{4}$, $\omega t = \frac{\pi}{2}$, $y = A$, $V = V_{\min} = 0$

$$\therefore K.E. = K E_{\min} = 0$$

$$\text{Kinetic energy in SHM, } K.E. = \frac{1}{2} m \omega^2 (a^2 - y^2)$$

$$= \frac{1}{2} m \omega^2 a^2 \cos^2 \omega t$$

Hence graph (b) correctly depicts kinetic energy time graph.

Question138

A block of mass 0.1kg is connected to an elastic spring of spring constant 640N m^{-1} and oscillates in a medium of constant 10^{-2}kgs^{-1} . The system dissipates its energy gradually. The time taken for its mechanical energy of vibration to drop to half of its initial value, is closest to :

[Online April 9,2017]

Options:

A. 2s

B. 3.5s

C. 5s

D. 7s

Answer: B

Solution:

Since system dissipates its energy gradually, and hence amplitude will also decreases with time according to

$$a = a_0 e^{-bt/m} \dots\dots(i)$$

$\therefore$ Energy of vibration drop to half of its initial value

(E_0), as $E \propto a^2 \Rightarrow a \propto \sqrt{E}$

$$a = \frac{a_0}{\sqrt{2}} \Rightarrow \frac{bt}{m} = \frac{10^{-2}t}{0.1} = \frac{t}{10}$$

$$\text{From eq}^n (i), \frac{a_0}{\sqrt{2}} = a_0 e^{-t/10}$$

$$\frac{1}{\sqrt{2}} = e^{-t/10} \text{ or } \sqrt{2} = e^{\frac{t}{10}}$$

$$\ln \sqrt{2} = \frac{t}{10} \therefore t = 3.5 \text{ seconds}$$

Question139

In an experiment to determine the period of a simple pendulum of length 1m, it is attached to different spherical bobs of radii r_1 and r_2 .

The two spherical bobs have uniform mass distribution. If the relative difference in the periods, is found to be 5×10^{-4} s, the difference in radii, $|r_1 - r_2|$ is best given by:

[Online April 9,2017]

Options:

- A. 1cm
- B. 0.1cm
- C. 0.5cm
- D. 0.01cm

Answer: B

Solution:

As we know, Time-period of simple pendulum,
 $T \propto \sqrt{l}$

differentiating both side, $\frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l}$

$\therefore$ change in length $\Delta l = r_1 - r_2$

$$5 \times 10^{-4} = \frac{1}{2} \frac{r_1 - r_2}{l}$$

$$r_1 - r_2 = 10 \times 10^{-4}$$

$$10^{-3} \text{m} = 10^{-1} \text{cm} = 0.1 \text{cm}$$

Question140

A 1kg block attached to a spring vibrates with a frequency of 1H z on a frictionless horizontal table. Two springs identical to the original spring are attached in parallel to an 8kg block placed on the same table. So, the frequency of vibration of the 8kg block is :

[Online April 8, 2017]

Options:

- A. $\frac{1}{4}$ H z

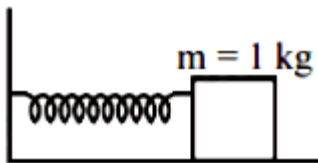
B. $\frac{1}{2\sqrt{2}} \text{ Hz}$

C. $\frac{1}{2} \text{ Hz}$

D. 2 Hz

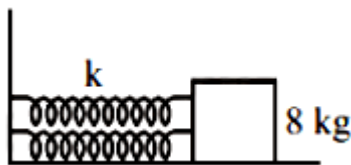
Answer: C

Solution:



$$\text{Frequency of spring (f)} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = 1 \text{ Hz}$$

$$\Rightarrow 4\pi^2 = \frac{k}{m}$$



If block of mass $m = 1 \text{ kg}$ is attached then, $k = 4\pi^2$

Now, identical springs are attached in parallel with mass $m = 8 \text{ kg}$. Hence,

$$k_{\text{eq}} = 2k$$

$$F = \frac{1}{2\pi} \sqrt{\frac{k \times 2}{m}} = \frac{1}{2} \text{ Hz}$$

Question141

A particle performs simple harmonic motion with amplitude A . Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is: [2016]

Options:

A. $A\sqrt{3}$

B. $\frac{7A}{3}$

C. $\frac{A}{3}\sqrt{41}$

D. $3A$

Answer: B

Solution:

We know that $V = \omega \sqrt{A^2 - x^2}$

Initially $V = \omega \sqrt{A^2 - \left(\frac{2A}{3}\right)^2}$

Finally $3V = \omega \sqrt{A'^2 - \left(\frac{2A}{3}\right)^2}$

Where A' = final amplitude (Given at $x = \frac{2A}{3}$, velocity to trebled)

On dividing we get

$$\frac{3}{1} = \frac{\sqrt{A'^2 - \left(\frac{2A}{3}\right)^2}}{\sqrt{A^2 - \left(\frac{2A}{3}\right)^2}}$$

$$9 \left[A^2 - \frac{4A^2}{9} \right] = A'^2 - \frac{4A^2}{9} \therefore A' = \frac{7A}{3}$$

Question142

Two particles are performing simple harmonic motion in a straight line about the same equilibrium point. The amplitude and time period for both particles are same and equal to A and T , respectively. At time $t = 0$ one particle has displacement A while the other one has displacement $-\frac{A}{2}$ and they are moving towards each other. If they cross each other at time t , then t is:

[Online April 9, 2016]

Options:

A. $\frac{5T}{6}$

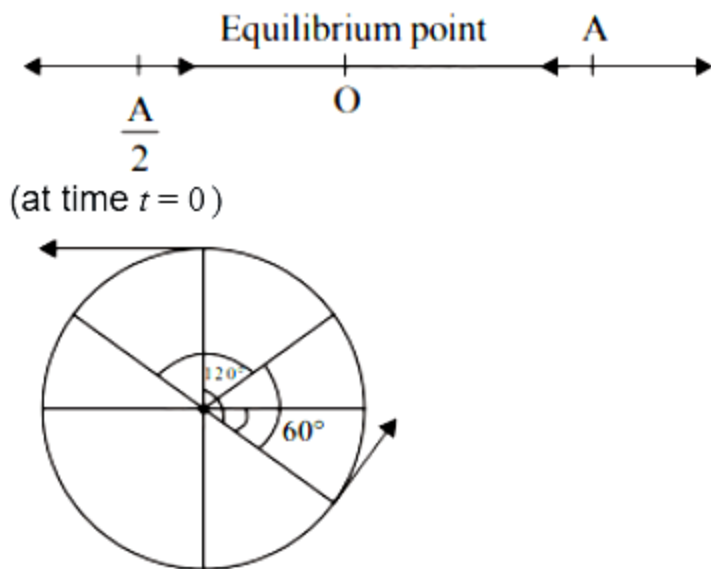
B. $\frac{T}{3}$

C. $\frac{T}{4}$

D. $\frac{T}{6}$

Answer: D

Solution:



Angle covered to meet $\theta = 60^\circ = \frac{\pi}{3}$ rad

If they cross each other at time t then

$$t = \frac{\theta}{2\pi} = \frac{\pi}{3 \times 2\pi} T = \frac{T}{6}$$

Question143

A pendulum clock loses 12s a day if the temperature is 40°C and gains 4s a day if the temperature is 20°C . The temperature at which the clock will show correct time, and the co-efficient of linear expansion (α) of the metal of the pendulum shaft are respectively: [2016]

Options:

A. 30°C ; $\alpha = 1.85 \times 10^{-3}/^\circ\text{C}$

B. 55°C ; $\alpha = 1.85 \times 10^{-2}/^\circ\text{C}$

C. 25°C ; $\alpha = 1.85 \times 10^{-5} / ^{\circ}\text{C}$

D. 60°C ; $\alpha = 1.85 \times 10^{-4} / ^{\circ}\text{C}$

Answer: C

Solution:

$$\text{Time lost/gained per day} = \frac{1}{2} \alpha \Delta\theta \times 86400 \text{ second}$$

$$12 = \frac{1}{2} \alpha (40 - \theta) \times 86400 \dots\dots(i)$$

$$4 = \frac{1}{2} \alpha (\theta - 20) \times 86400 \dots\dots(ii)$$

$$\text{On dividing we get, } 3 = \frac{40 - \theta}{\theta - 20}$$

$$3\theta - 60 = 40 - \theta$$

$$4\theta = 100 \Rightarrow \theta = 25^{\circ}\text{C}$$

Question144

In an engine the piston undergoes vertical simple harmonic motion with amplitude 7cm. A washer rests on top of the piston and moves with it. The motor speed is slowly increased. The frequency of the piston at which the washer no longer stays in contact with the piston, is close to :

[Online April 10,2016]

Options:

A. 0.7H z

B. 1.9H z

C. 1.2H z

D. 0.1H z

Answer: B

Solution:

Washer contact with piston $\Rightarrow N = 0$ Given Amplitude $A = 7\text{cm} = 0.07\text{m}$

$$a_{\max} = g = \omega^2 A$$

The frequency of piston

$$f = \frac{\omega}{2\pi} = \sqrt{\frac{g}{A}} \frac{1}{2\pi} = \sqrt{\frac{1000}{7}} \frac{1}{2\pi} = 1.9\text{Hz}$$

Question 145

A simple harmonic oscillator of angular frequency 2rad s^{-1} is acted upon by an external force $F = \sin t \text{ N}$. If the oscillator is at rest in its equilibrium position at $t = 0$, its position at later times is proportional to:

[Online April 10, 2015]

Options:

A. $\sin t + \frac{1}{2} \cos 2t$

B. $\cos t - \frac{1}{2} \sin 2t$

C. $\sin t - \frac{1}{2} \sin 2t$

D. $\sin t + \frac{1}{2} \sin 2t$

Answer: C

Solution:

As we know,

$$F = ma \Rightarrow a \propto F$$

or, $a \propto \sin t$

$$\Rightarrow \frac{dv}{dt} \propto \sin t$$

$$\Rightarrow \int_0^t dv \propto \int_0^t \sin t dt$$

$$v \propto -\cos t + 1$$

$$\int_0^x dx = \int_0^t (-\cos t + 1) dt$$

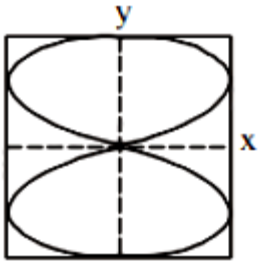
$$x = \sin t - \frac{1}{2} \sin 2t$$

Question146

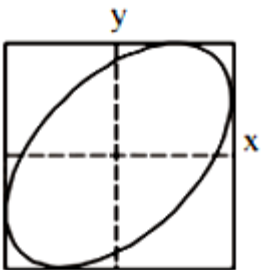
x and y displacements of a particle are given as $x(t) = a \sin \omega t$ and $y(t) = a \sin 2 \omega t$. Its trajectory will look like:
[Online April 10, 2015]

Options:

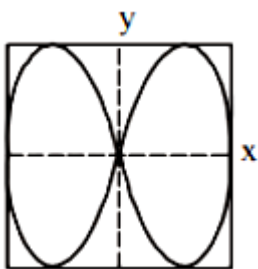
A.



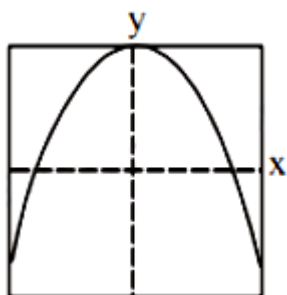
B.



C.



D.



Answer: C

Solution:

At $t = 0$, $x(t) = 0$; $y(t) = 0$

$x(t)$ is a sinusoidal function At

$t = \frac{\pi}{2\omega}$; $x(t) = a$ and $y(t) = 0$

Hence trajectory of particle will look like as (c).

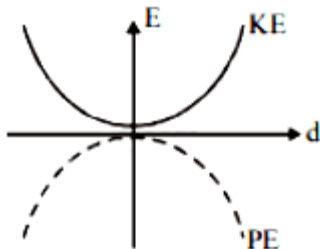
Question147

For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement d . Which one of the following represents these correctly? (graphs are schematic and not drawn to scale)

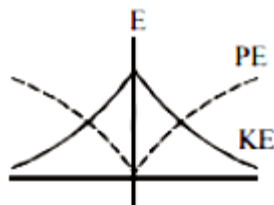
[2015]

Options:

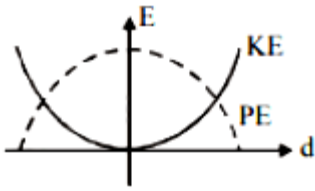
A.



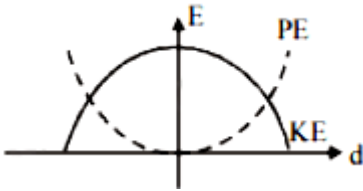
B.



C.



D.



Answer: D

Solution:

$$K.E = \frac{1}{2}k(A^2 - d^2)$$

$$\text{and } P.E = \frac{1}{2}kd^2$$

At mean position $d = 0$. At extreme positions $d = A$

Question148

**A pendulum with time period of 1s is losing energy. At certain time its energy is 45J . If after completing 15 oscillations, its energy has become 15J , its damping constant (in s^{-1}) is :
[Online April 11, 2015]**

Options:

A. $\frac{1}{2}$

B. $\frac{1}{30} \ln 3$

C. 2

D. $\frac{1}{15} \ln 3$

Answer: D

Solution:

As we know, $E = E_0 e^{-\frac{bt}{m}}$

$$15 = 45 e^{-\frac{b15}{m}}$$

[As no. of oscillations = 15 so $t = 15\text{sec}$]

$$\frac{1}{3} = e^{-\frac{b15}{m}}$$

Taking log on both sides

$$\frac{b}{m} = \frac{1}{15} \ln 3$$

Question149

A pendulum made of a uniform wire of cross sectional area A has time period T . When an additional mass M is added to its bob, the time period changes to T_M . If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to:

(g = gravitational acceleration)

[2015]

Options:

A. $\left[1 - \left(\frac{T_M}{T} \right)^2 \right] \frac{A}{Mg}$

B. $\left[1 - \left(\frac{T}{T_M} \right)^2 \right] \frac{A}{Mg}$

C. $\left[\left(\frac{T_M}{T} \right)^2 - 1 \right] \frac{A}{Mg}$

D. $\left[\left(\frac{T_M}{T} \right)^2 - 1 \right] \frac{Mg}{A}$

Answer: C

Solution:

As we know, time period, $T = 2\pi \sqrt{\frac{l}{g}}$

When additional mass M is added then

$$T_M = 2\pi \sqrt{\frac{l + \Delta l}{g}}$$

$$\frac{T_M}{T} = \sqrt{\frac{l + \Delta l}{l}}$$

$$\Rightarrow \left(\frac{T_M}{T}\right)^2 = \frac{l + \Delta l}{l}$$

$$\text{or, } \left(\frac{T_M}{T}\right)^2 = 1 + \frac{Mg}{AY} \left[\because \Delta l = \frac{Mgl}{AY} \right]$$

$$\therefore \frac{1}{Y} = \left[\left(\frac{T_M}{T}\right)^2 - 1 \right] \frac{A}{Mg}$$

Question150

A body is in simple harmonic motion with time period half second ($T = 0.5s$) and amplitude one cm($A = 1cm$). Find the average velocity in the interval in which it moves from equilibrium position to half of its amplitude.

[Online April 19,2014]

Options:

- A. 4cm/s
- B. 6cm/s
- C. 12cm/s
- D. 16cm/s

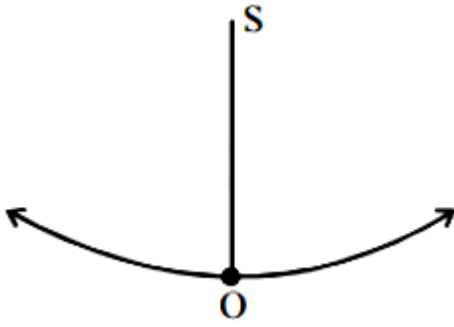
Answer: C

Solution:

Given: Time period, $T = 0.5 \text{ sec}$

Amplitude, $A = 1cm$

Average velocity in the interval in which body moves from equilibrium to half of its amplitude, $v = ?$



Time taken to a displacement $A/2$ where A is the amplitude of oscillation from the mean position 'O' is $\frac{T}{12}$

Therefore, time, $t = \frac{0.5}{12}$ sec

Displacement, $s = \frac{A}{2} = \frac{1}{2}$ cm

$\therefore$ Average velocity, $v = \frac{\frac{A}{2}}{t} = \frac{\frac{1}{2}}{\frac{0.5}{12}} = 12 \text{ cm/s}$

Question151

Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants?

[Online April 12,2014]

Options:

A. $a + bx - cx^2$

B. bx^2

C. $a - bx + cx^2$

D. $-bx$

Answer: D

Solution:

In linear S.H.M., the restoring force acting on particle should always be proportional to the displacement of the particle and directed towards the equilibrium position.

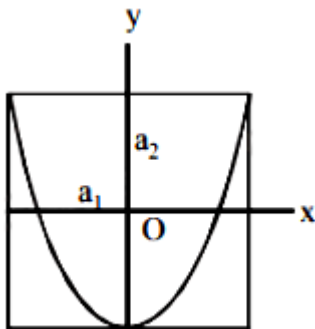
i.e., $F \propto x$
or $F = -bx$ where b is a positive constant.

Question152

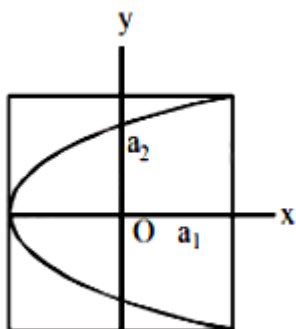
A particle which is simultaneously subjected to two perpendicular simple harmonic motions represented by; $x = a_1 \cos \omega t$ and $y = a_2 \cos 2 \omega t$ traces a curve given by:
[Online April 9, 2014]

Options:

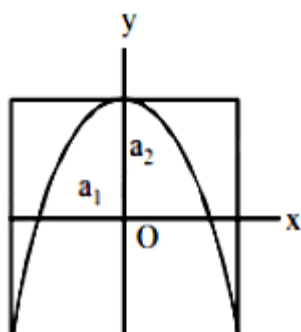
A.



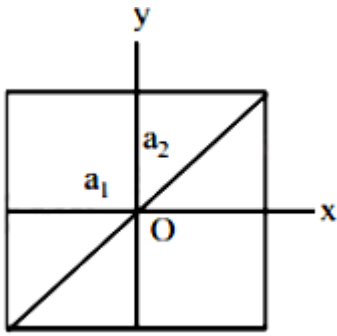
B.



C.



D.



Answer: B

Solution:

Two perpendicular S.H.Ms are

$$x = a_1 \cos \omega t \dots(1)$$

$$\text{and } y = a_2 2 \cos \omega t \dots(2)$$

From eqn (1)

$$\frac{x}{a_1} = \cos \omega t$$

and from eqn (2)

$$\frac{y}{a_2} = 2 \cos \omega t$$

$$\therefore y = 2 \frac{a_2}{a_1} x$$

Question153

**A particle moves with simple harmonic motion in a straight line. In first τ s, after starting from rest it travels a distance a , and in next τ s it travels $2a$, in same direction, then:
[2014]**

Options:

A. amplitude of motion is $3a$

B. time period of oscillations is 8τ

C. amplitude of motion is $4a$

D. time period of oscillations is 6τ

Answer: D

Solution:

In simple harmonic motion, starting from rest,

At $t = 0$, $x = A$

$x = A \cos \omega t$ (i)

When $t = \tau$, $x = A - a$

When $t = 2\tau$, $x = A - 3a$

From equation (i)

$A - a = A \cos \omega \tau$ (ii)

$A - 3a = A \cos 2\omega \tau$ (iii)

As $\cos 2\omega \tau = 2\cos^2 \omega \tau - 1$ (iv)

From equation (ii), (iii) and (iv)

$$\frac{A - 3a}{A} = 2 \left(\frac{A - a}{A} \right)^2 - 1$$

$$\Rightarrow \frac{A - 3a}{A} = \frac{2A^2 + 2a^2 - 4Aa - A^2}{A^2}$$

$$\Rightarrow A^2 - 3aA = A^2 + 2a^2 - 4Aa$$

$$\Rightarrow 2a^2 = aA$$

$$\Rightarrow A = 2a$$

$$\Rightarrow \frac{a}{A} = \frac{1}{2}$$

Now, $A - a = A \cos \omega \tau$

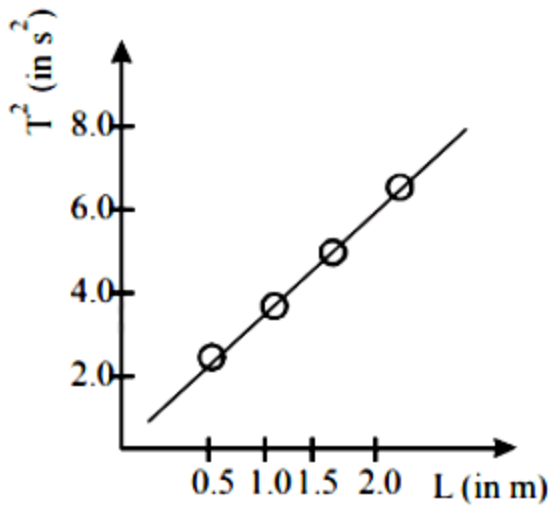
$$\Rightarrow \cos \omega \tau = \frac{A - a}{A}$$

$$\Rightarrow \cos \omega \tau = \frac{1}{2} \text{ or } \frac{2\pi}{T} \tau = \frac{\pi}{3}$$

$$\Rightarrow T = 6\tau$$

Question154

In an experiment for determining the gravitational acceleration g of a place with the help of a simple pendulum, the measured time period square is plotted against the string length of the pendulum in the figure.



What is the value of g at the place?
[Online April 19, 2014]

Options:

- A. 9.81m/s^2
- B. 9.87m/s^2
- C. 9.91m/s^2
- D. 10.0m/s^2

Answer: B

Solution:

From graph it is clear that when

$$L = 1\text{m}, T^2 = 4\text{s}^2$$

As we know,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\Rightarrow g = \frac{4\pi^2 L}{T^2}$$

$$= 4 \times \left(\frac{22}{7}\right)^2 \times \frac{1}{4} = \left(\frac{22}{7}\right)^2$$

$$\therefore g = \frac{484}{49} = 9.87\text{m/s}^2$$

Question155

The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10cm to 8cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3 . The time in which amplitude of this pendulum will reduce from 10cm to 5cm in carbon dioxide will be close to ($\ln 5 = 1.601$, $\ln 2 = 0.693$).

[Online April 9, 2014]

Options:

A. 231s

B. 208s

C. 161s

D. 142s

Answer: D

Solution:

As we know,

$$x = x_0 e^{-bt/2m}$$

From question,

$$8 = 10 e^{-\frac{40b}{2m}} \dots\dots(i)$$

$$\text{Similarly, } 5 = 10 e^{-\frac{bt}{2m}} \dots\dots(ii)$$

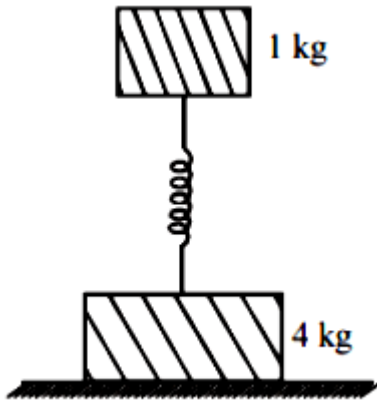
Solving eqns (i) and (ii) we get

$$t \cong 142s$$

Question 156

Two bodies of masses 1kg and 4kg are connected to a vertical spring, as shown in the figure. The smaller mass executes simple harmonic motion of angular frequency 25 rad/s , and amplitude 1.6cm while the bigger mass remains stationary on the ground. The maximum

force exerted by the system on the floor is (take $g = 10\text{ms}^{-2}$)



[Online April 9, 2014]

Options:

A. 20N

B. 10N

C. 60N

D. 40N

Answer: C

Solution:

Mass of bigger body $M = 4\text{kg}$

Mass of smaller body $m = 1\text{kg}$

Smaller mass ($m = 1\text{kg}$)

executes S.H.M of angular frequency $\omega = 25\text{ rad/s}$

Amplitude $x = 1.6\text{cm} = 1.6 \times 10^{-2}$

As we know,

$$T = 2\pi \sqrt{\frac{m}{K}}$$

$$\text{or, } 2\pi\omega = 2\pi \sqrt{\frac{m}{K}}$$

$$\text{or, } \frac{1}{25} = \sqrt{\frac{1}{K}} \quad [\because m = 1\text{kg}; \omega = 25\text{rad/s}]$$

$$\text{or, } K = 625\text{N m}^{-1}.$$

The maximum force exerted by the system on the floor $= Mg + Kx + mg$

$$= 4 \times 10 + 625 \times 1.6 \times 10^{-2} + 1 \times 10$$

$$= 40 + 10 + 10$$

$$= 60\text{N}$$

Question157

The angular frequency of the damped oscillator is given by,

$\omega = \sqrt{\left(\frac{k}{m} - \frac{r^2}{4m^2}\right)}$ where k is the spring constant, m is the mass of the oscillator and r is the damping constant.

If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period compared to the undamped oscillator is approximately as follows:

[Online April 11, 2014]

Options:

- A. increases by 1%
- B. increases by 8%
- C. decreases by 1%
- D. decreases by 8%

Answer: B

Solution:

The change in time period compared to the undamped oscillator increases by 8%.

Question158

The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10 s it will decrease to α times its original magnitude, where α equals
[2013]

Options:

- A. 0.7
- B. 0.81

C. 0.729

D. 0.6

Answer: C

Solution:

$$\therefore A = A_0 e^{-\frac{bt}{2m}}$$

(where, A_0 = maximum amplitude)

According to the questions, after 5 second,

$$0.9 \sim A_0 = A_0 e^{-\frac{b(5)}{2m}} \dots\dots(i)$$

After 10 more second,

$$A = A_0 e^{-\frac{b(15)}{2m}} \dots\dots(ii)$$

From eqⁿs (i) and (ii)

$$A = 0.729 A_0$$

$$\therefore \alpha = 0.729$$

Question159

An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M . The piston and the cylinder have equal cross sectional area A . When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0 . The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency
[2013]

Options:

A. $\frac{1}{2\pi} \frac{A\gamma P_0}{V_0 M}$

B. $\frac{1}{2\pi} \frac{V_0 M P_0}{A^2 \gamma}$

$$C. \frac{1}{2\pi} \sqrt{\frac{A^2 \gamma P_0}{M V_0}}$$

$$D. \frac{1}{2\pi} \sqrt{\frac{M V_0}{A \gamma P_0}}$$

Answer: C

Solution:

$$\frac{Mg}{A} = P_0$$

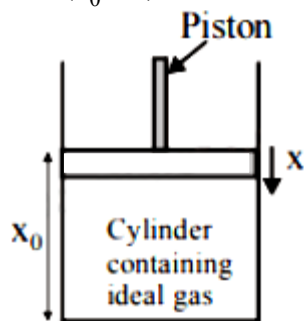
$$P_0 V_0^\gamma = P V^\gamma$$

$$Mg = P_0 A \dots (i)$$

Let piston is displaced by distance x

$$P_0 A x_0^\gamma = P A (x_0 - x)^\gamma$$

$$P = \frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma}$$



$$Mg - \left(\frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma} \right) A = F_{\text{restoring}}$$

$$P_0 A \left(1 - \frac{x_0^\gamma}{(x_0 - x)^\gamma} \right) = F_{\text{restoring}} [x_0 - x \approx x_0]$$

$$F = -\frac{\gamma P_0 A x}{x_0}$$

$\therefore$ Frequency with which piston executes SHM.

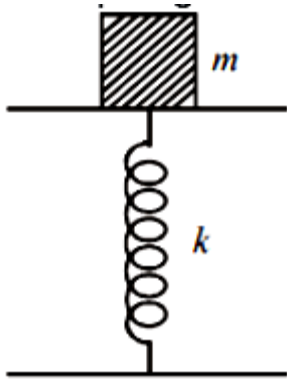
$$f = \frac{1}{2\pi} \sqrt{\frac{\gamma P_0 A}{x_0 M}} = \frac{1}{2\pi} \sqrt{\frac{\gamma P_0 A^2}{M V_0}}$$

Question160

A mass $m = 1.0\text{kg}$ is put on a flat pan attached to a vertical spring fixed on the ground. The mass of the spring and the pan is negligible. When pressed slightly and released, the mass executes simple harmonic motion. The spring constant is 500N/m . What is the amplitude A of the motion, so that the mass m tends to get detached from the pan?

(Take $g = 10\text{m/s}^2$).

The spring is stiff enough so that it does not get distorted during the motion.



[Online April 22, 2013]

Options:

- A. $A > 2.0\text{cm}$
- B. $A = 2.0\text{cm}$
- C. $A < 2.0\text{cm}$
- D. $A = 1.5\text{cm}$

Answer: C

Solution:

As $F = -kx$

Question161

Two simple pendulums of length 1m and 4m respectively are both given small displacement in the same direction at the same instant.

They will be again in phase after the shorter pendulum has completed number of oscillations equal to :
[Online April 9,2013]

Options:

A. 2

B. 7

C. 5

D. 3

Answer: A

Solution:

Let T_1 and T_2 be the time period of the two pendulums $T_1 = 2\pi \sqrt{\frac{l_1}{g}}$ and $T_2 = 2\pi \sqrt{\frac{l_2}{g}}$

As $l_1 < l_2$ therefore $T_1 < T_2$

Let at $t = 0$ they start swinging together. Since their time periods are different, the swinging will not be in unison always. Only when number of completed oscillations differ by an integer, the two pendulums will again begin to swing together

Let longer length pendulum complete n oscillation and shorter length pendulum complete $(n + 1)$ oscillation. For unison swinging

$$(n + 1)T_1 = nT_2$$

$$(n + 1) \times 2\pi \sqrt{\frac{l_1}{g}} = (n) \times 2\pi \sqrt{\frac{l_2}{g}}$$

$$\Rightarrow n = 1$$

$$\therefore n + 1 = 1 + 1 = 2$$

Question162

A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. When the cylinder is given a downward push and released, it starts oscillating vertically with a small amplitude. The time period T of the oscillations of the cylinder will be :

[Online April 25, 2013]

Options:

A. Smaller than $2\pi \left[\frac{M}{(k + A\sigma g)} \right]^{1/2}$

B. $2\pi \sqrt{\frac{M}{k}}$

C. Larger than $2\pi \left[\frac{M}{(k + A\sigma g)} \right]^{1/2}$

D. $2\pi \left[\frac{M}{(k + A\sigma g)} \right]^{1/2}$

Answer: A

Question163

Bob of a simple pendulum of length l is made of iron. The pendulum is oscillating over a horizontal coil carrying direct current. If the time period of the pendulum is T then :
[Online April 23, 2013]

Options:

A. $T < 2\pi \sqrt{\frac{l}{g}}$ and damping is smaller than in air alone.

B. $T < 2\pi \sqrt{\frac{l}{g}}$ and damping is larger than in air alone.

C. $T < 2\pi \sqrt{\frac{l}{g}}$ and damping is smaller than in air alone.

D. $T < 2\pi \sqrt{\frac{l}{g}}$ and damping is larger than in air alone.

Answer: D

Solution:

When the pendulum is oscillating over a current carrying coil, and when the direction of oscillating pendulum bob is opposite to the direction of current. Its instantaneous acceleration increases.

Hence time period $T < 2\pi \sqrt{\frac{l}{g}}$

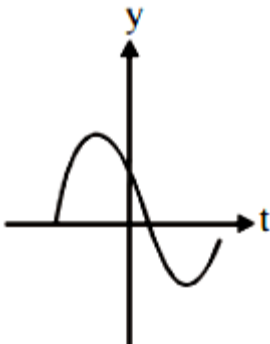
and damping is larger than in air alone due energy dissipation.

Question164

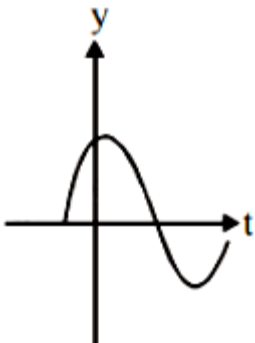
The displacement $y(t) = A \sin(\omega t + \phi)$ of a pendulum for $\phi = \frac{2\pi}{3}$ is correctly represented by
[Online May 19, 2012]

Options:

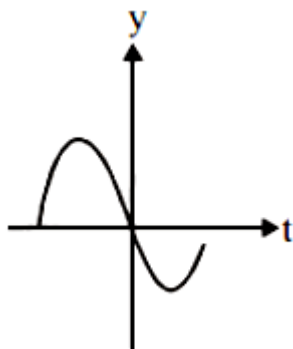
A.



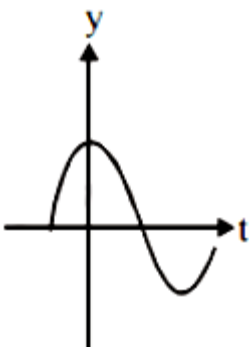
B.



C.



D.



Answer: A

Solution:

Displacement $y(t) = A \sin(\omega t + \phi)$ [Given]

For $\phi = \frac{2\pi}{3}$

at $t = 0$; $y = A \sin \phi = A \sin \frac{2\pi}{3}$

$= A \sin 120^\circ = 0.87A$ [$\because \sin 120^\circ \simeq 0.866$]

Graph (a) depicts $y = 0.87A$ at $t = 0$

Question165

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

If two springs S_1 and S_2 of force constants k_1 and k_2 respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2

Statement 1: If stretched by the same amount work done on S_1

Statement 2: $k_1 < k_2$

[2012]

Options:

A. Statement 1 is false, Statement 2 is true.

B. Statement 1 is true, Statement 2 is false.

C. Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation for Statement 1

D. Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation for Statement 1

Answer: B

Solution:

Work done, $w = \frac{1}{2}kx^2$

Work done by spring S_1 , $w_1 = \frac{1}{2}k_1x^2$

Work done by spring S_2 , $w_2 = \frac{1}{2}k_2x^2$

Since $w_1 > w_2$ Thus ($k_1 > k_2$)

Question166

If a simple pendulum has significant amplitude (up to a factor of $1/e$ of original) only in the period between $t = 0$ s to $t = \tau$ s, then τ may be called the average life of the pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity with b as the constant of proportionality, the average life time of the pendulum in second is (assuming damping is small)

[2012]

Options:

A. $\frac{0.693}{b}$

B. b

C. $\frac{1}{b}$

D. $\frac{2}{b}$

Answer: D

Solution:

The equation of motion for the pendulum, for damped harmonic motion

$$F = -kx - bv$$

$$\Rightarrow ma + kx + bv = 0$$

$$\Rightarrow m \frac{d^2 x}{dt^2} + kx + b \frac{dx}{dt} = 0$$

$$\Rightarrow \frac{d^2 x}{dt^2} + \frac{k}{m}x + \frac{b}{m} \frac{dx}{dt} = 0$$

$$\Rightarrow \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m}x = 0 \dots\dots(i)$$

Let $x = e^{\lambda t}$ is the solution of the equation (1)

$$\frac{dx}{dt} = \lambda e^{\lambda t} \Rightarrow \frac{d^2 x}{dt^2} = \lambda^2 e^{\lambda t}$$

Substituting in the equation (1)

$$\lambda^2 e^{\lambda t} + \frac{b}{m} \lambda e^{\lambda t} + \frac{k}{m} e^{\lambda t} = 0$$

$$\lambda^2 + \frac{b}{m} \lambda + \frac{k}{m} = 0$$

$$\lambda = \frac{-\frac{b}{m} \pm \sqrt{\frac{b^2}{m^2} - 4 \frac{k}{m}}}{2} = \frac{-b \pm \sqrt{b^2 - 4km}}{2m}$$

Solving the equation (1) for x , we have

$$x = e^{\frac{-b}{2m}t}$$

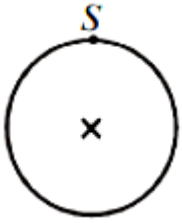
$$\omega = \sqrt{\omega_0^2 - \lambda^2} \text{ where } \omega_0 = \frac{k}{m}, \lambda = \frac{+b}{2}$$

$$\text{The average life} = \frac{1}{\lambda} = \frac{2}{b}$$

Question167

A ring is suspended from a point S on its rim as shown in the figure. When displaced from equilibrium, it oscillates with time period of 1

second. The radius of the ring is (take $g = \pi^2$)



[Online May 19, 2012]

Options:

A. 0.15m

B. 1.5m

C. 1.0m

D. 0.5m

Answer: A

Question168

Two particles are executing simple harmonic motion of the same amplitude A and frequency ω along the x -axis. Their mean position is separated by distance X_0 ($X_0 > A$). If the maximum separation between them is $(X_0 + A)$, the phase difference between their motion is:

[2011]

Options:

A. $\frac{\pi}{3}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{6}$

D. $\frac{\pi}{2}$

Answer: A

Solution:

Let, $x_1 = A \sin \omega t$ and $x_2 = A \sin(\omega t + \phi)$

$$x_2 - x_1 = 2A \cos\left(\omega t + \frac{\phi}{2}\right) \sin \frac{\phi}{2}$$

The above equation is SHM with amplitude $2A \sin \frac{\phi}{2}$

$$\therefore 2A \sin \frac{\phi}{2} = A$$

$$\Rightarrow \sin \frac{\phi}{2} = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{3}$$

Question169

A mass M , attached to a horizontal spring, executes S.H.M. with amplitude A_1 . When the mass M passes through its mean position then a smaller mass m is placed over it and both of them move together with amplitude A_2 . The ratio of $\left(\frac{A_1}{A_2}\right)$ is:

[2011]

Options:

A. $\frac{M+m}{M}$

B. $\left(\frac{M}{M+m}\right)^{\frac{1}{2}}$

C. $\left(\frac{M+m}{M}\right)^{\frac{1}{2}}$

D. $\frac{M}{M+m}$

Answer: C

Solution:

At mean position, $F_{\text{net}} = 0$

Therefore, by principle of conservation of linear momentum.

$$\therefore M v_1 = (M + m)v_2$$

$$M w_1 a_1 = (M + m)w_2 a_2$$

$$M A_1 \sqrt{\frac{k}{M}} = (M + m)A_2 \sqrt{\frac{k}{m + M}}$$

$$\therefore \left(V = A \sqrt{\frac{k}{M}} \right)$$

$$\Rightarrow A_1 \sqrt{M} = A_2 \sqrt{M + m}$$

$$\Rightarrow \frac{A_1}{A_2} = \sqrt{\frac{m + M}{M}}$$

Question 170

A wooden cube (density of wood ' d ') of side ' l ' floats in a liquid of density ' ρ ' with its upper and lower surfaces horizontal. If the cube is pushed slightly down and released, it performs simple harmonic motion of period ' T '

[2011 RS]

Options:

A. $2\pi \sqrt{\frac{ld}{\rho g}}$

B. $2\pi \sqrt{\frac{l\rho}{dg}}$

C. $2\pi \sqrt{\frac{ld}{(\rho - d)g}}$

D. $2\pi \sqrt{\frac{l\rho}{(\rho - d)g}}$

Answer: A

Solution:

Let the cube be at a depth x from the equilibrium position.

Force acting on the cube = up thrust on the portion of length x .

$$F = -\rho l^2 x g [\because \text{mass density} \times \text{volume}] \dots\dots (i)$$

Clearly $F \propto -x$, Hence it is a SHM.

$$\text{Equation of SHM is } F = -kx \dots\dots(ii)$$

Comparing equation (i) and (ii) we have

$$k = \rho l^2 g$$

$$\text{Now, Time period, } T = 2\pi \sqrt{\frac{m}{k}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{l^3 d}{\rho l^2 g}}$$

$$= 2\pi \sqrt{\frac{ld}{\rho g}}$$

Comparing the above equation with

$$a = -\omega^2 x, \text{ we get}$$

$$\therefore \omega = \sqrt{\frac{\rho g}{d l}} \Rightarrow T = 2\pi \sqrt{\frac{ld}{\rho g}}$$

Question171

If x , v and a denote the displacement, the velocity and the acceleration of a particle executing simple harmonic motion of time period T , then, which of the following does not change with time?
[2009]

Options:

A. aT / x

B. $aT + 2\pi v$

C. aT / v

D. $a^2 T^2 + 4\pi^2 v^2$

Answer: A

Solution:

For an SHM, the acceleration

$$a = -\omega^2 x \text{ where } \omega^2 \text{ is a constant.}$$

$$a = \frac{-4\pi^2 x}{T^2} \Rightarrow \frac{aT}{x} = \frac{-4\pi^2}{T}$$

The time period T is also constant. Therefore, $\frac{aT}{x}$ is a constant.

Question172

**A point mass oscillates along the x -axis according to the law $x = x_0 \cos(\omega t - \pi/4)$. If the acceleration of the particle is written as $a = A \cos(\omega t + \delta)$, then
[2007]**

Options:

A. $A = x_0 \omega^2, \delta = 3\pi/4$

B. $A = x_0, \delta = -\pi/4$

C. $A = x_0 \omega^2, \delta = \pi/4$

D. $A = x_0 \omega^2, \delta = -\pi/4$

Answer: A

Solution:

Given,

Displacement, $x = x_0 \cos\left(\omega t - \frac{\pi}{4}\right)$

$\therefore$ Velocity, $v = \frac{dx}{dt} = -x_0 \omega \sin\left(\omega t - \frac{\pi}{4}\right)$

Acceleration,

$$a = \frac{dv}{dt} = -x_0 \omega^2 \cos\left(\omega t - \frac{\pi}{4}\right)$$

$$= x_0 \omega^2 \cos\left[\pi + \left(\omega t - \frac{\pi}{4}\right)\right]$$

$$= x_0 \omega^2 \cos\left(\omega t + \frac{3\pi}{4}\right) \dots\dots(1)$$

Acceleration, $a = A \cos(\omega t + \delta) \dots\dots(2)$

Comparing the two equations, we get

$$A = x_0 \omega^2 \text{ and } \delta = \frac{3\pi}{4}$$

Question173

A particle of mass m executes simple harmonic motion with amplitude a and frequency ν . The average kinetic energy during its motion from the position of equilibrium to the end is [2007]

Options:

A. $2\pi^2 ma^2 \nu^2$

B. $\pi^2 ma^2 \nu^2$

C. $\frac{1}{4} ma^2 \nu^2$

D. $4\pi^2 ma^2 \nu^2$

Answer: B

Solution:

The kinetic energy of a particle executing S.H.M. at any instant t is given by

$$K = \frac{1}{2} m a^2 \omega^2 \sin^2 \omega t$$

where, m = mass of particle

a = amplitude

ω = angular frequency

t = time

The average value of $\sin^2 \omega t$ over a cycle is $\frac{1}{2}$.

$$\therefore K E = \frac{1}{2} m \omega^2 a^2 \left(\frac{1}{2} \right) \left(\because \langle \sin^2 \theta \rangle = \frac{1}{2} \right)$$

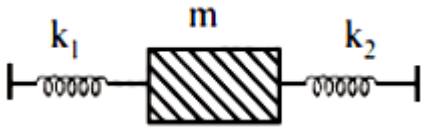
$$= \frac{1}{4} m \omega^2 a^2 = \frac{1}{4} m a^2 (2\pi\nu)^2 (\because \omega = 2\pi\nu)$$

$$\text{Or } \langle K \rangle = \pi^2 m a^2 \nu^2$$

Question174

Two springs, of force constants k_1 and k_2 are connected to a mass m as shown. The frequency of oscillation of the mass is f . If both k_1

and k_2 are made four times their original values, the frequency of oscillation becomes



[2007]

Options:

- A. $2f$
- B. $f/2$
- C. $f/4$
- D. $4f$

Answer: A

Solution:

(a) The two springs are in parallel.

$\therefore$ Effective spring constant,

$$k = k_1 + k_2$$

Initial frequency of oscillation is given by

$$v = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}} \dots\dots(i)$$

When both k_1 and k_2 are made four times their original values, the new frequency is given by

$$\begin{aligned} v' &= \frac{1}{2\pi} \sqrt{\frac{4k_1 + 4k_2}{m}} \\ &= \frac{1}{2\pi} \sqrt{\frac{4(k_1 + k_2)}{m}} = 2 \left(\frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}} \right) = 2v \end{aligned}$$

Question175

The displacement of an object attached to a spring and executing simple harmonic motion is given by $x = 2 \times 10^{-2} \cos \pi t$ metre. The time at which the maximum speed first occurs is

[2007]

Options:

A. 0.25s

B. 0.5s

C. 0.75s

D. 0.125s

Answer: B

Solution:

Here, Displacement, $x = 2 \times 10^{-2} \cos \pi t$

Velocity is given by

$$v = \frac{dx}{dt} = 2 \times 10^{-2} \pi \sin \pi t$$

For the first time, the when velocity becomes maximum,

$$\sin \pi t = 1$$

$$\Rightarrow \sin \pi t = \sin \frac{\pi}{2}$$

$$\Rightarrow \pi t = \frac{\pi}{2} \text{ or, } t = \frac{1}{2} = 0.5 \text{ sec}$$

Question176

A coin is placed on a horizontal platform which undergoes vertical simple harmonic motion of angular frequency ω . The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time
[2006]

Options:

A. at the mean position of the platform

B. for an amplitude of $\frac{g}{m^2}$

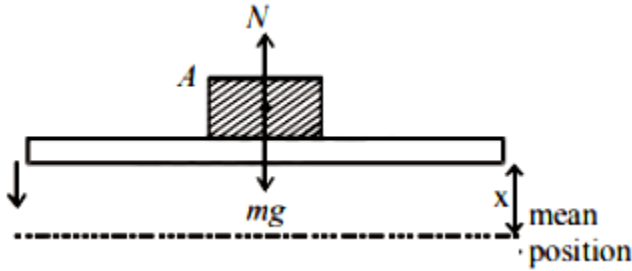
C. for an amplitude of $\frac{g^2}{m^2}$

D. at the highest position of the platform

Answer: B

Solution:

For block A to move in SHM .



$$mg - N = m\omega^2 x$$

where x is the distance from mean position For block to leave contact $N = 0$

$$\Rightarrow mg = m\omega^2 x \Rightarrow x = \frac{g}{\omega^2}$$

Question177

The maximum velocity of a particle, executing simple harmonic motion with an amplitude 7mm, is 4.4m/s. The period of oscillation is

[2006]

Options:

A. 0.01s

B. 10s

C. 0.1s

D. 100s

Answer: A

Solution:

Maximum velocity,

$$v_{\max} = a\omega$$

Here, a = amplitude of SHM

ω = angular velocity of SHM

$$v_{\max} = a \times \frac{2\pi}{T} \therefore \left(\because \omega = \frac{2\pi}{T} \right)$$

$$\Rightarrow T = \frac{2\pi a}{v_{\max}} = \frac{2 \times 3.14 \times 7 \times 10^{-3}}{4.4} \approx 0.01 \text{ s}$$

Question 178

Starting from the origin a body oscillates simple harmonically with a period of 2s. After what time will its kinetic energy be 75% of the total energy?

[2006]

Options:

A. $\frac{1}{6} \text{ s}$

B. $\frac{1}{4} \text{ s}$

C. $\frac{1}{3} \text{ s}$

D. $\frac{1}{12} \text{ s}$

Answer: A

Solution:

K.E. of a body undergoing SHM is given by,

$$K.E. = \frac{1}{2} m a^2 \omega^2 \cos^2 \omega t$$

Here, a = amplitude of SHM

ω = angular velocity of SHM

$$\text{Total energy in S.H.M} = \frac{1}{2} m a^2 \omega^2$$

Given $K.E. = 75\% T.E$

$$\frac{1}{2} m a^2 \omega^2 \cos^2 \omega t = \frac{75}{100} \times \frac{1}{2} m a^2 \omega^2$$

$$\Rightarrow 0.75 = \cos^2 \omega t \Rightarrow \omega t = \frac{\pi}{6}$$

$$\Rightarrow t = \frac{\pi}{6 \times \omega} \Rightarrow t = \frac{\pi \times 2}{6 \times 2\pi} \Rightarrow t = \frac{1}{6} \text{ s}$$

Question179

The function $\sin^2(\omega t)$ represents
[2005]

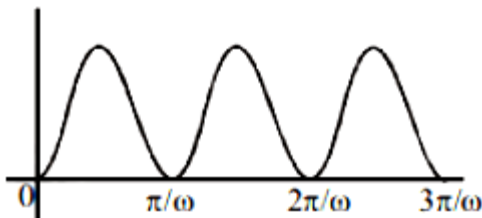
Options:

- A. a periodic, but not simple harmonic motion with a period $\frac{\pi}{\omega}$
- B. a periodic, but not simple harmonic motion with a period $\frac{2\pi}{\omega}$
- C. a simple harmonic motion with a period $\frac{\pi}{\omega}$
- D. a simple harmonic motion with a period $\frac{2\pi}{\omega}$

Answer: A

Solution:

Clearly $\sin^2 \omega t$ is a periodic function with period $\frac{\pi}{\omega}$



For SHM $\frac{d^2y}{dt^2} \propto -y$

$$y = \sin^2 \omega t = \frac{1 - \cos 2\omega t}{2}$$

$$= \frac{1}{2} - \frac{1}{2} \cos 2\omega t$$

$$v = \frac{dy}{dt} = \frac{1}{2} \times 2\omega \sin 2\omega t = 2\omega \sin \omega t \cos \omega t$$

$$= \omega \sin 2\omega t$$

Acceleration, $a = \frac{d^2y}{dt^2} = 2\omega^2 \cos 2\omega t$ which is not proportional to $-y$. Hence, it is not in SHM.

Question180

Two simple harmonic motions are represented by the equations

$y_1 = 0.1 \sin \left(100\pi t + \frac{\pi}{3} \right)$ and $y_2 = 0.1 \cos \pi t$ The phase difference of the velocity of particle 1 with respect to the velocity of particle 2 is [2005]

Options:

A. $\frac{\pi}{3}$

B. $\frac{-\pi}{6}$

C. $\frac{\pi}{6}$

D. $\frac{-\pi}{3}$

Answer: B

Solution:

Velocity of particle 1 ,

$$v_1 = \frac{dy_1}{dt} = 0.1 \times 100\pi \cos \left(100\pi t + \frac{\pi}{3} \right)$$

Velocity of particle 2

$$v_2 = \frac{dy_2}{dt} = -0.1\pi \sin \pi t = 0.1\pi \cos \left(\pi t + \frac{\pi}{2} \right)$$

$\therefore$ Phase difference of velocity of particle 1 with respect to the velocity of particle 2 is

$$= \phi_1 - \phi_2 = \frac{\pi}{3} - \frac{\pi}{2} = \frac{2\pi - 3\pi}{6} = -\frac{\pi}{6}$$

Question181

The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out, the time period of oscillation would [2005]

Options:

- A. first decrease and then increase to the original value
- B. first increase and then decrease to the original value
- C. increase towards a saturation value
- D. remain unchanged

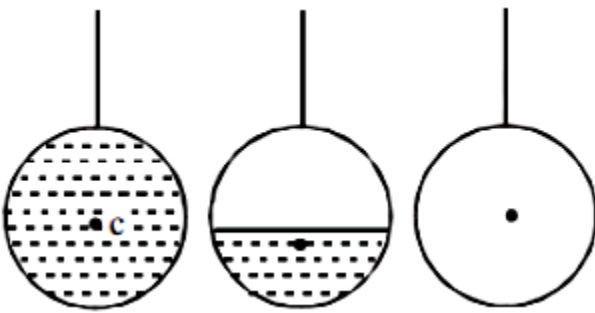
Answer: B

Solution:

When plugged hole near the bottom of the oscillating bob gets suddenly unplugged, centre of mass of combination of liquid and hollow portion (at position 1), first goes down (to 1 + Δl) and when total water is drained out, centre of mass regain its original position (to 1),

$$\text{Time period, } T = 2\pi \sqrt{\frac{l}{g}}$$

$\therefore$ 'T' first increases and then decreases to original value.



Question182

If a simple harmonic motion is represented by $\frac{d^2x}{dt^2} + \alpha x = 0$, its time period is
[2005]

Options:

- A. $\frac{2\pi}{\sqrt{\alpha}}$
- B. $\frac{2\pi}{\alpha}$
- C. $2\pi\sqrt{\alpha}$
- D. $2\pi\alpha$

Answer: A

Solution:

Standard differential equation of SHM is

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

Given equation is $\frac{d^2x}{dt^2} + \alpha x = 0 \therefore \omega^2 = \alpha$

or $\omega = \sqrt{\alpha}$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\alpha}}$$

Question 183

The total energy of a particle, executing simple harmonic motion is where x is the displacement from the mean position, hence total energy is independent of x [2004]

Options:

A. independent of x

B. $\propto x^2$

C. $\propto x$

D. $\propto x^{1/2}$

Answer: A

Solution:

At any instant the total energy in SHM is $\frac{1}{2}kA_0^2 = \text{constant}$,

where $A_0 = \text{amplitude}$

$k = \text{spring constant}$

hence total energy is independent of x .

Question184

The bob of a simple pendulum executes simple harmonic motion in water with a period t , while the period of oscillation of the bob is t_0 in air. Neglecting frictional force of water and given that the density of the bob is $\left(\frac{4}{3}\right) \times 1000 \text{ kg/m}^3$. Which relationship between t and t_0 is true?
[2004]

Options:

A. $t = 2t_0$

B. $t = t_0/2$

C. $t = t_0$

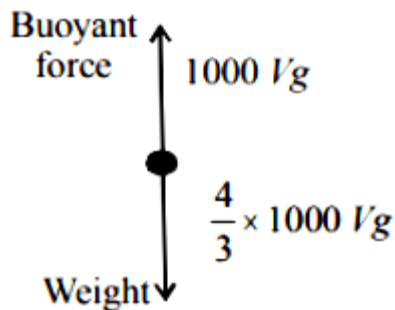
D. $t = 4t_0$

Answer: A

Solution:

Time period, $t = 2\pi \sqrt{\frac{1}{g_{\text{eff}}}}$

In air, $t_0 = 2\pi \sqrt{\frac{1}{g}}$



$$\text{Net force} = \left(\frac{4}{3} - 1\right) \times 1000Vg = \frac{1000}{3}Vg$$

$$g_{\text{eff}} = \frac{1000Vg}{3 \times \frac{4}{3} \times 1000V} = \frac{g}{4}$$

$$\therefore t = 2\pi\sqrt{l g/4} = 2 \times 2\pi\frac{1}{g}$$

$$t = 2t_0$$

Question 185

A particle at the end of a spring executes S.H.M with a period t_1 . while the corresponding period for another spring is t_2 . If the period of oscillation with the two springs in series is T then [2004]

Options:

A. $T^{-1} = t_1^{-1} + t_2^{-1}$

B. $T^2 = t_1^2 + t_2^2$

C. $T = t_1 + t_2$

D. $T^{-2} = t_1^{-2} + t_2^{-2}$

Answer: B

Solution:

Time period for first spring, $t_1 = 2\pi \sqrt{\frac{m}{k_1}}$,

Time period for second spring, $t_2 = 2\pi \sqrt{\frac{m}{k_2}}$

Force constant of the series combination $k_{\text{eff}} = \frac{k_1 k_2}{k_1 + k_2}$

$\therefore$ Time period of oscillation for series combination $T = 2\pi \sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$

$$\therefore T = 2\pi \sqrt{\frac{m}{k_2} + \frac{m}{k_1}} = 2\pi \sqrt{\frac{t_2^2}{(2\pi)^2} + \frac{t_1^2}{(2\pi)^2}}$$

$$\Rightarrow T^2 = t_1^2 + t_2^2$$

where x is the displacement from the mean position

Question186

In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force while the energy is maximum for a frequency ω_2 of the force; then

[2004]

Options:

- A. $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large
- B. $\omega_1 > \omega_2$
- C. $\omega_1 = \omega_2$
- D. $\omega_1 < \omega_2$

Answer: C

Solution:

As energy $\propto (\text{Amplitude})^2$, the maximum for both of them occurs at the same frequency and this is only possible in case of resonance.

In resonance state $\omega_1 = \omega_2$

Question187

A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0 . An external force $F(t)$ proportional to $\cos \omega t$ ($\omega \neq \omega_0$) is applied to the oscillator. The displacement of the oscillator will be proportional to

[2004]

Options:

A. $\frac{1}{m(\omega_0^2 + \omega^2)}$

B. $\frac{1}{m(\omega_0^2 - \omega^2)}$

C. $\frac{m}{\omega_0^2 - \omega^2}$

D. $\frac{m}{(\omega_0^2 + \omega^2)}$

Answer: B

Solution:

Equation of displacement in forced oscillation is given by

$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)^2}$$

$$= \frac{F_0}{m(\omega_0^2 - \omega^2)}$$

Here damping effect is considered to be zero

$$\therefore x \propto \frac{1}{m(\omega_0^2 - \omega^2)}$$

Question188

Two particles A and B of equal masses are suspended from two massless springs of spring constants k_1 and k_2 , respectively. If the maximum velocities, during oscillation, are equal, the ratio of amplitude of A and B is [2003]

Options:

A. $\sqrt{\frac{k_1}{k_2}}$

B. $\frac{k_2}{k_1}$

C. $\sqrt{\frac{k_2}{k_1}}$

D. $\frac{k_1}{k_2}$

Answer: C

Solution:

Maximum velocity during SHM, $V_{\max} = A\omega$ But $k = m\omega^2$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

$$\therefore \text{Maximum velocity of the body in SHM} = A \sqrt{\frac{k}{m}}$$

As maximum velocities are equal

$$\therefore A_1 \sqrt{\frac{k_1}{m}} = A_2 \sqrt{\frac{k_2}{m}}$$

$$\Rightarrow A_1 \sqrt{k_1} = A_2 \sqrt{k_2} \Rightarrow \frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}}$$

Question189

The displacement of a particle varies according to the relation $x = 4(\cos \pi t + \sin \pi t)$. The amplitude of the particle is [2003]

Options:

A. -4

B. 4

C. $4\sqrt{2}$

D. 8

Answer: C

Solution:

Displacement, $x = 4(\cos \pi t + \sin \pi t)$

$$= \sqrt{2} \times 4 \left(\frac{\sin \pi t}{\sqrt{2}} + \frac{\cos \pi t}{\sqrt{2}} \right)$$

$$= 4\sqrt{2}(\sin \pi t \cos 45^\circ + \cos \pi t \sin 45^\circ)$$

$$x = 4\sqrt{2} \sin(\pi t + 45^\circ)$$

On comparing it with standard equation $x = A \sin(\omega t + \phi)$

we get $A = 4\sqrt{2}$

Question190

A body executes simple harmonic motion. The potential energy (P.E), the kinetic energy (K.E) and total energy (T.E) are measured as a function of displacement x . Which of the following statements is true ?

[2003]

Options:

- A. K.E. is maximum when $x = 0$
- B. T . E is zero when $x = 0$
- C. K.E is maximum when x is maximum
- D. P.E is maximum when $x = 0$

Answer: A

Solution:

$$\text{K.E. of simple harmonic motion} = \frac{1}{2}m\omega^2(a^2 - x^2)$$

Question191

A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes SHM of time period T . If the mass is increased by m , the time period becomes $\frac{5T}{3}$. Then the ratio of $\frac{m}{M}$ is

[2003]

Options:

A. $\frac{3}{5}$

B. $\frac{25}{9}$

C. $\frac{16}{9}$

D. $\frac{5}{3}$

Answer: C

Solution:

With mass M , the time period of the spring.

$$T = 2\pi \sqrt{\frac{M}{k}}$$

With mass $M + m$, the time period becomes,

$$T' = 2\pi \sqrt{\frac{M + m}{k}} = \frac{5T}{3}$$

$$\therefore 2\pi \sqrt{\frac{M + m}{k}} = \frac{5}{3} \times 2\pi \sqrt{\frac{M}{k}}$$

$$\Rightarrow M + m = \frac{25}{9} \times M$$

$$\Rightarrow 1 + \frac{m}{M} = \frac{25}{9}$$

$$\Rightarrow \frac{m}{M} = \frac{25}{9} - 1 = \frac{16}{9}$$

Question192

The length of a simple pendulum executing simple harmonic motion is increased by 21%. The percentage increase in the time period of the pendulum of increased length is [2003]

Options:

A. 11%

B. 21%

C. 42%

D. 10%

Answer: D

Solution:

$$\text{Time period, } T = 2\pi \sqrt{\frac{l}{g}}$$

New length, $l' = 1 + 21\%$ of l

$$l' = 1 + 0.21l$$

$$\Rightarrow l' = 1.21l$$

$$T' = 2\pi \sqrt{\frac{1.21l}{g}}$$

$$\% \text{ increase in length} = \frac{T' - T}{T} \times 100$$

$$= \frac{\sqrt{1.21l} - \sqrt{l}}{\sqrt{l}} \times 100 = (\sqrt{1.21} - \sqrt{1}) \times 100$$

$$= (1.1 - 1) \times 100 = 10\%$$

Question193

**In a simple harmonic oscillator, at the mean position
[2002]**

Options:

- A. kinetic energy is minimum, potential energy is maximum
- B. both kinetic and potential energies are maximum
- C. kinetic energy is maximum, potential energy is minimum
- D. both kinetic and potential energies are minimum

Answer: C

Solution:

The kinetic energy (K . E .) of particle in SHM is given by,

$$K . E = \frac{1}{2}k(A^2 - x^2)$$

Potential energy of particle in SHM is $U = \frac{1}{2}kx^2$

Where A = amplitude and $k = m\omega^2$

x = displacement from the mean position

At the mean position $x = 0$

$$\therefore K.E. = \frac{1}{2}kA^2 = \text{Maximum and } U = 0$$

Question 194

If a spring has time period T , and is cut into n equal parts, then the time period of each part will be [2002]

Options:

A. $T\sqrt{n}$

B. $T/\sqrt{n}$

C. nT

D. T

Answer: B

Solution:

Let k be the spring constant of the original spring.

Time period $T = 2\pi\sqrt{\frac{m}{k}}$ where m is the mass of oscillating body.

When the spring is cut into n equal parts, the spring constant of one part becomes nk . Therefore the new time period,

$$T' = 2\pi\sqrt{\frac{m}{nk}} = \frac{T}{\sqrt{n}}$$

Question 195

A child swinging on a swing in sitting position, stands up, then the time period of the swing will [2002]

Options:

A. increase

B. decrease

C. remains same

D. increases if the child is long and decreases if the child is short

Answer: B

Solution:

The time period $T = 2\pi \sqrt{\frac{l}{g}}$ where l = distance between the point of suspension and the centre of mass of the child.

As the child stands up, her centre of mass is raised. The distance between point of suspension and centre of mass decreases ie length l decreases.

$\therefore l' < l$

$\therefore T' < T$ i.e., the period decreases.

